The exact extremal value in Erdős Problem 272 for $N \geq 4000$

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Abstract

Let $t(N)$ be the maximum size of a family of distinct subsets of $\{1, \ldots, N\}$ such that the intersection of any two distinct members is a nonempty arithmetic progression. We give a computer-assisted proof that

$$t(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor \qquad (N \geq 4000).$$

The upper bound combines an analytic argument for an infinite range with finite certificates for the remaining integers. The reductions retain the original family and a fixed maximum arithmetic-progression core in each non-progression member. A feedback argument couples singleton traces, member lengths, and short-progression counts to establish a minimum member size of fifteen in a putative non-starred extremal family for $N \geq 10000$. Separate arguments cover smaller minimum sizes and the entire unbounded minimum-size branches. We state the mathematical acceptance conditions for the finite certificates, describe exact verification and parameter coverage, and identify the accompanying mathematical and computational material. The lower bound is the construction of Szabó. No classification of all extremal families or determination of all values below 4000 is asserted.

Keywords: arithmetic-progression intersections; extremal set systems; computer-assisted proof; exact certificates.

Contents

1 Introduction and main result **3**

2 Dependency order and assembly of the ranges **4**

3 Arithmetic counting on the original family **6**

4 Small minimum members throughout $4000 \le N \le 6999$ **11**

5 A bridge at one fixed root when all members have size at least eight **19**

6 Excluding all minimum sizes at least eight **22**

7 Sufficiency and compression of the selected low-range certificates **26**

8 Completion of the proof **29**

A The exact bound with a prescribed common point **30**

B A refined position bound for selfclosed supports **32**

C An independent analytic tail without a minimum-member hypothesis **35**

D An acyclic minimum-size-nine reduction and its ambient domains **39**

E The minimum-size-fifteen reduction: joint feedback from an actual shortest member and the AP reservoir **41**

F A uniform finite proof for $10000 \le N \le 74999$ **49**

G A new finite connection from 75000 to the independent tail **53**

H The two complementary branches for 7000 to 9999 **58**

I Exact verification and reproducibility **64**

J Accompanying mathematical texts and certificate files **67**

References **69**

1 Introduction and main result

Write $[N] = \{1, \dots, N\}$. A family $\mathcal{F}$ of distinct subsets of $[N]$ is *admissible* if the intersection of any two distinct members is a nonempty arithmetic progression (AP). Singletons and two-element sets are APs. Members of $\mathcal{F}$ themselves need not be APs. Let $t(N)$ denote the largest cardinality of an admissible family, and put

$$B(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor. \tag{1}$$

A family containing the empty set has at most one member, so it does not affect the extremal values considered here. A family is *starred* if all its members contain a common point, called a root. A family is *near-extremal* in this paper if its size is at least $B(N)$; this terminology specifies an exact inequality, not an asymptotic estimate.

1.1 Background and related work

The problem appears in Erdős and Graham [1] and as Problem 272 in the Erdős problems database [4]. Simonovits and Sós proved a quadratic upper bound [2]. Szabó constructed a family of size $B(N)$ and proved an asymptotic formula with leading term $N^2/2$ [3]. The lower-bound construction is not new here. Yang proved the exact upper bound $B(N)$ under the common-point hypothesis and determined the unrestricted values for $3 \le N \le 12$ [5]. Kohlmeyer and Kruer reported the linear error bound $t(N) = N^2/2 + O(N)$ [6]. Li Xiang's draft dated October 30, 2026, and accompanying public submission report the exact formula for $N \ge 642498$ [7]. The present paper gives the explicit threshold 4000. These references distinguish the statements and ranges of the results; they do not constitute a determination of discovery priority.

1.2 The theorem and its computational component

Theorem 1.1 (Exact extremal value)**.** *For every integer $N \ge 4000$, one has $t(N) = B(N)$.*

The upper bound is computer-assisted. The combinatorial implications are proved below and in the accompanying technical notes. The remaining finite inequalities are evaluated using exact integer or rational arithmetic, or certified one-sided bounds. We specify both their acceptance conditions and their complete parameter ranges. The electronic mathematical texts and computational certificates are part of the proof material; this article alone is not a substitute for those files.

An important structural input is established in Appendix E, rather than adopted as an unproved threshold assertion.

Proposition 1.2 (Minimum member size)**.** *Let $N \ge 10000$. If $\mathcal{F}$ is an admissible non-starred family and $|\mathcal{F}| \ge B(N)$, then every member of $\mathcal{F}$ has at least fifteen elements.*

Proof. The independent infinite-range argument and small-transversal reductions first give minimum size at least nine. The feedback argument in Appendix E increases this to fifteen for $10000 \le N \le 11993$. The 506 integers $11994 \le N \le 12499$ are checked by the stated exact criterion, and an analytic argument applies for $N \ge 12500$. None of these steps uses Theorem 1.1 or the finite-range theorem whose proof requires the fifteen-element input. □

Proposition 1.3 (Finite verification)**.** *The selected finite certificates satisfy the acceptance inequalities, dependency conditions, strict terminal inequalities, and coverage conditions stated in Section 7 and Appendices F and G for all their stated parameters.*

Proof. The cited sections reduce the required conclusions to finite records with explicit sufficient acceptance conditions. The complete selected records, including the ancestors required by their recurrences, were evaluated over their declared integer ranges. Closed rational subdivisions verify the continuous gates; analytic monotonicity separately handles the rays. Appendix I records the coverage and arithmetic, and Appendix J identifies the reproducible inputs. This is a finite computation, not an enumeration of admissible set families. □

1.3 Organization and scope

The proof covers $[4000, 6999]$, $[7000, 9999]$, and $[10000, \infty)$. The last range splits into two finite ranges and an independent analytic tail. For $4000 \le N \le 6999$, let $k = \min_{A \in \mathcal{F}} |A|$.

Minimum k	Mechanism	Conclusion
$k \le 7$	An actual member selects a root; core and incompatible-trace counts	$\lvert\mathcal{F}\rvert \le B(N)$
$k = 8$	Population and length propagation at one fixed root	$\lvert\mathcal{F}\rvert < B(N)$
$9 \le k \le 63$	A conditional root argument followed by a global restart	$\lvert\mathcal{F}\rvert < B(N)$
$k \ge 64$	Monotonicity at the actual minimum of the tail family	$\lvert\mathcal{F}\rvert < B(N)$

The first branch excludes $|\mathcal{F}| > B(N)$; the other three exclude $|\mathcal{F}| \ge B(N)$. We preserve this distinction. Auxiliary subfamilies do not inherit the size hypothesis of the original family. Each original non-AP member retains the maximum AP core selected for it at the beginning. The argument does not determine all values for $N < 4000$ and does not classify every extremal family.

1.4 The matching lower bound

For a specified root $p \in [N]$, the starred extremal value is

$$B_p(N) = \binom{N}{2} + 1 + \left\lfloor \frac{\min(p-1, N-p)}{2} \right\rfloor.$$

The construction in Appendix A, with $p = \lfloor (N+1)/2 \rfloor$, attains $B(N)$. Thus only the upper bound remains. For the common-point problem, see also [5].

2 Dependency order and assembly of the ranges

The infinite-range argument comes first. Small transversals then give a minimum size of nine, and joint feedback gives a minimum size of fifteen. Only afterwards do we use the finite non-starred exclusion arguments that require this minimum size. The two smaller ranges use separate reductions. Historical thresholds 49000, 49100, and 39500 are not required junctions between the finite certificates in this proof.

2.1 The starred case

Proposition 2.1 (Exact value at a fixed root)**.** *For every $N \ge 1$ and $p \in [N]$, the largest admissible family all of whose members contain p has size $B_p(N)$. A central root attains $B(N)$.*

Proof. Appendix A gives the private incompatible-pair argument, staircase count, finite checks, analytic remainder, and construction. □

2.2 The structural reductions precede the finite exclusions

Appendix C excludes an original non-starred near-extremal family for $N \geq 1025001$, without assuming a minimum size of nine or fifteen. Appendix D then proves the minimum-nine reduction on its remaining local domain for $N \geq 10000$. Appendix E proves the minimum-fifteen reduction. These steps do not pass through any finite ROOT or MASS conclusion that already assumes minimum size fifteen.

2.3 The middle and upper ranges

Theorem 2.2. *For every integer $N \geq 10000$, one has $t(N) = B(N)$.*

Proof. Suppose that $|\mathcal{F}| > B(N)$. The fixed-root bound implies that $\mathcal{F}$ is non-starred. If $N \geq 1025001$, the independent analytic tail gives a contradiction directly. Otherwise the minimum-fifteen reduction applies. For $75000 \leq N \leq 1025000$, use the ROOT160 and three-stage MASS certificates of Appendix G. For $10000 \leq N \leq 74999$, use the complete minimum-size classification and ordinary-member feedback in Appendix F. These finite ranges meet at 74999/75000; the upper range meets the analytic tail at 1025000/1025001. Each unbounded minimum-size branch is handled at its actual minimum by monotonicity. The construction gives the reverse inequality. □

Proposition 2.3 (Distinguishing two threshold statements)**.** *For $N \geq 49000$, one has $t(N) = B(N)$. The minimum-fifteen statement applies already for $N \geq 10000$.*

Proof. The first assertion is a restriction of Theorem 2.2; the second is Proposition 1.2. Neither is obtained by extending the other's parameter range downwards. □

2.4 The range from seven thousand to ten thousand

Theorem 2.4. *Let $7000 \leq N \leq 9999$. If an admissible family has actual minimum member size at most ten, its size is at most $B(N)$. If its actual minimum size is at least eleven, its size is strictly less than $B(N)$.*

The proof is in Appendix H and the accompanying local mathematical texts. Its finite component includes 27000 small-member parameter cases, all closed boxes and real rays, and all required ancestors and physical resources for 816 large-member certificates. Appendix H.4 supplies a uniform replacement for the CORE gate. The small-member proof retains the condition that every actual two-element member contains the same root. The large-member proof separates the conditional ROOT stage from the unconditional GLOBAL stage; the entire minimum-size branch beginning at 133 is handled by monotonicity. This range does not use the minimum-fifteen proposition.

Theorem 2.5. *For every integer $N \geq 7000$, one has $t(N) = B(N)$.*

Proof. For $7000 \leq N \leq 9999$, empty families and families containing the empty set are immediate. Otherwise the actual minimum is attained and is either at most ten or at least eleven. Apply Theorem 2.4 and the lower-bound construction. For $N \geq 10000$, use Theorem 2.2. □

It remains to cover $4000 \leq N \leq 6999$, with actual minimum size at most seven, equal to eight, between nine and sixty-three, or at least sixty-four.

3 Arithmetic counting on the original family

We state the counting interfaces used in the final range, together with the arguments that turn them into finite certificates. The underlying capacity theorems, including their individual hypotheses, are given in the technical sources [26, 27, 28, 29, 30]. A numerical bound proved for a larger value of N is never extrapolated to a smaller one. The refined physical position kernels and typed fringe inventories used by these interfaces are proved in Appendix B; they are not assumed merely because their formulas can be evaluated.

3.1 Fixed maximum progressions and the full AP inventory

An *ordinary member* is an original member that is an interval of integers. For every non-AP member A of the original family, fix, once and for all, a maximum-cardinality arithmetic progression $I_A \subseteq A$, and write

$$k_A = |I_A| \geq 2, \qquad j_A = |A| - k_A \geq 1, \qquad d_A = \mathrm{step}(I_A).$$

The maximum is taken over progressions of all steps. Ties may be resolved arbitrarily, but the same choice of I_A is retained in every partition, recurrence and cost calculation. Let C be the number of all original non-AP members, and let O be the number of all ordinary members.

For integers $v \geq a \geq 2$, define

$$f_a(v) = \sum_{h=a}^{v} \min(h, v+1-h), \qquad f_a(v) = 0 \quad (v < a),$$

and

$$R_a(N) = \sum_{d=1}^{\lfloor (N-1)/(a-1) \rfloor} f_a\Big(\Big\lfloor \frac{N-1}{d} \Big\rfloor + 1\Big). \tag{2}$$

Lemma 3.1 (Full AP inventory)**.** *If every member of the original family has size at least $a \geq 2$, its number of AP members is at most $R_a(N)$, and $O \leq f_a(N)$. If also $|\mathcal{F}| \geq B(N)$, then*

$$C \geq B(N) - R_a(N), \qquad O \geq B(N) - R_a(N) + f_a(N) - C. \tag{3}$$

Proof. Fix a step d. All actual AP members of step d lie in a single residue class modulo d, since members in different classes would be disjoint. On that residue-class chain they are pairwise intersecting intervals, and therefore have a common chain point. The chain has length at most $v = \lfloor (N-1)/d \rfloor + 1$. At any specified chain point, at most $\min(h, v+1-h)$ intervals of length h contain that point; a central chain point attains these maxima simultaneously. Summing proves the inventory bound. Separating the $d = 1$ term gives $|\mathcal{F}| \leq C + O + R_a(N) - f_a(N)$ and hence the second inequality in (3). □

On an integer box $L \leq N \leq H$, set

$$D_a = \min_{L \leq N \leq H} \{B(N) - R_a(N)\}, \qquad E_a = \min_{L \leq N \leq H} \{B(N) - R_a(N) + f_a(N)\}. \tag{4}$$

Both minima are evaluated exactly at every integer N in the box; they may occur at different integers. No endpoint monotonicity is assumed, and inventory terms belonging to different values of N are never subtracted. Thus $C \geq D_a$ and $O \geq E_a - C$ throughout the box.

3.2 Truncated-defect resources on the full ambient interval

All capacities for a box are evaluated in the original ambient interval $[H]$. Put $\Gamma = H - 2$. Write $\tau(g)$ for the divisor function, $d_3(g)$ for the number of ordered positive triples with product g, and φ for Euler's totient function. The truncated original-defect pair theorem gives, for every actual pair e and every positive integer q,

$$\sum_{\substack{A \text{ an original non-AP member} \\ e \subseteq A}} \min(j_A, q) \le \Gamma + \beta_q(\mathrm{gap}(e)), \tag{5}$$

where

$$\begin{aligned} C_0(g) &= 2(d_3(g) - \tau(g)), \\ \beta_q(g) &= \min\{H_q(g), (q-1)C_0(g)\}, \\ H_q(g) &= \sum_{\substack{b \mid g \\ b>1}} \max\{0, q[\rho_g(b) + 2\eta_g(b)] - \varphi(b)\}. \end{aligned} \tag{6}$$

Here $\rho_g(b)$ counts the distinct prime divisors of g that are smaller than the least prime divisor of b; $\eta_g(b)$ counts the distinct prime divisors of b that are larger than the least prime divisor of g. Set $\beta_q(1) = 0$. In particular, (5) concerns the original defects j_A; it does not replace them by defects of modified sets.

The full vertex resource is

$$\mathrm{BH}_q(H) = \max_{0 \le x \le H-1} \left\{ \sum_{g=1}^{x} \beta_q(g) + \sum_{g=1}^{H-1-x} \beta_q(g) \right\}.$$

For a different purpose, form a fixed multiset containing one copy of $\beta_q(g)$ for every $g \le H-1$, and a second copy exactly when $2g \le H-1$. Let $\mathrm{Top}_q(s)$ be the sum of its s largest entries, with zero padding when necessary. At every actual vertex, the incident resource list is a submultiset of this fixed multiset. Consequently Top_q is nondecreasing and discretely concave. This construction does not take the maximum of separately optimized functions associated with different vertices.

A separate resource controls the number of non-AP members through an arbitrary point:

$$S(n) = 2 \sum_{d \le n} (\tau(d) - 1) \left\lfloor \frac{n}{d} \right\rfloor, \qquad n = H - 1.$$

The reserved-unit pair-cost argument yields this resource, rather than a bound on an unweighted simple graph. Superadditivity of the nonnegative divisor convolution allows the resources on both sides of a point to be bounded by a single $S(n)$. Hence the number of original non-AP members of size at least $a \ge 3$ through any specified point is at most

$$\mathrm{Rank}_a(H) = \left\lfloor \frac{(H-1)(H-2) + S(H-1)}{2(a-2)} \right\rfloor. \tag{7}$$

This applies to an arbitrary subfamily through that point and requires no near-extremal density assumption on the subfamily. The resource in (7) is distinct from that of the mixed-cut argument below. In the low-range calculations with $H \le 7500$, the latter may use the full-ambient constants $D^*_{5500} = 245873$ for $H \le 5500$ and $D^*_{7500} = 365931$ for $5500 < H \le 7500$. These constants are valid in their stated ambient intervals, including for original sets embedded in those intervals; they are not asserted for $H > 7500$.

3.3 Common-fringe geometry and a basic capacity

We give a derivation of one basic capacity, in addition to specifying the computational interface. Consider a subfamily $\mathcal{G}$ of original non-AP members with a common fixed-core step d. Suppose every member has size at least $r+1$, its original defect lies in $[\ell, u]$, and $k_0 = r+1-u \geq 4$. Write $D_A = A \setminus I_A$, and, for each ambient point z, put

$$a_z = |\{A \in \mathcal{G} : z \in D_A\}|.$$

This is a common-fringe degree, not the total number C of non-AP members.

Lemma 3.2 (Common-fringe geometry; CORECAP)**.** *Fix z, and consider the original members with $z \in D_A$. Within each residue class modulo d, their fixed cores do not contain one another; ordering their first points strictly orders their last points in the same direction. In any one residue class, at most one pair of these cores has an intersection of size at least two. Every ambient point belongs to at most two such cores, and every pairwise core intersection has size at most $u+1$.*

Proof. If $I_A \subseteq I_B$, the AP intersection $A \cap B$ contains I_A and z, contradicting the maximum cardinality of I_A within A. This proves the strict ordering of both endpoints.

Suppose two ordered cores I_i, I_j overlap in $w \geq 2$ points. The step h of the original intersection $P = A_i \cap A_j$ divides d. If z is in the same residue class, it lies outside the overlap interval; the progression P then contains an entire one of the two cores together with z, again a contradiction. Thus $h < d$, so $h \leq d/2$, and

$$2w - 1 \leq |P| \leq \min(|I_i|, |I_j|).$$

The corresponding span bounds imply $m_i \leq z \leq m_j$, where m_i, m_j are the physical midpoints of the two cores.

A core cannot have such an overlap on both sides: its midpoint would have to equal z. Both span inequalities would then be equalities, and both intersections would have step $d/2$ and the size of the middle core. Together they would form a larger progression inside the middle original member. Thus a long overlap cannot pass over an intermediate core, and two long overlaps cannot share a core. Two separated long overlaps are also impossible, by the strict midpoint ordering and the inequalities $m_i \leq z \leq m_j$. If three cores contained a common point, the middle one would overlap each of its neighbours in at least two points, which has just been excluded. Finally, if an intersection has step d/q with $q \geq 2$, each gap between consecutive overlapping core points contains $q-1$ points in each original fringe. Hence $(w-1)(q-1) \leq u$, giving $w \leq u+1$. □

Define

$$\mathrm{pack}(H, d, t) = \sum_{c \bmod d} \left\lfloor \frac{|[H] \cap (c + d\mathbb{Z})|}{t} \right\rfloor.$$

Lemma 3.2 gives the following four bounds:

$$\begin{aligned}
a_z &\leq \mathrm{pack}(H, d, \lfloor k_0/2 \rfloor), \\
a_z &\leq \left\lfloor \frac{2(H-1)}{k_0} \right\rfloor, \\
a_z &\leq \mathrm{pack}(H, d, k_0 - 1) + d - 1, \\
a_z &\leq \mathrm{pack}(H, d, k_0 - u - 1) \qquad (k_0 > 2u + 2).
\end{aligned}$$

For the first, the initial half-cores are disjoint. The second counts core-point incidences, using depth at most two and the absence of z from every core. For the third, remove at most one core

from each residue class in which a long overlap is possible, then delete the last point of each remaining core and pack. The residue class of z contributes no exception, giving $d-1$. The last bound follows by trimming the possible overlap of at most $u+1$ points. Sharper residue-class and repeated-edge inventory bounds are given in [25]. The minimum of all applicable bounds provides an integer C_{fr} satisfying $a_z \le C_{\mathrm{fr}}$.

Lemma 3.3 (Basic full-band capacity)**.** *Put $M = \lceil H/d \rceil$, and let C_{fr} be any valid uniform common-fringe degree bound. Then $m = |\mathcal{G}|$ satisfies*

$$m \le \left\lfloor \frac{A_0 + \sqrt{A_0^2 - 16H^2}}{2\ell} \right\rfloor, \qquad A_0 = 4H + \ell u C_{\mathrm{fr}} + \min(4uH, 2\ell C_{\mathrm{fr}} M).$$

Proof. Concatenate the ambient residue-class chains. Each original core is an interval in the resulting total order. If $m > 0$, let $\omega \ge 1$ be the actual clique number of their interval graph. A clique has an actual common point. The common-fringe geometry bounds the total fringe incidence within such a clique by $2(H-1)$, so $\ell\omega \le 2(H-1)$ and in particular $\omega \le W = 2H/\ell$. An interval graph with m vertices and clique number ω has at most $(\omega-1)m - \omega(\omega-1)/2$ edges, as follows by removing intervals in increasing order of their right endpoints. It therefore has at least $(m-\omega)^2/2$ unordered pairs with disjoint cores.

The corresponding original members must still intersect, either between one fringe and the other core or between the two fringes. With $b_z = |\{A : z \in I_A\}|$, the incidence bounds are

$$\sum_z a_z b_z \le \min(u\omega, C_{\mathrm{fr}} M)m, \qquad \sum_z \binom{a_z}{2} \le u C_{\mathrm{fr}} m/2.$$

Consequently

$$(m-\omega)^2 \le [uC_{\mathrm{fr}} + 2\min(u\omega, C_{\mathrm{fr}} M)]m.$$

First solve this inequality using the actual ω. Put $b(x) = uC_{\mathrm{fr}} + 2\min(ux, C_{\mathrm{fr}} M)$. Its upper root is

$$R(x) = x + \tfrac{1}{2} b(x) + \tfrac{1}{2}\sqrt{b(x)^2 + 4xb(x)}.$$

Since b is nonnegative and nondecreasing, R is nondecreasing on $x \ge 0$. Thus $m \le R(\omega) \le R(W)$. Substituting $W = 2H/\ell$ gives the claimed expression. In particular, the argument does not substitute an upper bound for ω directly into a negative quadratic term. For an empty family the asserted nonnegative bound holds directly. □

3.4 Precise conditions on the capacity interface

Write

$$\mathrm{Cap}(H; r, d, \ell, u; P, b)$$

for a valid upper bound on the entire original subfamily with size at least $r+1$, fixed maximum-core step d, and original defect in $[\ell, u]$. The parameters satisfy

$$1 \le \ell \le u, \qquad k_0 = r + 1 - u \ge 4.$$

Here P must already be a proved lower bound for the number of members of the *entire original ordinary subfamily*, all of length at least b. When $P = 0$, no additional owner, interval-union or central-location information is available.

We use the proved full bounds in [27, 26, 30]. The following conditions specify the mathematical scope of the calls made here.

1. If $d(k_0-1) > H-1$, the class is empty. Packing, two-layer contact, endpoint and repeated-edge bounds use the original ambient interval. The packing option with denominator k_0-u-1 requires $k_0 > 2u+2$.

2. The basic quadratic bound uses

$$A_0 = 4H + \ell u C_{\mathrm{fr}} + \min(4uH, 2\ell C_{\mathrm{fr}} M), \qquad M = \lceil H/d \rceil,$$

and

$$\left\lfloor \frac{A_0 + \sqrt{A_0^2 - 16H^2}}{2\ell} \right\rfloor.$$

Here C_{fr} bounds a common-fringe degree, not C. Parameterized refinements retain the same complete second-moment account and total resource $H(H-1)$. Every additional condition on a heavy-residue or high-density branch must also hold.

3. An actual ordinary owner is inferred only from a previously proved $P > 0$. Let λ be the least integer in $[b, H]$ such that

$$f_b(H) - f_{\lambda+1}(H) \geq P.$$

Then some actual ordinary member has length at least λ. Put $\theta = 1$ when $d = 1$, and otherwise put $\theta = 1 + \lfloor u/(\mathrm{spf}(d) - 1) \rfloor$. The direct-exclusion branch requires all four conditions

$$k_0 > \theta, \quad k_0 - 1 - \theta > 0, \quad d(k_0 - 1 - \theta) > H - \lambda - 1, \quad d(\theta + 1) \leq \lambda.$$

4. An interval union has its own population inverse. For $m \geq b$, if $P - [f_b(H) - f_m(H)] > 0$, the union of the original ordinary members of length at least m has length at least the least v satisfying $f_m(v) \geq P - [f_b(H) - f_m(H)]$. Such a union is not automatically an actual member and cannot receive an additional reserve available only to an actual owner.

5. The exact-defect-one options are used only when $\ell = u = 1$, within their domain $r \geq 4$. In particular the calls with $r = 7$ below lie in this domain.

6. The central branch requires $\ell \geq 2$ and $8P \geq (H+1)^2$, and uses

$$h = H + 1 - \left\lceil \frac{4P}{H+1} \right\rceil, \qquad w = H + 1 - \left\lceil 2\sqrt{P} \right\rceil.$$

The small and large classes retain their respective complete resources and additional hypotheses. Signed constants are not replaced by zero; floors and ceilings of negative numerators have their mathematical meanings.

7. Complete upper bounds may be minimized. Bounds may be added only over a proved disjoint, exhaustive partition of the original members. No option is evaluated on a substitute family obtained by deleting points, recolouring, or reselecting maximum AP cores.

These are the call conditions used in this proof. The complete formulas and proofs belong to the cited capacity sources, which are explicitly identified in the source guide. A returned numerical value is not itself a proof of the corresponding capacity theorem.

3.5 Finite prefixes and forward-only inference

For a fixed d, let $X_{d,j}$ count the original non-AP members under consideration with defect at most j. Nonnegativity gives

$$X_{d,j} \le X_{d,p} + \mathrm{Cap}(H; r, d, p+1, u; P, b), \qquad 0 \le p < j \le u \le J. \tag{8}$$

The selected low-range certificates use three sufficient choices: the direct band $[1, j]$, the preceding prefix plus $[j, j]$, and the defect-one prefix plus $[2, j]$. Their band endpoints satisfy $u = j$. The middle- and high-range certificates may use the more general $u \ge j$ in (8); in that case the actual band endpoint u is retained in every capacity call and ancestor check. In either case the parent index strictly decreases, so induction on j proves every selected bound. Optimality among all possible choices is irrelevant. All calls here satisfy $r - J \ge 3$. Every $d \le \lfloor (H-1)/(r-J) \rfloor$ is represented; larger steps are excluded by the strict physical-span condition $d(r-j) > H-1$.

A lower bound P on the ordinary population may be increased only after a complete counting step has been justified, and the new value is used only in the next step. Values obtained under a temporary contradiction assumption retain that assumption. Once the assumption is discharged, only the proved structural conclusion is retained, and its conditional population and capacity information is discarded. Both phases below observe this rule.

4 Small minimum members throughout $4000 \le N \le 6999$

All families in this section are original admissible families: members are distinct, intersections of distinct members are nonempty integer APs, and the members themselves need not be APs. Write

$$n = N - 1, \qquad b(N) = \binom{N}{2} + 1, \qquad B(N) = b(N) + \left\lfloor \frac{n}{4} \right\rfloor, \qquad \beta = \frac{7}{80}.$$

The minimum member is an actual original member of minimum full cardinality, not a trace, replacement or compressed set.

Theorem 4.1 (Small minimum members)**.** *For every integer* $4000 \le N \le 6999$*, an admissible family* $\mathcal{F}$ *that is empty or has an actual minimum member of size at most seven satisfies* $|\mathcal{F}| \le B(N)$*.*

We give the argument using the complete V130 reconstruction of the V128 small-minimum theorem [15, 16, 17]. Its numerical constants are weaker but sufficient, and its certificates are kept as one coherent set. No minimum-fifteen premise from $N \ge 10000$ is used, and no classification of all extremal families as stars is asserted.

4.1 Structural inputs on the original family

Fix a root $p \in [N]$. Let $\mathcal{E}$ be the entire class of actual members avoiding p, let $\mathcal{I} \subseteq \mathcal{E}$ consist of its AP members, put $I = |\mathcal{I}|$, and let c count its non-AP members. Let b_p count actual pairs through p and H_3 count actual rooted members of size at least three. The local structural lemmas and their precise finite scopes are as follows; their full proofs are supplied with the V130 technical material [17].

1. **Actual-contact injection and arbitrary-root rank.** Translate p to zero. For distinct nonzero u, v, define

$$\mathrm{Hull}(u, v) = \{jg : \min(0, u, v) \le jg \le \max(0, u, v)\}, \qquad g = \gcd(|u|, |v|).$$

An intrinsic pair of X is a pair $\{u, v\} \subset X$ whose hull is not contained in X. A rooted non-AP member has an intrinsic pair, private among all other actual rooted members: an AP intersection containing $0, u, v$ would otherwise contain the missing hull. A normalized pair is primitive when its gcd is one, and is BAD when $\{0, u, v\}$ is not an AP. There is no privacy assertion for arbitrary non-AP triples.

If $A \in \mathcal{E}$ has size a, the sum of rooted non-AP members and rooted AP triples is at most

$$K_N(a) = an - \frac{a(a+1)}{2} + T_N(a). \tag{9}$$

The first two terms count exactly the nonroot pairs touching A. A row without such an intrinsic pair is paid at an actual contact point by the physical selfclosed-support cost $T_N(a)$. Selfclosed at p, v means that every hull $\mathrm{Hull}(p, v, y)$ generated by a point of the member is contained in it. The physical hosts 5500 and 7500 have complete vertex-sum and pair-cost bounds

$$(D_*, C_*) = (245873, 709), \qquad (365931, 798),$$

respectively. Use the first host for $N \le 5500$ and the second otherwise. For $T_N(a)$ take the minimum of that host's full sum and the fixed 7500-host largest-a sum. Above table length 2048, the latter is bounded by

$$T_{2048} + (a - 2048)\lfloor T_{2048}/2048 \rfloor.$$

This is the average bound for decreasing order statistics; K_N is nondecreasing. At any specified root, a non-AP subfamily whose members all have full size at least $m \ge 3$ also satisfies

$$\#\{\text{members of the subfamily}\} \le \left\lfloor \frac{n(n-1) + D_*}{2(m-2)} \right\rfloor. \tag{10}$$

The physical arrays are explicit computational resources; their complete recomputation is recorded separately in the computation supplement.

2. **Finite rooted hole loss (GFL).** For $2 \le N \le 10000$, all actual rooted members of size at least three and each compatible actual avoider A satisfy

$$H_3 \le \binom{n}{2} - \beta h_A(h_A - 1) + \frac{18}{5}n, \qquad h_A = n - |A|. \tag{11}$$

If there is no actual rooted triple, then $H_3 \le (33/40)\binom{n}{2} + (18/5)n$. The input includes its primitive-pool form, its finite rectangle certificate for spans $0, \dots, 9999$, and the six-shape classification of long members with exactly one intrinsic pair.

3. **Owner bounds and actual bad-trace packing.** These apply at integers $4 \le N \le 10000$, with no empty or singleton member, and with every actual pair containing the same p. The class $\mathcal{I}$ must contain all actual AP avoiders and have an actual witness wholly to the right of p; its extremal coordinates are defined from all of $\mathcal{I}$. We use the zero-span, positive-span error and packing statements in the precise forms below (V88 and V132). If the entire nonempty avoider class shares an adjacent pair, the interval-core theorem, LOCAL90B, gives $|\mathcal{F}| \le b(N)$; it does not require the entire family to be a star.

The AP stock of the whole family is charged once. For integers $k \ge 3$, $m \ge 1$, put

$j = \lfloor (m+1)/2 \rfloor$ and define

$$L_k(m) = 0 \quad (m < k),$$
$$L_k(m) = j(m-j+1) - \sum_{\ell=1}^{k-1} \min(j, m-j+1, \ell, m-\ell+1) \quad (m \geq k),$$
$$R_k(N) = \sum_{d=1}^{\lfloor n/(k-1) \rfloor} L_k(\lfloor n/d \rfloor + 1). \tag{12}$$

Actual APs of one common difference lie in one residue chain; their pairwise intersecting chain intervals have a common point. On a chain of m points the center maximizes the number of intervals of length at least k. Subtracting the shorter lengths gives L_k; in particular $L_4(4) = 1$, so the boundary term is retained. Thus R_k bounds all AP members of size at least k, without being charged again at each contact point.

4.2 Three fixed-root closure cases

We first prove tools independent of a unique-pair conclusion. Fix $4000 \leq N \leq 6999$, assume there are no empty or singleton members, $\mathcal{E} \neq \varnothing$, every actual pair contains p, and every avoider has full size greater than $n/K + 1$. Put $e = c + b_p$. The three cases are

Case	K	Exception budget	AP-avoider difference bound
CORE	15	$e + \lfloor n/4 \rfloor \leq (3/2)n$	14
SMALL	13	$e \leq (50408/4005)n$	12
NO3	14	$e \leq 41n$, no actual rooted triples	13

CORE excludes $|\mathcal{F}| > b(N)$; the other cases exclude $|\mathcal{F}| > B(N)$. Only for CORE define the formal count

$$f = |\mathcal{F}| + \lfloor n/4 \rfloor = H_3 + I + e_{\text{eff}}, \qquad e_{\text{eff}} = e + \lfloor n/4 \rfloor.$$

In the other cases let $f = |\mathcal{F}|$ and $e_{\text{eff}} = e$. No new family is formed. Every contradiction hypothesis gives $f > B(N)$, and exceptions are restored exactly once; below write $e_{\text{eff}} \leq Cn$.

No AP avoiders. If $I = 0$, (11) gives, for every actual avoider,

$$\beta h_A(h_A - 1) < (C + 47/20)n - 1/4.$$

For CORE and SMALL the quadratic

$$\frac{\beta}{16}(n-7)(n-11) - (C + 47/20)n + 1/4$$

has positive leading coefficient, value and derivative at 3999, and is therefore positive on the complete real ray $n \geq 3999$. It follows that every avoider has size greater than $(3N+4)/4$. Two distinct avoiders intersect in an AP of size greater than $\lceil N/2 \rceil$, hence in an ordinary interval containing a fixed central adjacent pair of $[N]$. Fixing one avoider makes this pair common to the whole class. A single avoider has an adjacent pair directly by its density. LOCAL90B gives $|\mathcal{F}| \leq b(N)$. For NO3 the positive quadratic

$$\beta n^2 + (93/80 - 41 - 18/5)n + 1/4 > 0$$

gives the contradiction directly. No fictitious AP witness is introduced.

All AP avoiders straddle the root. Suppose $I \neq 0$ and no AP avoider is wholly on one side of p. Their common differences are at least two. A witness of minimum actual difference d has nonzero root-coordinate residue modulo d and forces every rooted member to contain a nonmultiple of d. Put

$$M_d = \lfloor (p-1)/d \rfloor + \lfloor (N-p)/d \rfloor \geq n/d - 2.$$

If every intrinsic pair of a rooted non-AP member lay in the pool whose two coordinates are divisible by d, a signed nonmultiple gcd anchor could be combined repeatedly with the other points. Every new anchor divides the original nonmultiple, and closure would fill both extremes, a contradiction. Thus some intrinsic pair lies outside that pool. The common difference of a rooted AP cannot be divisible by d either; its physical primitive-pair gcd is its own difference, so its pool also avoids the multiples pool.

The required primitive matching statement is for every actual subcollection of normalized long rooted intervals: its covered primitive BAD pairs match the intervals with at most one dummy place, and that place is allowed only if an actual interval of the group contains $[-2, 2]$. After excluding positive odd–odd pairs, at most four dummy places are needed. This is a statement about every subcollection, not merely its enclosing rectangle. Downward closure and missing-cell injections reduce it to coprime counts. All 50086 ordered rectangles $L, R \geq 2$, $L + R < 320$ are checked exactly; the range $L + R \geq 320$ follows from a positive quadratic obtained by a common-prime union bound, and thin boundaries have direct injections [17]. Distinct physical differences give disjoint gcd pools; intrinsic privacy separates them from the non-AP pools. Consequently

$$H_3 \leq \binom{n}{2} - \binom{M_d}{2} + n/4.$$

At $d = 3$ retain the complete parity split. If every avoider difference is odd, use only odd differences in its stock. Otherwise choose an actual even-difference avoider. Odd coordinates permit the stronger $d = 2$ reserve. Even coordinates, together with the step-three witness, permit reserving the union of the multiples-of-three pool and the positive odd–odd pool. It contains at least $n^2/12 - 23n/24 + 25/8$ pairs, with at most n matching dummy places. The even point and the nonmultiple of three may come from different actual witnesses; the closure argument does not assume they are the same specified point.

For the complete finite comparisons, write

$$J(s,v) = \begin{cases} 0 & s \leq v, \\ (s-v)^2/2 & v \leq s \leq 2v, \\ s^2/4 - v^2/2 & s \geq 2v, \end{cases} \quad S = \sum_{e \in U} J(1/e, 1/K), \quad H_U = \sum_{e \in U} 1/e, \quad a = 1 - 1/d.$$

At each actual difference e, all AP avoiders share a chain point. Their endpoint unit squares lie above $i + j > n/K$ in a rectangle whose side sum is at most $n/e + 2$. Balancing the sides maximizes this area. Thus the complete inventory satisfies

$$I \leq Sn^2 + 2H_U n + 2|U|.$$

For CORE and SMALL the loss polynomial $An^2 + B_1 n + C_1$ has the following complete list of coefficients:

Actual case (U)	A	B_1	C_1
$d = 2$ $(2, \ldots, K-1)$	$1/8 - S$	$1 - 5/4 - C - 2H_U$	$13/4 - 2\|U\|$
$d = 3$, all odd $(3, 5, \ldots)$	$1/18 - S$	$1 - 5/6 - C - 2H_U$	$13/4 - 2\|U\|$
$d = 3$, even anchor $(3, \ldots, K-1)$	$1/12 - S$	$1/4 - 23/24 - C - 2H_U$	$27/8 - 2\|U\|$
$d \geq 4$ $(d, \ldots, K-1)$	$\beta a^2 - S$	$-3\beta a - C - 47/20 - 2H_U$	$2\beta + 1/4 - 2\|U\|$

The last row uses the actual witness's $h \geq an - 1$ and (11). For NO3, complete long non-AP intrinsic pools are charged at rate $1/2$; the six single-intrinsic-pair shapes add at most $11n/24$. Long AP pools have rate $33/40$ and avoid the multiples pool, retaining a saving of $13/40$ there. Set

$$c_W = \frac{1}{40}\frac{285139977338150579}{2079149797771200} + \frac{11}{24}.$$

For every $d = 2, \ldots, 13$, $U = \{d, \ldots, 13\}$, the coefficients are

$$A = \beta + \frac{13}{80d^2} - S, \qquad B_1 = \frac{93}{80} - \frac{13}{16d} - 41 - c_W - 2H_U, \qquad C_1 = \frac{49}{40} - 2|U|.$$

There are $14 + 12 + 12 = 38$ explicit rational quadratics. The complete V130 coefficient certificate gives $A > 0$, $P(3999) > 0$ and $P'(3999) > 0$ for each; even the tightest NO3 case, $d = 2$, has base gap greater than 7172. Each polynomial is therefore positive on the entire real ray. All straddling differences and parity cases have been exhausted, rather than sampled at finitely many integers.

A one-sided witness and zero span. The remaining case has an actual AP avoider wholly on one side. Reflect the whole original configuration and its same root if necessary to place the witness on the right. Define from all of $\mathcal{I}$

$$t = \max_{A \in \mathcal{I}} \min A, \quad u = \min_{A \in \mathcal{I}} \max A, \quad q = t - p \geq 1, \quad r = u - t, \quad x = N - u, \quad w = t + x. \tag{13}$$

Actual AP members intersect pairwise, so do their convex hulls, and $t \leq u$. At $r = 0$, the owner ledger gives

$$H_3 + I \leq \frac{329}{800}N^2 + 4NH_N + (46/63 + 4)N.$$

Even after restoring the largest budget $41n$, the right-hand side is strictly below $n^2/2 < B(N)$ at $n = 3999$; use $H_N \leq 1 + \log 10000 < 11$ for this rational comparison. After expansion and division by n^2, all positive $1/n, 1/n^2$ terms decrease, covering the full finite range. In CORE this bounds the formal count f, without changing the original family.

Positive span: step feedback on the same inventory. For $r \geq 1$, $w = N - r \leq n$, and the actual coordinates lie in the closed simplex $p, q, x \geq 0$, $p + q + x \leq n$. The hull kernel $K_0 = [t, u]$ is initially only a convex-hull kernel; it is not assumed to be contained in a nonordinary AP. For a currently proved common-difference cutoff D, put

$$B_D = \sum_{e=2}^{D} e^{-2}, \qquad H_D = \sum_{e=1}^{D} e^{-1},$$

$$I_D^{\mathrm{up}} = (B_D t + q)(x + 1) + (w + 1)(H_D - 1) + (D - 1), \tag{14}$$

$$W_D = I_D^{\mathrm{up}} + (C + 47/20)n - 1/4. \tag{15}$$

An ordinary avoider is wholly on the right, with at most q starts and $x + 1$ ends. For a fixed nonunit difference there is one residue, and at most $(t/e + 1)(x/e + 1)$ endpoint choices. Thus (14) bounds every actual member of $\mathcal{I}$, including nonunit members that may still straddle the root. From $f > B(N)$ and (11), every actual avoider satisfies $\beta h_A(h_A - 1) < W_D$.

For $1 \leq d < D$, let $T_d = nd/(d + 1) - 1$. Then

$$T_d > 1, \quad W_D < \beta T_d(T_d - 1) \quad \Longrightarrow \quad \text{all differences in the same actual } \mathcal{I} \text{ are at most } d. \tag{16}$$

Indeed, an AP of difference at least $d + 1$ has $h_A \geq T_d$. The old W_D must first establish this fact; only then is W_d used. The objects $p, \mathcal{F}, \mathcal{E}, \mathcal{I}, t, u$ never change. At $D = 1$,

$$W_1 = q(x + 1) + (C + 47/20)n - 1/4. \tag{17}$$

The negative constant is retained; its ray argument cannot simply claim that every coefficient is nonnegative.

There are three terminals. First, a common adjacent pair: put $k_0 = |K_0| = n - w + 2$, $T_K = (n - w)/2$. If $T_K > 1$ and $W_D < \beta T_K(T_K - 1)$, every avoider has more than $\lceil k_0/2 \rceil$ points in K_0. Actual AP avoiders must therefore be ordinary intervals. An actual such interval then shows that every trace is an ordinary interval, so all share a central adjacent pair, giving $|\mathcal{F}| \le b(N)$. CORE can use this terminal at its initial difference cutoff; no ordinary kernel was assumed in advance.

Second, actual bad-trace packing. In the certificates used here this terminal occurs only in SMALL. Its conclusion is $|\mathcal{F}| \le B(N)$, which would not by itself close the stronger CORE hypothesis $|\mathcal{F}| > b(N)$; CORE uses only owner and COMMON/HOLE leaves. The exact badness criterion is

$$\boxed{|A \cap K_0| \le \lceil k_0/2 \rceil},$$

not omission of an arbitrary kernel point. After reaching difference one, let h_0 bound the holes of every avoider, and put

$$A_\ell = t - 2, \quad R = x, \quad \nu = \lfloor (p-1)/2 \rfloor + \lfloor (t-p-1)/2 \rfloor + \lfloor x/2 \rfloor, \quad \delta = h_0 - \lfloor k_0/2 \rfloor.$$

Each bad row occupies at least $(A_\ell - \delta)(R - \delta)$ cells in the outside rectangle. At one cell with M bad-row owners, each of the ν disjoint adjacent pairs in the three outside intervals can be fully contained in at most one owner; otherwise an AP intersection would contain the entire kernel. Hence $M(\nu - \delta) \le \nu$. This gives the cell-multiplicity bound for every integer $m \ge 1$. Take $J = 999$, $1 \le m \le J$, $\eta = m/(2m+2)$, and define

$$z_m(A, R; J) = \frac{A + R - \sqrt{(A-R)^2 + 4mAR/(J+1)}}{2},$$

$$T_m = \min\left\{ \frac{n}{2} - 1, \frac{n-w}{2} + z_m(t-2, x; J), \frac{n}{2} - (1/2 - \eta)w - 5\eta \right\}. \tag{18}$$

Under $t > 2, x > 0, w \ge 5, T_m > 1$ and $W_1 < \beta T_m(T_m - 1)$, the exact rounding bounds of the packing lemma give at most J actual bad rows. Good avoiders include the actual class $\mathcal{I}$, so are nonempty, and share a central adjacent pair. Only now delete the bad rows, apply LOCAL90B and restore them once:

$$|\mathcal{F}| \le b(N) + 999 \le B(N), \qquad 999 \le \lfloor n/4 \rfloor \quad (n \ge 3999).$$

No density, owner budget or new causal root is asserted for the deleted family.

The third terminal is specific to NO3. The direct bound with no rooted triples forces

$$I > L_0 = \beta n^2 + (93/80 - C - 18/5)n + 1/4.$$

If $L_0 > I_D^{\mathrm{up}}$, the whole actual AP inventory is contradictory.

For completeness, the owner terminal has the following exact formula. Put $b_0 = 129/200$, $a_0 = 329/200$, $\gamma = 16641/131600$, and

$$Q_p = (t^2 + x^2 - b_0^2 p^2)/2 - b_0 t x,$$

$$g(z) = \begin{cases} z/2 & 0 \le z \le 2, \\ \min(z/2, \log z + 3/5) & z > 2, \end{cases} \qquad s(z) = \min(9/2, 1 + \log \max(1, z)),$$

$$H_b = \log(n/2) + 3/5 + 1/n, \quad Z = -n/2 + (a_0 + 3/2)t + (46/63 + 1)q + (9/2)x,$$

$$E_- = pg(q) + (2 - \gamma)pg(p) + ps(q) + 2\sqrt{n+1} + \max(0, Z) - 1/2 + (w+1)(H_b - 1). \tag{19}$$

The strengthened owner lemma gives $H_3 + I \le N^2/2 - Q_p + E_-$. Thus $Q_p > E_- + Cn + N/4$ contradicts $f > B(N)$, using

$$B(N) \ge n^2/2 + 3n/4 + 1/4 = N^2/2 - N/4. \tag{20}$$

4.3 Closed-simplex certificates and complete real rays

These terminals are inequalities on the whole closed real simplex, not a menu inferred from geometric samples. The V130 certificates at $n_0 = 3999$ are [17]

Case	Nodes	Owner leaves	Common-pair leaves	Feedback leaves	Inventory leaves
CORE	455	198	30	0	0
SMALL	455	181	0	47	0
NO3	487	179	0	0	65

The SMALL paths $12 \to 1$, $12 \to 2 \to 1$ and $12 \to 3 \to 2 \to 1$ occur 9, 35 and three times. Their terminals comprise 11 common-pair leaves and 36 packing leaves, with fixed multiplicity $m = 1$ or two.

Here are the acceptance rules. Normalized endpoints have denominator 2^{22}. Clip each box's ith upper coordinate to $\min(u_i, 1 - \sum_{j \neq i} \ell_j)$, preserving its intersection with the simplex. Two closed children share their splitting face and cover the parent. Discarding a box requires $\sum \ell_i > 1$ strictly. The complete preorder stream records every node; replay reconstructs clipping and splitting from the initial box and rejects missing or extra nodes. Directed rational bounds certify a strict inequality on each whole leaf box. The quadratic remainder satisfies

$$Q_p(v) = \frac{(v_p + v_q - b_0 v_x)^2 + (1 - b_0^2)v_x^2 - b_0^2 v_p^2}{2} \geq -b_0^2 R_p^2/2.$$

Positive error terms use their whole-box upper bounds. The packing radicand is separately convex, so its enclosing rectangle's corners give its maximum. Positive atanh series on $[1, 2]$ with explicit tails enclose logarithms; integer square roots with exact comparisons enclose radicals. Zero-coordinate, sloping, minimum and truncation faces are therefore retained. The computation checks these inequalities, not the underlying reduction from families to inequalities.

To propagate to actual N, fix arbitrary real $P, Q, X \geq 0$, $P+Q+X \leq 1$, and put $(p, q, x) = n(P, Q, X)$, $T = P+Q$, $W = P+Q+X$. Then Q_p/n^2 is constant, and $(E_- + Cn + N/4)/n^2$ is nonincreasing for $n \geq 16$, including all zero and piecewise faces. The full direct proof, retaining the negative constant, appears in Section H.4. Moreover

$$\frac{W_D}{n^2} = (B_D T + Q)X + \frac{B_D T + Q + C + 47/20 + W(H_D - 1)}{n} + \frac{H_D + D - 9/4}{n^2}. \tag{21}$$

For $D \geq 2$ the last two coefficients are nonnegative. For $D = 1$ the final term is $-1/(4n^2)$, but the derivative of the nonconstant part is $-(Q + C + 47/20)/n^2 + 1/(2n^3) < 0$. Thus every W_D/n^2 is nonincreasing. On the other side $T_d/n = d/(d+1) - 1/n$ increases, $T_K/n = (1-W)/2$ is constant, and every positive threshold has $T(T-1)/n^2$ nondecreasing. The same finite feedback path therefore propagates one edge at a time.

The packing function z_m is homogeneous and coordinatewise nondecreasing on the nonnegative quadrant. For $k = m/(J+1) \leq 1$, its derivative sign follows from

$$[(A - R)^2 + 4kAR] - [A + (2k-1)R]^2 = 4k(1-k)R^2 \geq 0;$$

continuity handles zero boundaries. Thus all three terms of T_m/n are nondecreasing, and every strict guard persists on the same ray. For NO3, $(L_0 - I_D^{\mathrm{up}})/n^2$ is a constant minus positive $1/n, 1/n^2$ terms and is likewise nondecreasing. Every actual positive-span configuration normalizes to a certified base leaf; that leaf and its fixed path cover the complete ray. This propagates only scalar inequalities. The structural inputs are invoked only at actual integers $4000 \leq N \leq 6999 \leq 10000$. The three fixed-root cases are proved.

4.4 The true core precedes the choice of a causal root

Lemma 4.2 (The true avoider core)**.** *For $4000 \le N \le 6999$, if the nonempty actual avoider class at a fixed p has true intersection T of size at least two, then $|\mathcal{F}| \le b(N)$.*

Proof. Suppose $|\mathcal{F}| > b(N)$. An empty member permits at most one row. An avoiding singleton makes the true core have at most one point; the singleton $\{p\}$ forbids any avoider. These cases are excluded. Every actual pair through p has its other endpoint in T, so $b_p \le |T|$. If $|\mathcal{E}| \ge 2$, non-AP avoiders with private points inject into $[N] \setminus (T \cup \{p\})$; the others are selfclosed at two fixed core points and are paid by C_*. With a single avoider, $c \le 1$ suffices. In both cases $c + b_p \le n + C_*$.

Let $a_* = \lfloor n/15 \rfloor + 1$. If an actual avoider has size at most a_*, then (9), the global R_4 stock and this budget give

$$|\mathcal{F}| \le R_4 + a_* n - a_*(a_*+1)/2 + D_* + n + C_* + 4 \le R_4 + a_* n - a_*(a_*-1)/2 + N - 2 + D_* + C_*.$$

The extra four accounts for the pair and at most three AP triples containing the two fixed core points. All 3000 entrance records put the last expression strictly below $b(N)$, with minimum gap 121783 at $N = 4006$. Hence every actual avoider has size greater than $n/15 + 1$. In particular all actual pairs contain p. The same records give $c + b_p \le (6/5)n$, and therefore $c + b_p + \lfloor n/4 \rfloor \le (3/2)n$. The already proved CORE closure contradicts $|\mathcal{F}| > b(N)$. No unique-pair assumption has been used. □

If $|\mathcal{F}| > B(N)$ and two actual pairs occur, they intersect at some p. With no avoider the prescribed-root bound gives a contradiction. Otherwise every avoider contains the two other endpoints, and Lemma 4.2 gives a contradiction again. Thus only after the true-core lemma do we conclude that a super-$B(N)$ family has at most one actual pair; the triangle case is included.

Now choose an actual full minimum member S, with $k \in \{2, \dots, 7\}$, and set $\epsilon = 1$ for $k = 2$, zero otherwise. Every actual member meets S. Assign non-AP members and AP triples disjointly to one first actual contact in S; pay other APs once by the global stock. If each point of S has an actual avoider of size at most r, (9) yields

$$|\mathcal{F}| \le R_{\max(4,k)} + kK_N(r) + \epsilon. \tag{22}$$

Choose the largest $r \in \{2, \dots, n\}$ for which this is at most $B(N)$; every channel also certifies that such an r exists. The super-$B(N)$ hypothesis selects one fixed $p \in S$ all of whose actual avoiders have size at least $r + 1$. An empty avoider class is excluded by the star bound. For $k = 2$, p is an endpoint of the unique actual pair; for $k \ge 3$ there are no pairs. Thus the actual-pair guard precedes every owner application.

The root is never reselected. Every non-AP avoider meets $S \setminus \{p\}$. By (10) and a union bound,

$$\begin{aligned} c &\le (k-1)\left\lfloor \frac{n(n-1) + D_*}{2(r-1)} \right\rfloor, \\ m &= B(N) + 1 - \epsilon - R_{\max(3,k)} - c, \quad a_{\min} = \min\{a : K_N(a) \ge m\}, \\ r_{\text{new}} &= \max(r, a_{\min} - 1). \end{aligned} \tag{23}$$

Here m is a lower bound for the actual rooted non-AP population. For the computable recurrence, replace c in its definition by the displayed integer upper budget for c. This only weakens the lower bound; the recorded inverse tests and subsequent updates use that conservative value throughout. Every actual avoider must satisfy $K_N(|A|) \ge m$, so the inverse set is nonempty. Infer the new length bound before recomputing c; the inverse bound is $a_{\min} - 1$,

not $a_{\min}$. The integer iteration fixes $S, p, \mathcal{F}$ and the physical host scope, and terminates at a fixed point.

All $6 \cdot 3000 = 18000$ channels and 78168 states check the largest initial r, both inverse boundaries and every transition. They yield

Actual minimum	$e = c + b_p$	All actual avoider sizes	AP differences
2,3	$\leq (50408/4005)n$	$> n/13 + 1$	≤ 12
4,5,6,7	$\leq 41n$	$> n/14 + 1$	≤ 13

The last four minima also exclude all actual triples directly. Apply the complete SMALL and NO3 closures, respectively, to obtain the final contradiction.

Conclusion of Theorem 4.1. The empty family and a family containing an empty member are immediate. A singleton forces a star and the prescribed-root bound applies. The remaining counterexample has actual minimum two to seven. The true-core argument first rules out multiple pairs; the actual minimum member then supplies one causal root. The iteration and the exhaustive fixed-root closures exclude it. Original members are restored through the exact identities whenever needed. □

For certificate provenance, the original V128 SMALL tree has 821 nodes, 348 owner leaves and 63 feedback leaves. Its three feedback paths occur 11, 48 and four times, ending in 15 common-pair and 48 packing leaves. The complete V130 tree used above has different counts, and each tree has its own full replay and real-ray argument. The V138 comparison records the true-core entrance, all 18000 channels, all 3000 entrances, and the algebraic passage from original families to loss polynomials [15, 16, 17]. These numerical records accompany the explicit mathematical reductions. This section does not treat minimum at least eight.

5 A bridge at one fixed root when all members have size at least eight

This section supplies a conditional bridge. It does not itself produce a root: the existence of a root with the required budgets is established in the next section. Conclusions obtained at different roots cannot be combined here.

Theorem 5.1 (The minimum-eight bridge at a fixed root)**.** *Let* $4000 \leq N \leq 6999$*,* $n = N - 1$*, and let* $\mathcal{F}$ *be admissible with every actual member of size at least eight. Suppose the same fixed* $p \in [N]$ *has at most* $50n$ *non-AP avoiders, and every AP avoider has size strictly greater than* $n/14 + 1$*. Then* $|\mathcal{F}| < B(N)$*. The empty family is included.*

We use the complete V140 proof and its $D = 13$ certificate [18, 19, 20]. The V137 proof uses the weaker inventory cutoff $D = 14$. Both invoke GFL, the owner ledger and the strengthened structure lemma only inside their physical domain $N \leq 10000$.

5.1 A strict saving for rooted non-AP members

Let H count all actual members containing p, let $\mathcal{I}$ be the family of all AP avoiders, put $I = |\mathcal{I}|$, let c count all non-AP avoiders, and let m count rooted non-AP members. Throughout,

$$|\mathcal{F}| = H + I + c, \qquad c \leq Cn, \quad C = 50. \tag{24}$$

The minimum-size assumption excludes actual singletons, pairs and triples. In particular the condition that every actual pair contain p, required by the owner lemmas, is automatically satisfied.

Use the intrinsic-pair definition from the preceding section. Complete intrinsic pools are private among actual rooted members. The six-shape classification used in GFL applies to rooted sets of size at least four with exactly one intrinsic pair: they are positive dilates of

$$\{0,1,2,4\}, \quad \{0,-1,-2,-4\},$$
$$\{0,1,2,3,4,6\}, \quad \{0,-1,-2,-3,-4,-6\},$$
$$\{-2,0,1,2\}, \quad \{-2,-1,0,2\}.$$

These have only four or six points. Thus every rooted non-AP member here has at least two pairs in its complete intrinsic pool. The classification is not being applied to triples, nor may the full pool be replaced by an arbitrarily chosen single pair when claiming this saving.

The primitive-pool bound for actual long AP members has coefficient $33/40$ and error less than $(18/5)n$. Pools of different actual common differences have different physical gcds, and are disjoint from the intrinsic non-AP pools. For a non-AP member with pool size $k \geq 2$, $1 \leq (33/40)k - 13/20$. Summing in the same physical pair pool, of size at most $\binom{n}{2}$, gives

$$H \leq \frac{33}{40}\binom{n}{2} + \frac{18}{5}n - \frac{13}{20}m. \tag{25}$$

The saving concerns rooted non-AP members, not all non-AP members indiscriminately.

Suppose $|\mathcal{F}| \geq B(N)$. The global AP inventory is at most $5N^2/12 + 4N(\log_2 N + 1)$. For completeness, the common-chain count at each step gives at most $(n/d+2)^2/4$ members. Summing over $1 \leq d \leq n$, using $\sum_{d\geq 1} d^{-2} \leq \sum_{d=1}^{5} d^{-2} + 1/5 < 5/3$ and $H_n \leq 1 + \log n$, gives $5n^2/12 + n\log n + 2n$, which is bounded by the displayed, deliberately larger inventory. Since $N \leq 10000 < 2^{14}$, (20) yields

$$m + c \geq n^2/12 - 61n, \qquad m \geq n^2/12 - (C+61)n. \tag{26}$$

The first comparison holds for $n \geq 66$, with exact residual $(11n - 722)/12 \geq 0$. We subtract the actual non-AP avoider count c before applying (25) to the rooted pool. Its resulting cost is $(1 + 13/20)c$, not merely c.

5.2 Empty, straddling and zero-span cases

If $I = 0$, discarding the saving in (25) already gives

$$B(N) - |\mathcal{F}| \geq P_{\varnothing}(n) = \frac{7}{80}n^2 - (C + 39/16)n + 1/4.$$

At 3999 its value is $47584111/40 > 0$ and its derivative is $51791/80 > 0$. The leading coefficient is positive, so the inequality is strict on the entire real ray. This also covers an empty total avoider class, without a fictitious AP witness.

Now assume $I \neq 0$. The strict AP-length hypothesis gives actual common differences $e < 14$, hence $e \in \{1, \ldots, 13\}$. If all AP avoiders straddle p, choose a witness of minimum actual common difference $d \in \{2, \ldots, 13\}$. It forces every rooted member to contain a coordinate not divisible by d. The signed-gcd closure argument from the preceding section supplies, in every rooted non-AP member, an intrinsic pair outside the pool of pairs whose two coordinates are divisible by d. If the complete pool has size $k \geq 2$ and its outside part has size $o \geq 1$, then

$$1 \leq \frac{7}{40}k + \frac{13}{20}o.$$

The primitive pools of rooted AP members lie entirely outside the multiples pool. Intrinsic privacy prevents double charging. With $M_d \geq n/d - 2$ the number of nonroot multiples, we obtain

$$H \leq \frac{33}{40}\binom{n}{2} - \frac{13}{20}\binom{M_d}{2} + \frac{18}{5}n. \tag{27}$$

This is a weighted account: the complete pool and its outside part may overlap; they are not treated as disjoint resources. Since $n/d-2>1$, $\binom{M_d}{2}\ge(n/d-2)(n/d-3)/2$ is legitimate.

Using $J(s,v)$ from the preceding section, put

$$S_d=\sum_{e=d}^{13}J(1/e,1/14),\qquad H_d=\sum_{e=d}^{13}1/e,\qquad k_d=14-d.$$

The whole actual AP-avoider inventory satisfies $I\le S_dn^2+2H_dn+2k_d$. Combining this with (24), (27) and (20) gives twelve explicit loss polynomials

$$P_d(n)=\left[\frac{7}{80}+\frac{13}{40d^2}-S_d\right]n^2-\left[C+\frac{39}{16}+\frac{13}{8d}+2H_d\right]n+\frac{11}{5}-2k_d. \tag{28}$$

For every $d=2,\dots,13$, exact coefficient checks give a leading coefficient greater than $1/20$, $P_d(3999)>600000$ and $P_d'(3999)>360$. The smallest base value is $17902703898460013/28857628800$, at $d=2$. Thus all twelve real rays are strictly positive. No numerical conclusion for the different budget $C=41$ is reused here.

Otherwise there is an actual one-sided witness. Reflect the entire configuration and its fixed root so that the witness lies on the right, and define t,u,q,r,x,w by (13) from all of $\mathcal{I}$. Actual intersection gives $t\le u$. If $r=0$, the owner ledger gives $H+I\le 329N^2/800+4NH_N+(46/63+4)N$. Using $H_N<15$ and restoring $c\le 50n$, we have

$$B(N)-|\mathcal{F}|\ge P_0(n)=\frac{71}{800}n^2-\frac{2893027}{25200}n-\frac{3270527}{50400}.$$

Its base value is $60488150/63>0$ and its derivative is $149945/252>0$. The real ray therefore closes. The actual-pair guard was already ensured by the minimum-size assumption.

5.3 Complementary owner and inventory bounds at positive span

Let $r\ge1$. The actual coordinates lie in the whole closed simplex $p,q,x\ge0$, $p+q+x\le n$. The strengthened owner bound uses exactly Q_p,E_- in (19). Its first sufficient strict terminal is

$$Q_p-E_--Cn-N/4>0. \tag{29}$$

Together with $H+I\le N^2/2-Q_p+E_-$ and the original counting identity, this contradicts $|\mathcal{F}|\ge B(N)$.

For the second terminal use the rooted non-AP saving. The established step cutoff is $D=13$, and the full inventory is I_D^{up} from (14), with

$$B_{13}=\frac{14827177181}{25971865920},\qquad H_{13}=\frac{1145993}{360360}.$$

Subtracting c in (26) before using (25) gives

$$B(N)-|\mathcal{F}|\ge\frac{17}{120}n^2-\frac{9967}{80}n+\frac{1}{4}-I_{13}^{\mathrm{up}}. \tag{30}$$

Positivity of the right-hand side is the inventory terminal. Its linear coefficient is exactly $9967/80=39/16+50+(13/20)(50+61)$. Nonunit AP avoiders may still straddle the root. Ordinary avoider intervals lie on the right because they intersect the actual right-hand witness; we have not assumed that every AP avoider has become ordinary.

The V140 certificate at $n_0=3999$ covers the entire closed simplex with 249 nodes: 124 closed splits, 89 owner leaves and 36 inventory leaves, with no discarded exterior leaf [20]. Owner leaves use (29), and inventory leaves use (30). Each whole box is checked by separate

directed rational bounds. The minimum inventory gap is $899322107017021/79152353280 > 0$; all owner gaps exceed 900. Replay reconstructs every clipping and closed split, rejecting missing or extra records. Logarithms and radicals use rigorous enclosures of a different precision from those used in generation. Zero faces, closed faces and piecewise joins are covered for the same reasons as in the preceding section.

Persistence on the complete real rays is also required. Fix arbitrary $P, Q, X \geq 0$, $P+Q+X \leq 1$, and put $(p, q, x) = n(P, Q, X)$, $T = P + Q$, $W = P + Q + X$. Then Q_p/n^2 is constant and $(E_- + Cn + N/4)/n^2$ is nonincreasing for every real $n \geq 16$. The complete proof, including the negative constant $-1/2$, the zero faces and the truncation faces, is given in Section H.4; it applies to every $C \geq 0$. Equivalently, the combination of $-1/2$ and $N/4$ is $(n-1)/4$ before division by n^2, and the remaining harmonic terms must be retained when checking monotonicity. For inventory leaves,

$$\frac{I_D^{\mathrm{up}}}{n^2} = (B_D T + Q)X + \frac{B_D T + Q + W(H_D - 1)}{n} + \frac{H_D + D - 2}{n^2}.$$

After division by n^2, (30) is a constant minus positive $1/n$ terms and $(H_D + D - 2 - 1/4)/n^2$, hence is nondecreasing. Every actual configuration normalizes into a closed base leaf and then follows that leaf along its entire ray. Combinatorial inputs are invoked only in their licensed physical domains.

Conclusion of Theorem 5.1. The empty family is immediate. Otherwise suppose $|\mathcal{F}| \geq B(N)$ and keep the same fixed p in (24). If $\mathcal{I}$ is empty, use the empty-class polynomial. If all AP avoiders straddle p, use the twelve minimum-step polynomials. Otherwise reflect the whole configuration to obtain a one-sided witness; use P_0 at zero span and a closed-simplex owner or inventory leaf at positive span. Every branch gives a strict contradiction. □

For comparison, V137 uses the larger cutoff $D = 14$; its V136 certificate has 427 nodes, 175 owner leaves and 39 inventory leaves. The $D = 13$ V140 tree is a complete alternative. The supporting records distinguish the 13/20 saving, the subtraction of actual non-AP avoiders, the twelve cross rays, zero span and the physical scopes of the two inventories [18, 19, 20]. The bridge alone neither establishes the required root for an arbitrary minimum-eight family nor proves an unconditional threshold. It is combined below with a root entrance and the remaining minimum-size branches.

6 Excluding all minimum sizes at least eight

Theorem 6.1 (The large-minimum branch)**.** *For every* $4000 \leq N \leq 6999$*, no admissible original family satisfies both* $|\mathcal{F}| \geq B(N)$ *and* $\min_{A \in \mathcal{F}} |A| \geq 8$*.*

We use the V155 version of the argument [12, 21]. The cases are actual minimum eight, actual minimum between nine and 63, and actual minimum at least 64. They exhaust all possibilities.

6.1 The conditional common-point phase

Restrict the actual minimum to an integer window $a \in [A, \overline{A}]$, and take $R \geq \overline{A}$. Let S be an actual minimum member, $|S| = a$, and set

$$\mathcal{K}_R = \{X \in \mathcal{F} : |X| \leq R\}.$$

Temporarily suppose $\mathcal{K}_R$ has no common point. Since $S \in \mathcal{K}_R$, there is $T \in \mathcal{K}_R$ with $t = |S \cap T| \in \{1, \ldots, a-1\}$; otherwise every small member would contain S.

Order the points of $S \cap T$ first. For each of these points choose an actual small member omitting it; for the other points use T. From each partner set remove previously listed points of S. Every original member meets S and, at its first contact with S, also meets the corresponding surviving partner set. Thus the resulting outgoing pair buckets contain no repeated unordered pair. Each bucket has size at most R, and their total size is at most

$$Rt + (a-t)(R-t) \le (R-1)a + 1.$$

This controls outgoing buckets, not total degrees in an undirected graph.

Pad to $\overline{A}$ buckets with zeros. Discrete concavity of the single fixed dominating multiset bounds the total β_q cost by $(\overline{A}-1)\mathrm{Top}_q(R-1) + \mathrm{Top}_q(R)$. For all original non-AP members, use prefixes with $r = A-1$, $J \le A-4$. The truncated identity gives

$$C \le U_{\mathrm{root}} = \left\lfloor \frac{[(R-1)\overline{A}+1]\Gamma + (\overline{A}-1)\mathrm{Top}_q(R-1) + \mathrm{Top}_q(R) + \sum_{j<q} \mathrm{CDF}_j}{q} \right\rfloor. \tag{31}$$

Here CDF_j is an already justified prefix bound summed over every physical common difference. Under the same temporary hypothesis start with $P = 0$, and after each completed state set

$$P_{\mathrm{next}} = \max\{P, 0, E_A - U_{\mathrm{root}}\}.$$

Each selected certificate chain ends with $U_{\mathrm{root}} < D_A$, contradicting Lemma 3.1. Hence $\mathcal{K}_R$ is nonempty and has a common point.

All populations, owners, central intervals and prefixes established only under the temporary hypothesis must now be discarded. Only the common-point conclusion is retained.

6.2 Actual minimum eight: an entrance at the same root

For the 30 intervals $[L, L+99]$, $L = 4000, 4100, \ldots, 6900$, take

$$R = \left\lfloor \frac{49L}{4000} \right\rfloor, \qquad r_{\mathrm{root}} = 7, \qquad J_{\mathrm{root}} = 4.$$

All 30 conditional chains give strict contradictions. Since an actual eight-point member S belongs to $\mathcal{K}_R$, choose $p \in \bigcap \mathcal{K}_R \subseteq S$. Keep this same S and p. Every avoider initially has size at least $R+1$; put $r = R$.

At the actual integer N, write $n = N-1$ and let

$$D^* = \begin{cases} 245873, & N \le 5500, \\ 365931, & N > 5500. \end{cases}$$

Every non-AP avoider meets one of the seven points of $S \setminus \{p\}$. Apply the arbitrary-point rank bound with its actual size lower bound $r+1$ and then a union bound. Thus

$$c = 7 \left\lfloor \frac{n(n-1) + D^*}{2(r-1)} \right\rfloor \tag{32}$$

is an upper budget for the number of non-AP avoiders, rather than an assertion that this number attains the budget. The original inequality $W_1 \le C^*$ licenses the complete C^* vertex resource D^* in this rank account; a truncated β resource is not substituted for it. Pay the total AP population once using $R_8(N)$. The number of rooted non-AP members is therefore at least

$$m = B(N) - R_8(N) - c. \tag{33}$$

The contradiction hypothesis is $|\mathcal{F}| \ge B(N)$; there is no extra $+1$ here.

The actual-contact pair injection says that an actual compatible avoider of size a bounds the rooted non-AP population by

$$K_N(a) = an - \frac{a(a+1)}{2} + \operatorname{top}(a). \tag{34}$$

Rows with an intrinsic pair touching the avoider use their private pairs. The other rows have an actual isolated contact and are paid by the complete C^* resource. The number of nonroot pairs touching an a-point set is exactly $an - a(a+1)/2$.

Use the complete checked array $T_a(7500)$, $0 \le a \le 2048$. Above 2048 use

$$\min\left\{365931, T_{2048} + (a-2048)\left\lfloor \frac{T_{2048}}{2048} \right\rfloor\right\},$$

and for $N \le 5500$ take the additional minimum with 245873. These are simultaneous upper bounds; their maximizers need not coincide. Both top and K_N are nondecreasing on $0 \le a \le n$.

If there is no avoider, the bridge theorem applies immediately with vacuous length conditions. Otherwise an actual avoider ensures that the set defining

$$a_{\min} = \min\{a : K_N(a) \ge m\}$$

is nonempty. Every actual avoider has size at least $a_{\min}$. First update $r \leftarrow \max(r, a_{\min}-1)$, and only then recompute (32); the new length bound cannot retroactively sharpen the preceding state. For every $N = 4000, \dots, 6999$, the complete certificates check both inverse boundaries, every state and the final fixed point, for 18002 states in total. They give

$$\max \frac{c}{n} = \frac{196021}{4000} = 49.00525 < 50, \qquad \min(14r - n) = 74 > 0,$$

both attained at $N = 4001$. At the same actual root p we therefore have

$$c \le 50n, \qquad |A| \ge r + 1 > n/14 + 1 \quad \text{for every actual avoider } A.$$

In particular the AP avoiders satisfy this length condition. The bridge [18] applies at this same point and gives $|\mathcal{F}| < B(N)$, a contradiction. Empty subclasses cause no omission: their universal conditions are vacuous.

6.3 Actual minima nine to 63: a fresh global phase

Both phases use the same radius rule:

Actual minimum	9	10	11–39	40–55	56–63
R	59	56	49	55	63

The N-boxes and minimum-size windows may differ between phases, but the radius must agree at every actual pair (N, a).

For a global window $[A, \overline{A}]$, fix a common point p of the established $\mathcal{K}_R$. All non-AP members of size at most R contain p, so their number is at most $\operatorname{Rank}_A(H)$ from (7). Let

$$\mathcal{T}_R = \{X \in \mathcal{F} : X \text{ is non-AP}, \ |X| > R\}.$$

Since $\overline{A} \le R$, the actual minimum member S is outside this tail.

The same-account cut theorem [29] uses, for integers $t \ge 2$,

$$g_t(k,j) = \max\big\{(k-1)\min(j,t) + j\min(\max(j, \lfloor k/2 \rfloor), t), \ 2(k+j-2)\big\}. \tag{35}$$

The two expressions charge the same resource and combine by maximum, not addition. For each $X \in \mathcal{T}_R$, the intersection $X \cap S$ is nonempty and proper in X; equality with X would make X an AP by admissibility. Charge across $S \times ([N] \setminus S)$ and use the complete mixed resource:

$$\sum_{X \in \mathcal{T}_R} g_t(k_X, j_X) \le K_t, \qquad K_t = \max_{A \le a \le \overline{A}} a\big[(H-a)\Gamma + (t-1)D^*\big]. \tag{36}$$

There is no restoration term, since S is not a counted tail member.

Put $v_j = g_t(\max(R+1-j, 2), j)$ and $w_j = \min_{z \ge j} v_z$. The fee is nondecreasing in core size. For $z \ge \max(R-1, t)$, $v_z = t(z+1)$ and increases, so a finite head followed by this linear tail determines the suffix minima over all defects. Nonnegative telescoping gives

$$|\mathcal{T}_R| \le U_{\mathrm{mix}} = \left\lfloor \frac{K_t + \sum_{j<Q}(w_{j+1} - w_j)\mathrm{CDF}_j}{w_Q} \right\rfloor, \qquad 1 \le Q \le J+1. \tag{37}$$

Members with defect greater than J remain paid by the tail fee.

There is a separate complete truncated-pair bound. If the tail is nonempty, choose its own actual minimum member B_0, of size $s \ge R+1$. Every other tail row gives at least $s-1$ cut pairs relative to B_0. Restore B_0 once in the truncated identity to obtain

$$|\mathcal{T}_R| \le \left\lfloor 1 + \frac{s[(H-s)\Gamma + \mathrm{BH}_q]}{(s-1)q} + \frac{\sum_{j<q} \mathrm{CDF}_j}{q} \right\rfloor.$$

Before substituting a size lower bound, note that

$$\frac{s[(H-s)\Gamma + \mathrm{BH}_q]}{s-1} = (H-1-s)\Gamma + \mathrm{BH}_q + \frac{(H-1)\Gamma + \mathrm{BH}_q}{s-1}$$

is nonincreasing for every real $s \ge 2$, since $\Gamma, \mathrm{BH}_q \ge 0$. Hence the safe whole-tail bound is

$$U_\beta = \left\lfloor 1 + \frac{(R+1)[(H-R-1)\Gamma + \mathrm{BH}_q]}{Rq} + \frac{\sum_{j<q} \mathrm{CDF}_j}{q} \right\rfloor. \tag{38}$$

The empty tail also satisfies this nonnegative bound. The cuts in (37) and (38) are separate complete alternatives; let U be their minimum.

Every global chain starts afresh at $P = 0$. Its prefixes use $r = R$ on the original tail and depend only on previous global populations. Thus

$$C \le \mathrm{Rank}_A + U, \qquad O \ge E_A - \mathrm{Rank}_A - U.$$

For $A \le 31$ use ordinary lengths at least $b = 31$ and deduct a complete upper bound for the shorter ordinary intervals; for $A \ge 32$ put $b = A$. Explicitly,

$$P_{\mathrm{next}} = \max\left\{ P, 0, E_A - \mathrm{Rank}_A - U - \sum_{h=A}^{b-1} h \right\}. \tag{39}$$

Only a completed state licenses the next state's additional structure. Every selected global window ends with $\mathrm{Rank}_A + U < D_A$, contradicting (3). The full cover checks every $4000 \le N \le 6999$, $9 \le a \le 63$ exactly once in each phase, with matching radii.

6.4 Minimum size at least 64: the entire remaining branch

For each width-100 N-box start an independent unconditional context

$$P = 0, \qquad r = 63, \qquad J = 30, \qquad b = 64.$$

All non-AP members have at least 64 points. If they exist, their own actual minimum is some $s \geq 64$, not necessarily 64 and not necessarily the minimum of the whole family or of its ordinary subfamily. The monotonicity above licenses (38) with $R = 63$ for every such s simultaneously. No common point or conditional ROOT data are used.

The complete stocks give $C \geq D_{64}$ and $O \geq E_{64} - C$. After each completed U_β bound, update

$$P_{\text{next}} = \max\{P, 0, E_{64} - U_\beta\}$$

for the next state only. All 30 independent chains end with $U_\beta < D_{64}$. This excludes every actual minimum at least 64, rather than testing only minimum 64. Together with the preceding cases it proves Theorem 6.1.

7 Sufficiency and compression of the selected low-range certificates

The preceding arguments associate a valid integer-parameter state with every hypothetical counterexample and exclude it by complete capacities and strict inequalities. The computation checks necessary conditions that hold uniformly for all such families; it does not test a sample of concrete families. We explain why the retained finite data suffice and state the scope of the completed verification. The mathematical specification is the reconciled low-range proof, and the selected covering certificates are those identified in [21, 12, 24].

7.1 Coverage of the original parameter domain

Each selected record specifies an integer box $[L, H]$, a window $[A, AA]$ for the attained minimum size of the full original family, a radius R, a prefix cutoff J, a mode, and the corresponding result file. The latter records, state by state, the lower bound P on the original ordinary population, original-defect prefix bounds, the selected scalar upper bounds, and a strict terminal inequality. Every chain starts with $P = 0$. A subsequent population bound must follow from the preceding completed state; it cannot strengthen the premises of that same state retroactively. [21]

Proposition 7.1 (Sufficiency of a finite covering certificate)**.** *Suppose that the complete inventory, capacity and accounting formulas above apply at every parameter in a selected rectangle, and its record satisfies the following conditions.*

1. *Every original-defect prefix used is bounded by the saved valid capacity recurrence.*
2. *The initial population bound holds unconditionally, and each subsequent population bound follows from the preceding state under the same hypothesis.*
3. *A conditional ROOT argument retains only the conclusion that the small-member subfamily has a common point; all numerical information specific to the contrary assumption is then discarded.*
4. *A terminal state yields an upper bound strictly contradicting the complete inventory lower bound.*

Then no original family in that rectangle satisfies the hypothesis being excluded. If such rectangles cover the required parameters exactly once, they exclude the entire parameter domain.

Proof. Fix an actual integer N in a rectangle and a hypothetical original family, with its fixed cores and defects. The inventory bounds apply at this same N, while all ambient resources are valid on the full interval $[H]$. The first condition proves the complete initial upper bound. The second then inductively proves the hypotheses and bounds at every subsequent state. After the conditional chain, the third condition leaves only the common-point conclusion. A new GLOBAL chain starting with its own $P = 0$ is therefore valid. The fourth condition contradicts the inventory lower bound for the same original family. Every actual parameter lies in a covering rectangle, so this pointwise argument covers all parameters without enumerating concrete families. □

There are 1138 selected certificates: 630 ROOT certificates, 478 GLOBAL certificates, and 30 certificates for the unbounded minimum-size branch. Of the ROOT certificates, 30 concern attained minimum size 8 and the remaining 600 concern minimum sizes 9 through 63. The two finite phases may partition the parameters differently, but each must cover every

$$(N, a) \in \{4000, \dots, 6999\} \times \{9, \dots, 63\}$$

exactly once, with the same required radius. The radius assignment is $9 \mapsto 59$, $10 \mapsto 56$, $11{:}39 \mapsto 49$, $40{:}55 \mapsto 55$, and $56{:}63 \mapsto 63$. Thus each phase has $3000 \cdot 55 = 165000$ integer parameter cells, or 330000 cells in total. The coverage check traverses all these cells and verifies that there are no overlaps or gaps and that the radii agree. The minimum-eight branch and the minimum-at-least-64 branch each use 30 boxes of width 100, covering every integer $4000 \le N \le 6999$ exactly once. [22, 24]

The minimum-at-least-64 branch does not merely check attained minimum size 64. If the original non-AP tail is nonempty, choose a member of its actual minimum size $s \ge 64$. The size-dependent term in its full truncated account is

$$\frac{s((H-s)\Gamma + BH_q)}{s-1} = (H-1-s)\Gamma + BH_q + \frac{(H-1)\Gamma + BH_q}{s-1}, \qquad \Gamma = H-2.$$

Since $\Gamma, BH_q \ge 0$, the right-hand side is nonincreasing for all real $s \ge 2$. Substituting 64 therefore gives a safe upper bound for every $s \ge 64$. The selected actual member is restored exactly once in the truncated account. An empty tail satisfies the nonnegative bound directly. This monotonicity handles the entire unbounded branch without postulating an actual member of size 64. [12, 21]

7.2 Closure of the retained prefix heads

The original complete stream contains more than two hundred million records. A reduction in semantic recomputation requires a precise sufficiency argument; a record cannot be ignored merely because it was not inspected. The following lemma gives the closure rule used for the low-range certificates. [21, 24]

Lemma 7.2 (Closure of selected heads and their capacity ancestors)**.** *Fix an original-family state (H, r, P, b) and a fixed-core step d. Let $0 \le h \le J < r$; when $h = 0$, no prefix is required. Let $X_{d,j}$ count the corresponding original members with defect at most j, and let $B_{d,j}$ be a saved sufficient upper bound for $X_{d,j}$. Suppose the full-band capacity* $\mathrm{Cap}(H, r, d, l, u; P, b)$*, including its hypotheses, is independent of the unused outer cutoff J. Suppose also that at position j the selected recurrence uses one of the following three complete bounds:*

$$\mathrm{Cap}(1, j), \qquad B_{d,j-1} + \mathrm{Cap}(j, j), \qquad B_{d,1} + \mathrm{Cap}(2, j),$$

where the common fixed parameters have been suppressed and $B_{d,0} = 0$. Then every recurrence ancestor of every retained position $j \le h$ is still within that head. Moreover, the entire

original-defect prefix is empty whenever

$$d > \left\lfloor \frac{H-1}{r-h} \right\rfloor, \qquad j \le h.$$

Thus retaining just these heads and their physical step range loses no positive-coefficient contribution to the selected scalar inequalities.

Proof. Every selected recurrence has band endpoint $u = j$ and a parent index $p < j$. Tracing backwards from $j \le h$ therefore keeps every position and band endpoint at most h. A complete capacity depends only on its displayed context and actual band endpoints. Removing unused positions $j > h$ does not change an actual-owner inverse, a union inverse, the full ambient length, or any capacity hypothesis.

An original member of size at least $r+1$ and defect at most j has a fixed core of length at least $r+1-j$. Its physical core span is therefore at least $d(r-j)$. For an omitted step and every $j \le h$,

$$d(r-j) \ge d(r-h) > H-1,$$

which is impossible in $[H]$. The selected ROOT or beta inequality uses only prefixes $j \le q-1$. The mixed account uses only positions with $w_{j+1} - w_j > 0$ in its telescoping sum. Let h be the largest required position, or 0 if there is none. All positive-coefficient positions and all their ancestors are then retained. Original defects above J are still paid by the complete tail cost; no such members are removed from the original family. □

This rule proves sufficiency of the saved valid witnesses. It does not require a shorter search to select the same optimum, or require the search itself to be optimal. In the original study, the complete-stream checker read all 214151175 records, checking their order, physical zeros, parents, $u = j$, recurrence values and all original cumulative defect sums, and bound the complete and head streams by SHA256 hashes. A subsequent semantic replay read all 14512299 retained records, compared their sums exactly with the corresponding original cumulative heads, and evaluated all capacity ancestors required by the selected positive coefficients. These two checks have different roles: complete recurrence structure does not replace semantic capacity verification, and conversely. [21, 24]

7.3 Complete semantic replay and resource identity

The present replay uses the closure lemma to treat all selected heads directly as self-contained sufficient certificates. It recomputes every required capacity ancestor, every causal population update, all actual integer inventories, and every terminal strict inequality. The 1138 certificates contain 11731 states and 14512299 required cumulative defect records; 12837651 capacity values are reevaluated. The minimum strict terminal margin is 68. The ROOT and GLOBAL phases separately cover all 330000 phase-parameter cells, with matching radii.

The minimum-eight computation also recomputes all 3000 actual integers and 18002 reverse-inference states, retaining the same actual eight-point member and root. The largest ratio is $196021/4000 < 50$, and the smallest value of $14r - (N-1)$ is $74 > 0$, both at $N = 4001$. These are results of the new executions, not inferences from historical success labels or summaries. Full run details are given in the companion computation report.

The arbitrary-point rank uses the full W_1 master resource

$$S(n) = 2 \sum_{d \le n} (\tau(d) - 1) \left\lfloor \frac{n}{d} \right\rfloor, \qquad n = H - 1,$$

whereas the mixed account retains its own permitted D^* resource. The resources of these two accounts are not interchanged. The minimum-eight reverse inference uses the permitted

$T_a(7500)$ and its concave tail envelope; the full D^*_{5500} may be used where applicable. Omitting an optional strengthening only weakens the upper bound. All full-ambient pair resources, universal incident-resource lists and selected vertex–q resources are rebuilt in this execution. After the conditional ROOT hypothesis is discharged, GLOBAL starts with its own zero population. A capacity cache that depends on the ordinary population includes its entire context, rather than only the ambient interval.

7.4 Electronic certificates and implementation scope

The new self-contained low-range package retains every required head, selected result, resource, causal entry point, closed-simplex tree, real-ray parameter, source program and runner. The underlying arguments are included in its 31 technical mathematical texts; the common foundation supplement supplies the global-fringe-loss bound, ownership bound and general pair theorem. The package does not require the old unretained stream tail, unsuccessful searches, or the original private desktop paths. A separate existence check located the original directory of 73841 files and 2887146521 bytes. This is provenance information, not a substitute for the present mathematical replay.

The numerical capacity menu reuses the reviewed implementation identified in [21]. The replay reevaluates every necessary value; it is not an independent reimplementation of every capacity formula. The new verification layer checks complete ancestry, causal premises, coverage, inventories, resources and strict termination. Cross-checks within the same model family, complete semantic recomputation and file integrity are reported separately. None is presented as an independent-model review, external human peer review or formal proof certification. The main paper, complete technical supplement and necessary scientific certificates together form the checkable computer-assisted proof.

8 Completion of the proof

Proof of Theorem 1.1. The construction gives $t(N) \geq B(N)$. Fix $4000 \leq N \leq 6999$ and an admissible family $\mathcal{F}$. Empty families and families containing the empty set satisfy the upper bound immediately. Otherwise the minimum member size k is attained. The small-member argument gives $|\mathcal{F}| \leq B(N)$ when $k \leq 7$. For $k \geq 8$, the branches $k = 8$, $9 \leq k \leq 63$, and $k \geq 64$ exclude $|\mathcal{F}| \geq B(N)$. These branches exhaust the possibilities. For $N \geq 7000$, apply Theorem 2.5. The adjacent integer endpoints 6999/7000, 9999/10000, 74999/75000, and 1025000/1025001 leave no uncovered integer. □

Remark 8.1. In the lowest range the exclusion of $k \geq 8$ implies that an extremal family has a member of size at most seven. It does not, by itself, imply the common-point property or classify the extremal families.

Research provenance and computational assistance

The author states that the threshold-improvement research was conducted independently of the cited concurrent work. Private research dates and the date of first public release are distinct; private conversation timestamps are not used here to adjudicate public priority. AI tools assisted with mathematical exploration, coding, manuscript preparation, and checks. The verification described in this paper is computer-assisted checking of explicit mathematical conditions. It is neither a claim of external human peer review nor a proof-assistant certification of the whole theorem. The author reports partial formalization work that is not part of the present public proof material; no claim about its coverage or axioms is used in this paper.

A The exact bound with a prescribed common point

We give a proof of the prescribed-root formula. Its arithmetic component consists of 8001 rectangles and a rational quadratic tail; it is independent of the large-parameter capacity calculations.

Theorem A.1 (Exact prescribed-root bound)**.** *For $m \in [N]$, the maximum size of an admissible family all of whose members contain m is*

$$\binom{N}{2} + 1 + \left\lfloor \frac{\min(m-1, N-m)}{2} \right\rfloor.$$

Consequently the maximum size of a star is $B(N)$, and $t(N) \geq B(N)$ for every $N \geq 1$.

Proof. Translate m to zero. Put $A = m - 1$ and $B = N - m$, so the ground interval becomes $[-A, B] \cap \mathbb{Z}$. For distinct nonzero integers u, v, let $H(u, v)$ be the full arithmetic progression between $\min(0, u, v)$ and $\max(0, u, v)$ with common difference $\gcd(|u|, |v|)$. Call $\{u, v\}$ a bad pair if $\{0, u, v\}$ is not an arithmetic progression. A member with at least four points is called long.

Private bad pairs of non-AP members. Suppose a set X containing zero contains $H(u, v)$ for every pair of distinct nonzero points $u, v \in X$. Then X is an AP. This is immediate for $|X| \leq 2$. Otherwise, after reflection if necessary, start with a positive point and repeatedly take its gcd with the absolute values of the other points. The hull condition places the positive gcd d of all nonzero coordinates in X. The hulls $H(d, x)$ then fill every multiple of d between zero and each extreme point. Thus a non-AP member X has a pair with $H(u, v) \not\subseteq X$. This pair is bad and belongs to no other actual rooted member Y: otherwise the AP $X \cap Y$ would contain $0, u, v$ and hence all of $H(u, v)$.

Interval counts on an arbitrary staircase. A pair is primitive if $\gcd(|u|, |v|) = 1$. Let

$$\varnothing \neq \mathcal{S} \subseteq \{[-\ell, r] \cap \mathbb{Z} : \ell, r \geq 0,\ \ell + r \geq 3\}$$

be any finite family of distinct intervals. Write $\mathcal{P}(\mathcal{S})$ for the primitive bad pairs covered by its members, and put $L = \max \ell$, $R = \max r$. We claim

$$|\mathcal{S}| \leq |\mathcal{P}(\mathcal{S})| + \mathbf{1}_{\{L \geq 2,\ R \geq 2\}}. \tag{40}$$

Take the coordinatewise downward closure D of the selected endpoint pairs (ℓ, r). Adding its other long intervals introduces no new covered pair, since each added interval lies in an original one. Same-sign primitive bad pairs depend only on L, R. Opposite-sign primitive bad pairs correspond to cells $(a, b) \in D$ with $a, b \geq 1$ and $\gcd(a, b) = 1$, except for $(1, 1)$. The number of long intervals minus this opposite-sign supply is a sum of cell weights: weight one on long axis cells and noncoprime long mixed cells, and zero otherwise. All weights are nonnegative. Filling D to its enclosing rectangle can therefore only increase this difference, while leaving the same-sign supply fixed.

Set $T(n) = \sum_{j=1}^{n} \varphi(j)$, where $\varphi(1) = 1$, put $T(0) = 0$, and let

$$C(L, R) = \#\{1 \leq a \leq L,\ 1 \leq b \leq R : \gcd(a, b) = 1\}.$$

For $L, R \geq 2$ the rectangle contains $(L+1)(R+1) - 6$ long intervals and $(T(L) - 2) + (T(R) - 2) + (C(L, R) - 1)$ primitive bad pairs. Hence (40) reduces exactly to

$$T(L) + T(R) + C(L, R) \geq LR + L + R - 1. \tag{41}$$

When one side is zero or one, the required inequalities reduce respectively to $R - 2 \leq T(R) - 2$ and $2R - 3 \leq T(R) + R - 3$, which follow from $T(R) \geq R$. Reflection and configurations containing no long interval cover the remaining boundary cases.

For the unbounded range, put

$$q = \frac{1}{4} + \sum_{k=1}^{10} \frac{1}{(2k+1)^2} + \frac{1}{42} = \frac{82101180126023}{169351174972980} < \frac{1}{2}, \qquad c = 1 - q.$$

A noncoprime pair has a common prime divisor. The odd-prime square-reciprocal tail is bounded by the odd-integer sum and $\int_{10}^{\infty} (2x+1)^{-2}\,dx = 1/42$, so $C(L,R) \geq cLR$. Since $C(n,n) = 2T(n) - 1$, for $s = L + R$ we obtain

$$T(L) + T(R) + C(L,R) \geq \frac{cs^2}{2} + 1.$$

Using $LR \leq s^2/4$, it is enough that $f(s) = (c/2 - 1/4)s^2 - s + 2 \geq 0$. Exact arithmetic gives

$$f(130) = \frac{45438934559}{100207795842} > 0, \qquad f(131) - f(130) = \frac{37024219015103}{37633594438440} > 0.$$

Its leading coefficient is positive, so its integer forward differences increase; this proves every integer $s \geq 130$. The remaining domain $L, R \geq 2$, $L + R < 130$ contains exactly 8001 ordered rectangles. The complete integer check below verifies (41) on all of them, with minimum slack zero. Together with the downward-closure argument, this proves the assertion for every staircase, not merely for rectangle corners.

Paying for long members by missing triples. There are $\binom{N}{2} + 1$ subsets containing zero of sizes one, two or three. Every bad pair in a long member excludes the corresponding triple from the family. Choose a private bad pair for each non-AP long member. Every long AP has a unique positive common difference d and, after division by d, is an interval $[-\ell, r]$. Apply (40) in each d-class. The resulting physical primitive pairs have gcd exactly d, so different classes use disjoint pairs. They are also disjoint from the private pairs of non-AP members. Each class with $L, R \geq 2$ incurs at most one additional member. Such a class requires $2d \leq A$ and $2d \leq B$, so there are at most $\lfloor \min(A,B)/2 \rfloor$ such additional charges. Subtracting the missing triples and restoring the long members proves the upper bound.

Construction attaining the bound. Start with all subsets containing zero of sizes one, two and three. For every $1 \leq d \leq \lfloor \min(A,B)/2 \rfloor$, remove

$$\{-d, 0, 2d\}, \qquad \{-2d, 0, d\},$$

and add

$$\{-d, 0, d, 2d\}, \quad \{-2d, -d, 0, d\}, \quad \{-2d, -d, 0, d, 2d\}.$$

The removed and added sets are distinct for different d, and each replacement increases the size by one. Intersections between added sets are nonempty APs containing zero. Their non-AP triple intersections with the original small sets are precisely the triples removed above. Thus the construction is admissible. At a central root $\lfloor \min(A,B)/2 \rfloor = \lfloor (N-1)/4 \rfloor$. □

Complete finite check. The following standard-library Python code computes Euler's function and the coprime cells directly. Its domain is the full 8001-rectangle domain, and the rational tail values above are checked separately from all large-range calculations.

```
from fractions import Fraction as F
from math import gcd
T = [0]
for n in range(1, 130):
    T.append(T[-1] + sum(gcd(a, n) == 1
                         for a in range(1, n + 1)))
```

```
margins = []
for L in range(2, 128):
    for R in range(2, 130 - L):
        C = sum(gcd(a, b) == 1
                for a in range(1, L + 1)
                for b in range(1, R + 1))
        margins.append(T[L]+T[R]+C-(L*R+L+R-1))
assert len(margins) == 8001 and min(margins) >= 0
q = F(1,4) + sum((F(1,(2*k+1)**2)
                  for k in range(1,11)), F()) + F(1,42)
f = lambda s: ((1-q)/2-F(1,4))*s*s-s+2
assert q < F(1,2) and f(130) >= 0
assert f(131)-f(130) >= 0
```

B A refined position bound for selfclosed supports

This section proves the arithmetic interface used for the physical position resources in the lower ranges. In particular, evaluating the formula below is a finite arithmetic task only after the following combinatorial bound has been established. The consecutive grid component used in this proof is an auxiliary object; it is not the fixed maximum AP core used to define the defect of an original non-AP member.

Fix integers $1 \le u < v \le N$, and put

$$g = v - u, \qquad \ell = u - 1, \qquad r = N - v.$$

Let $\mathcal{C}$ be a family of distinct non-AP subsets of $[N]$ containing u, v. Assume that every intersection of two distinct members is an AP, and that every $A \in \mathcal{C}$ is selfclosed at $\{u, v\}$: for every $y \in A$, the smallest AP containing u, v, y is contained in A. All sets and all coordinates in this statement are actual supports. For such a support define

$$\alpha(A) = |A| - \max\{|P| : P \subseteq A \text{ is an AP containing } u, v\}.$$

Thus $\alpha(A) \ge 1$. Write $\tau(n)$ for the divisor function and $\Omega(n)$ for the number of prime factors counted with multiplicity, and set

$$J(g) = \tau(g) - 1 - \Omega(g).$$

For $f \mid g$, $f > 1$, define

$$m(f) = \max_{p \mid f} v_p(f), \qquad b(f) = \left\lfloor \frac{\tau(f)}{m(f)+1} \right\rfloor,$$

$$z_f(x) = \#\{1 \le i \le \lfloor x/f \rfloor : \gcd(i, g/f) = 1\}.$$

The quotient in the definition of $b(f)$ is in fact an integer: a prime whose exponent is $m(f)$ contributes the factor $m(f)+1$ to $\tau(f)$. Set

$$C^*(g;\ell,r) = \max\Bigg\{ \sum_{\substack{f \mid g \\ f>1}} \left[\min\{\tau(f)-1, b(f)z_f(\ell)\} + \min\{\tau(f)-1, b(f)z_f(r)\}\right], \\ J(g) + \sum_{\substack{f \mid g \\ f>1}} \left[\min\{\tau(f)-2, b(f)z_f(\ell)\} + \min\{\tau(f)-2, b(f)z_f(r)\}\right] \Bigg\}. \quad (42)$$

Proposition B.1. *Under the preceding assumptions,*

$$|\mathcal{C}| \le C^*(g;\ell,r).$$

Moreover, for every proper divisor $d \mid g$ put

$$F_d = \#\{1 \le z \le \lfloor \ell/d \rfloor : \gcd(z, g/d) > 1\} + \#\{1 \le z \le \lfloor r/d \rfloor : \gcd(z, g/d) > 1\},$$
$$A_d = \#\{f \mid g : d < f,\ d \mid f,\ f \le \ell\} + \#\{f \mid g : d < f,\ d \mid f,\ f \le r\}, \qquad A'_d = \min\{A_d, F_d\}.$$

For integers $Q \ge 1$ and $t \ge 2$, define

$$W_Q^{\mathrm{typ}} = \sum_{\substack{d \mid g \\ 1<d<g}} \min\{QA'_d, F_d\} + \max\{\min\{QA'_1, F_1\}, QJ(g)\},$$
$$K_t^{\mathrm{typ}} = \sum_{\substack{d \mid g \\ 1<d<g}} \min\{tA'_d, F_d + A'_d\} + \max\{\min\{tA'_1, F_1 + A'_1\}, tJ(g)\},$$
$$W_Q = \min\{QC^*, W_Q^{\mathrm{typ}}\}, \qquad K_t = \min\{tC^*, K_t^{\mathrm{typ}}\}.$$

Then

$$\sum_{A \in \mathcal{C}} \min\{\alpha(A), Q\} \le W_Q, \qquad \sum_{A \in \mathcal{C}} \min\{t, \max\{\alpha(A), 2\}\} \le K_t.$$

For $g = 1$ all these kernels are defined to be zero.

Proof. Translate u to 0, so that $v = g$ and the ground interval is $[-\ell, g + r]$. For $d \mid g$ write $T_d = \{0, d, \dots, g\}$. Selfclosure implies that the trace $A \cap [0, g]$ is a union of the grids T_d. In terms of the order layers

$$L_a = \{x \in [0, g] \cap \mathbb{Z} : g/\gcd(g, x) = a\} \qquad (a \mid g),$$

where $\gcd(g, 0) = g$, this trace is the union of the layers indexed by a divisor-closed set S_A containing 1. Put $q_A = \operatorname{lcm} S_A$. Every prime power dividing q_A belongs to S_A: some element of S_A attains that exponent, and S_A is divisor-closed. The trace is contained in T_{g/q_A}.

For two distinct supports A, B, let their actual AP intersection have step e. It contains $0, g$, so $e \mid g$, and its trace is exactly T_e. Consequently

$$S_A \cap S_B = \{a : a \mid g/e\}, \qquad \frac{g}{e} = \gcd(q_A, q_B). \tag{43}$$

For the second equality, one divisibility follows because g/e belongs to both divisor-closed sets. Conversely every prime power dividing $\gcd(q_A, q_B)$ belongs to both sets, hence divides g/e.

Call A principal if its trace is T_{g/q_A}, and nonprincipal otherwise. Two nonprincipal supports cannot have the same q_A: the preceding intersection identity would force the full grid T_{g/q_A} into both traces. A nonprincipal label q_A is neither 1 nor a prime power, since in either case divisor closure and the attained maximal prime power would already give the full grid. There are therefore at most $\tau(g) - 1 - \Omega(g) = J(g)$ nonprincipal supports.

For a principal support put $d = g/q_A$. Its trace is T_d, and every coordinate of A is divisible by d. Indeed, for $y \in A$, selfclosure places $T_{\gcd(g,y)}$ in the trace, forcing $d \mid \gcd(g, y)$. Let K_A be the maximal consecutive d-grid component of A containing T_d. As A is not an AP, there is a point $y_A \in A \setminus K_A$. Such a point is outside $[0, g]$, and

$$f_A = \gcd(g, |y_A|), \qquad d \mid f_A, \qquad d < f_A.$$

The strict inequality follows because $f_A = d$ would make the forced hull connect y_A to K_A. In particular, $d < g$. This proves also that a principal support with $d = 1$ cannot coexist with

a nonprincipal support: its trace is all of $[0, g] \cap \mathbb{Z}$, so the intersection trace would be the nonprincipal trace, which is not an AP. If a trace containing $0, g$ were an AP, it would itself be a full divisor grid, and thus principal.

Two distinct principal supports of the same type d have an actual intersection of step exactly d. Its step divides d because it contains T_d, and a smaller step would place a positive point below d in their traces. It follows that a point in $A \setminus K_A$ cannot belong to any other principal support of type d: the AP intersection would connect it to T_d. This is the same-type fringe privacy property.

Choose one point $y_A \in A \setminus K_A$ for each principal support. Fix d, f and one side of $[0, g]$. At most one support can have these three labels. To see this on the left, suppose two supports A, B have left endpoints $a_A \le a_B$ for their respective components. Both selected witnesses are multiples of f below a_B. The largest multiple w of f strictly below a_B belongs to both forced hulls from the witnesses through $0, g$. It is outside K_B, contrary to same-type fringe privacy. The right side follows by reflection. Thus, for fixed f and side, at most $\tau(f) - 1$ labels d occur. When a nonprincipal support is present, type 1 is absent, and this bound improves to $\tau(f) - 2$.

There is a second bound for fixed f and side. Fix the actual coordinate y chosen by some of these supports. Their types form an antichain under divisibility. For if $d \mid d'$ for two such types, both supports contain $T_{d'}$ and y. Its step divides d', and a smaller step would contradict the second support's trace $T_{d'}$. Its step is therefore exactly d'. The AP intersection connects y to $T_{d'}$ on this grid, contradicting that y is outside the corresponding maximal consecutive component. The same argument excludes two supports of the same type. Choose a prime whose exponent in f is $m(f)$. Fixing all other prime exponents partitions the divisors of f into exactly $b(f)$ chains, by varying that prime's exponent from 0 to $m(f)$. Thus this antichain has at most $b(f)$ members. The number of possible left witnesses of gcd label f is exactly $z_f(\ell)$, and the right count is $z_f(r)$. Combining the two bounds for each f and side gives the first branch of (42) when there are no nonprincipal supports. Otherwise the principal contribution uses $\tau(f) - 2$, and the nonprincipal contribution is at most $J(g)$, giving the second branch. This proves the count bound. When $g = 1$, the trace is necessarily principal of type $1 = g$, which is impossible for a non-AP selfclosed support, so the family is empty.

It remains to verify the weighted bounds. A principal support of type d satisfies

$$\alpha(A) \le |A \setminus K_A|.$$

Every point of $A \setminus K_A$ is a multiple of d outside $[0, g]$, and its gcd with g is strictly larger than d. A left point $y < 0$ is counted by the coordinate $-y/d$, and a right point $y > g$ by $(y - g)/d$; each coordinate has gcd greater than one with g/d. These are precisely positions counted by F_d. Same-type fringe privacy gives

$$\sum_{A \text{ principal of type } d} \alpha(A) \le F_d.$$

The side-label collision bound gives at most A_d such supports, and each has a nonempty private fringe, so their number is at most $A'_d = \min\{A_d, F_d\}$. Hence their two weighted contributions are at most $\min\{QA'_d, F_d\}$ and $\min\{tA'_d, F_d + A'_d\}$, respectively; here the latter uses $\max\{\alpha, 2\} \le \alpha + 1$ for integral $\alpha \ge 1$. Nonprincipal supports contribute at most $QJ(g)$ and $tJ(g)$. Their coexistence exclusion with principal type 1 gives precisely the two displayed maximum terms, rather than a sum of both alternatives. Finally, each summand is at most Q or t, so the already proved count bound also gives QC^* and tC^*. Taking minima completes the proof. □

Lemma B.2. *For $h = g/d \geq 2$ and $M = \lfloor (\ell + r)/d \rfloor$, the preceding inventory can safely be replaced by*

$$\overline{F}_d = \min \left\{ M, \sum_{p \mid h \text{ prime}} \lfloor M/p \rfloor, \#\{1 \leq z \leq M : \gcd(z,h) > 1\} + 2^{\omega(h)-1} - 1 \right\},$$

where $\omega(h)$ counts distinct prime factors. That is, $F_d \leq \overline{F}_d$.

Proof. Put $a = \lfloor \ell/d \rfloor$, $b = \lfloor r/d \rfloor$, so $a + b \leq M$, and let $B_h(x) = \#\{1 \leq z \leq x : \gcd(z,h) > 1\}$. Then $F_d = B_h(a) + B_h(b)$. The bounds by M and by $\sum_{p|h} \lfloor M/p \rfloor$ follow respectively by counting all positions and by the union bound, using $\lfloor a/p \rfloor + \lfloor b/p \rfloor \leq \lfloor M/p \rfloor$. For the last bound, inclusion–exclusion gives

$$B_h(a) + B_h(b) - B_h(a+b) = \sum_{\varnothing \neq S \subseteq \{p : p \mid h\}} (-1)^{|S|+1} \left(\left\lfloor \frac{a}{p_S} \right\rfloor + \left\lfloor \frac{b}{p_S} \right\rfloor - \left\lfloor \frac{a+b}{p_S} \right\rfloor \right), \qquad p_S = \prod_{p \in S} p.$$

Each parenthesis is 0 or -1. Only even nonempty subsets can contribute positively, and there are $2^{\omega(h)-1} - 1$ of them. Now use $B_h(a+b) \leq B_h(M)$. □

Remark B.3. The proposition concerns a distinct family with its actual AP intersections. When it is applied to retained supports associated with original members, the application must first establish that every shared point has been retained, so that the supports still have the required intersection property. Non-AP supports cannot then coincide, since their common support would be their AP intersection. The proposition does not license deletion of arbitrary points or replacement of the original maximum AP core and its defect by the auxiliary component K_A.

C An independent analytic tail without a minimum-member hypothesis

This section treats $N \geq 1025001$. Suppose that an admissible family $\mathcal{F}$ is nonstar and has $|\mathcal{F}| \geq B(N)$. No minimum member size is assumed. The argument uses neither a minimum-size reduction nor the finite high-range ROOT/MASS conclusion. Defects always refer to the fixed maximum AP cores of original members. This order allows the tail to enter the minimum-size-nine proof without a cycle. The full proofs of the F/P/I/O input results are included in the technical supplement [32]; this section gives their composition and the definitions needed for its numerical certificates. Their finite obligations and the present complete recomputations are specified in the computational report.

C.1 Mass bounds and propagation of one actual common point

Put

$$N_0 = 1025001, \quad n_0 = 1025000, \quad \gamma = \frac{51861}{50000}.$$

Result F09 bounds the number of original non-AP members through any ambient pair by γn for every real parameter $n \geq 1000000$. Its proof uses a complete finite prefix together with a prime-factor analytic tail. It is therefore not being described as an estimate free of finite computation.

For precision, the fixed polynomial convention used below is the following. Write

$$U(z) = \frac{z^3}{6} + \frac{7z^2}{10} + \frac{74z}{25} + \frac{568}{125},$$
$$V(z) = \frac{z^3}{6} + \frac{16z^2}{25} - \frac{5181z}{2500} - \frac{13514}{3125}.$$

F06 gives $XV(\log X) \le D_4(X) \le XU(\log X)$ for all real $X \ge 10000$. For rational $1 \le y \le 2$, let $r = (y-1)/(y+1)$ and use

$$2\sum_{j=0}^{79} \frac{r^{2j+1}}{2j+1} \le \log y \le 2\sum_{j=0}^{79} \frac{r^{2j+1}}{2j+1} + \frac{2r^{161}}{161(1-r^2)}.$$

Reduce positive rational arguments by powers of two and round outward to multiples of 10^{-18}. Denote the resulting lower and upper bounds for $\log 2, \log 3$ by L_2, U_2, L_3, U_3, respectively, and define

$$C(z) = U(z) + \frac{U(z-2(L_2+L_3))}{36} - \frac{V(z-2U_2)}{4} - \frac{V(z-2U_3)}{9}.$$

The signed two-prime sieve is

$$\sum_{g \le X} \tau(g)^2 \le D_4(X) - D_4(X/4) - D_4(X/9) + D_4(X/36).$$

The signs prescribe the stated directed substitutions. Exact checks establish positivity of the coefficients of C, the derivative guards for U, V, and $C(12) > 346$. Together with the full finite prefix in F07, this proves

$$\sum_{g \le X} \tau(g)^2 \le XC(\log N) \qquad (0 \le X \le N,\ N \ge 360000).$$

For $X \le 360000$ the finite statement is checked at every integer prefix; the remaining real range follows from the analytic sieve. Both complementary sides of each first-contact point are included.

Take the five fixed parameter triples

R_i	q_i	ρ_i
2969999/10000	39	39/125000
307	40	143/50000
2800	323	191/2500
75000	7700	877/10000
85000	8700	351/4000

and let $\mathcal{K}_i$ be all original non-AP members of size at most R_i. The radii are fixed real numbers. The initial integer first-contact threshold is $\lfloor R_0 \rfloor = 296$, not R_0.

Here is the closed low-defect bound used for each entire bin $[l,u] = [1, q_i - 1]$, with $R = R_i > 3u+2$. Define

$$V_R = R - u - 1, \quad W = R - 2u - 1, \quad L_R = 1/V_R, \quad A_R = 4/l + (u+2)/W,$$

$$B_R = 4u/l, \quad D_R = 2/W, \quad t_0 = D_R/B_R, \qquad \Psi(v) = \begin{cases} v, & v \le 1, \\ 1 + \log v, & v > 1. \end{cases}$$

The input result F03 supplies the upper bound

$$\ell_i = A_R L_R + D_R \Psi(B_R L_R / D_R) - 4I_R/l^2,$$

where

$$I_R = \begin{cases} L_R/(A_R+B_R), & L_R \le t_0, \\ t_0/(A_R+B_R) + (L_R - t_0)/A_R - \dfrac{D_R}{A_R^2} \log \dfrac{A_R L_R + D_R}{A_R t_0 + D_R}, & L_R > t_0. \end{cases}$$

This is an upper bound on original members with size greater than R_i and defect in the stated bin. Its original-member capacity proof is an explicit input from the technical supplement, not numerical quadrature of a stronger integral.

Subtracting the complete AP reservoir, the actual pair members, the high-defect class and this large low-defect class from the original density gives

$$\frac{|\mathcal{K}_i|}{N^2} \ge \frac{71}{800} - \frac{a_i}{q_i} - \ell_i - \frac{a_i C(\log N) + \log N + 3}{N}, \qquad a_i = \frac{q_i + 2}{q_i + 1}.$$

Subtracting an entire high-defect class may also subtract short members; that only weakens the lower bound. Directed rational logarithms verify at N_0 that the right side is strictly greater than ρ_i for all five triples. Logarithms in the subtracted integral are rounded with their actual signs. Since C has positive coefficients and degree three, $C(\log N)/N$ decreases when $\log N > 3$; so does $(\log N + 3)/N$. All parameters remain fixed. Thus $|\mathcal{K}_i| > \rho_i N^2$ for every integer $N \ge N_0$, by monotonicity on the entire real ray.

Let $\mathcal{W}_i$ contain all actual members of size at most R_i. The positive initial mass makes $\mathcal{W}_0$ nonempty. If it had empty total intersection, the first-contact theorem F04, with integer threshold 296, would use exactly $296^2 - \lceil 296/2 \rceil = 87468$ pair slots and give

$$|\mathcal{F}| \le \frac{329}{800} n^2 + n(1 + \log(n/2)) + \gamma n \cdot 87468 + 296.$$

The lower bound $B(N) \ge n^2/2 + n/2 + 1$ contradicts this at n_0. Explicitly the gap

$$G(n) = \frac{71}{800} n^2 - \gamma n \cdot 87468 - n(1/2 + \log(n/2)) + 1 - 296$$

satisfies $G(n_0) > 237332288$ and $(G(n)/n)' = 71/800 - 1/n + 295/n^2 > 0$ for $n \ge 12$. This establishes one actual common point p of $\mathcal{W}_0$ on the entire tail.

If all $\mathcal{W}_i$ members contain p, every member A avoiding p meets each member of $\mathcal{K}_i$. Charging to its first contact $x \in A$ gives $|\mathcal{K}_i| \le |A|\gamma n$. Therefore $|A| > \rho_i n/\gamma$. The four exact bridge conditions

$$\frac{\rho_i n_0}{\gamma} \ge R_{i+1} \qquad (0 \le i \le 3)$$

retain the same p at all stages. Equality in a bridge is safe because the mass inequality is strict. Finally every avoiding original member satisfies

$$|A| > \frac{2925}{34574}\, n > \frac{n}{15}.$$

For $a = 2925/34574$, counting all nonroot pairs in non-AP avoiders gives, when their number c is positive,

$$\frac{c}{n} < \frac{\gamma(n-1)}{a(an-1)} < 145.$$

The last bound holds at n_0 by rational arithmetic and thereafter by the negative derivative of $(n-1)/(an-1)$. The case $c = 0$ is immediate. All actual pair members are already rooted and number at most n. Restoring them gives a total original exception allowance at most $146n$. This allowance is derived before the one-sided closure is invoked.

C.2 All branches and the elliptic closure

No avoiding member means that the family is a star. One or two avoiding members give $|\mathcal{F}| \le \binom{N}{2} + 1 < B(N)$ by the actual-exception theorem. For at least three avoiders, their actual AP members are either absent, all straddling p, or include an actual one-sided member. An empty class is not provided with an invented AP witness. If the AP class is empty, the sharp hole estimate gives fewer than $n/5$ holes in each avoider. All pair intersections are then ordinary intervals of more than $N/2$ points. Their common interval exists by interval Helly, and every point outside an avoider's consecutive component through this interval is private among all avoiders. The private-fringe counting theorem closes this case. If all actual AP avoiders straddle p, their differences are in $\{2, \dots, 14\}$. The complete parity and nonmultiple classification P07 has fourteen paid ledgers: ODD, A, B, C and OFFROOT D for $5 \le D \le 14$. Each is applied with the original-member hitting premise supplied by an actual AP avoider, and with $c \le 145n$. Its positive quadratic gap and positive derivative at n_0 establish the contradiction on the full real tail. These ledger statements and their proofs remain input results of the technical supplement.

It remains to treat an actual one-sided AP avoider. Reflect if necessary so a one-sided witness lies to the right of p. Let t be the largest left endpoint of all actual AP avoiders, let u be their smallest right endpoint, and put $x = N - u$. These extrema refer to the same original AP class, and $p < t \le u$. The retained numerator in the one-sided estimate O02 and the endpoint estimate O05 yield

$$Q(t,x) \le T, \qquad \beta h^2 \le \theta(t^2 + x^2) + Ln^2,$$

for every original avoider, where $h = n - |A|$, $\beta = 7/80$, $\theta = 329/400$, and

$$Q(t,x) = \frac{23359}{80000}t^2 + \frac{1}{2}x^2 - \frac{129}{200}tx.$$

To specify every certificate parameter, set

$$E(z) = C(z) + z^2 + \frac{62}{5}z + \frac{393}{25}, \qquad T = 146n + (n+1)E(\log(n+1)) + 5 + (n+1)/2.$$

In the hole inequality one may take the directed upper bound

$$L = \frac{146 + 71/16}{n} + \frac{13}{4n^2} + \frac{3(n+1)(\log(n+1) + 3/5)}{n^2}.$$

This uses $H_N \le \log N + 3/5 + 1/(2N)$ and retains the additive constant outside the one-sided error. No factor θ multiplies T.

Put $y = (t/n, x/n)^{\mathsf{T}}$, $Q(t,x) = n^2 y^{\mathsf{T}} M y$ and $S = T/n^2$. Fix

$$\mu = \frac{8849}{500}, \qquad \varepsilon = \frac{1}{20000}, \qquad \chi = 1 - 2\varepsilon = \frac{9999}{10000},$$

and define

$$A = \frac{\beta}{4}\mathbf{1}\mathbf{1}^{\mathsf{T}} - \theta I + \mu M,$$
$$J = \frac{\beta\chi^2}{4} - L - \mu S - \frac{\beta^2\chi^2}{16}\mathbf{1}^{\mathsf{T}}A^{-1}\mathbf{1}.$$

The rational certificate proves that A is positive definite and $J > 0$. Independently, with $\kappa = 57/1000$, it verifies

$$Q(t,x) \ge \kappa(t^2 + x^2), \qquad 2S/\kappa < \chi^2, \qquad \varepsilon n_0 = 205/4 > 1.$$

For example, the determinant needed for the first bound is $7457/80000000 > 0$. Thus $\mathbf{1}^{\mathsf{T}}y < \chi$, independently of any squaring in the hole inequality. On the ellipse $y^{\mathsf{T}}My \le S$,

$$\frac{\beta}{4}(\chi - \mathbf{1}^{\mathsf{T}}y)^2 - \theta\|y\|^2 - L + \mu(y^{\mathsf{T}}My - S) \le \frac{\beta}{4}(\chi - \mathbf{1}^{\mathsf{T}}y)^2 - \theta\|y\|^2 - L.$$

Completing the square on the left gives a nonnegative quadratic form plus J, so the right side is strictly positive. Both this sign and the independent positive square-root branch are necessary. For $0 \le j \le 3$, $(n+1)\log^j(n+1)/n^2$ decreases when $\log(n+1) > 3$; for $j > 0$ its derivative has sign $jn - (n+2)\log(n+1) < 0$. Positive $1/n$ and $1/n^2$ terms decrease as well. Consequently S, L decrease, J increases and all sign guards improve on the whole real ray. No large-integer sampling supplies this continuation.

Every avoider therefore has

$$h < H := \frac{n - t - x}{2} - \varepsilon n.$$

In particular each actual AP avoider has more than $\lceil N/2 \rceil$ points and must be an ordinary interval. This conclusion precedes the construction of a common consecutive core. Their common interval $[t, u]$ has cardinality $W = n + 2 - t - x = 2H + 2 + 2\varepsilon n$. Choose one actual interval avoider P_0. For any non-AP avoider A, the trace $A \cap [t, u]$ is the restriction of the AP $A \cap P_0$, so is an AP or empty. An empty, singleton or nonunit-difference trace omits at least $\lfloor W/2 \rfloor$ points, contradicting $h < H < W/2 - 1$. Thus every trace is an ordinary nonempty interval. With $m = \lceil H \rceil$, all traces contain

$$K = [t + m, u - m] \cap \mathbb{Z}, \qquad |K| = W - 2m > W - 2H - 2 = 2\varepsilon n > 2.$$

Every two avoiders now intersect in an AP containing adjacent points, hence in an ordinary interval. A point outside a member's maximal consecutive component through K cannot be shared with another avoider: their interval intersection would connect it to K. These exterior points are private among all original avoiders. Each noninterval avoider has such a point. The actual interval class is nonempty, contains K, and lies on one side of p. The actual-interval and private-fringe results I01–I06, with all rooted pair members restored, give $|\mathcal{F}| \le \binom{N}{2} + 1$, contradicting $B(N)$. This completes the independent tail for every $N \ge 1025001$.

D An acyclic minimum-size-nine reduction and its ambient domains

We establish the structural reduction before using minimum size fifteen. Actual members and covering sets of ambient points retain their separate roles.

Theorem D.1 (Minimum-size-nine reduction)**.** *Let $N \ge 10000$. If an admissible family $\mathcal{F}$ is not a star and $|\mathcal{F}| \ge B(N)$, then every member has at least nine points.*

We assemble the local results C03 and C04 of the technical supplement on their stated domains; the endpoint in the title of the earlier 75000 paper is not used as a substitute for those domains. Complete proofs of their F/P/I/O foundations, finite reductions and the versions used in the present computations are provided in the electronic technical supplement [32]. The new checks cover the low integer bands, the full drawup domains, ambient labels, visible-edge geometry and closed-simplex tests. Historical acceptance records are not mathematical premises.

Proof. The independent tail theorem already excludes nonstar families of size at least $B(N)$ for $N \ge 1025001$, without assuming minimum size nine or fifteen. It remains to consider $10000 \le N \le 1025000$, entirely inside the local one-sided domain $10000 \le N \le 1080001$. An empty member cannot occur in a family of size greater than one, and a singleton member would force its point into every member.

Closure with fewer than $5n$ non-AP avoiders. For a previously specified ambient root p, C03(A) proves the following implication. If there are $c < 5n$ non-AP members avoiding p, where $n = N - 1$, the complete AP reservoir and the mass of rooted non-AP members first force every avoiding member to have size greater than $n/15 + 2$. Only after this step may one conclude that all actual two-point members contain p and that every actual AP avoider has its own difference at most 14. If there is no avoider, the family is a star. One or two avoiders, and the branch with no AP avoider, are handled by the actual-exception and interval results C02. An actual one-sided AP avoider permits O07, with its allowance $e \le 6n$. If all actual AP avoiders straddle p, P09 applies. These cases are exhaustive, and an empty AP class is never supplied with a fictitious witness. This closure precedes the stronger allowance for the empty-AP branch below, so the latter is not used circularly.

Excluding actual pairs, including the triangle case. If there is exactly one two-point member, C01 selects one of its points as p and gives $c < 12n$ and avoiding-member size greater than $n/15 + 2$. That actual pair is restored only once. The independent empty-class improvement in C03 handles the empty-AP branch; the one-sided branch uses the complete $12n + 1$ test O08; the straddling branch uses P09. If there are at least two actual pairs, choose two of them. They intersect and therefore have the form $\{p, x\}, \{p, y\}$. Every member avoiding p contains both x and y. The common-pair bound gives at most $N - 2 < 5n$ non-AP avoiders, so the preceding closure applies. This does not assume that all actual pairs have a common point: the triangle $\{p, x\}, \{p, y\}, \{x, y\}$ is included and subsequently excluded.

Excluding three-point covering sets before excluding actual triples. Suppose three ambient points meet every member. Result O09b selects one fixed root p with $c < (263/10)n$ and every avoiding member larger than $n/16 + 2$. The covering set need not be an AP or a member of $\mathcal{F}$. No avoider means that the family is already a star. With no AP avoider use the separate empty-class improvement. An actual one-sided AP avoider permits the two successive mass estimates of O09 at the same p, reducing to $c < 5n$ and the preceding closure. Otherwise all actual AP avoiders straddle p, have their own differences at most 15, and satisfy P10. Thus no set of three or fewer ambient points meets every member. Every actual nonempty member meets every member; only now do we conclude that there are no actual three-point members anywhere in the family.

Covering sets of four to eight points, with one fixed root. Given an r-point ambient covering set, $r = 4, \ldots, 8$, C04 selects the root only once and gives the following bounds. Every avoiding member is larger than $n/d + 2$.

r	strict upper bound for c/n	d	maximum AP-avoider difference
4	409/10	16	15
5	294/5	16	15
6	78	17	16
7	199/2	17	16
8	1243/10	18	17

All five values of r and every integer $10000 \le N \le 13999$ have been checked afresh. Derivative estimates for the original expressions cover $N \ge 14000$; these integer ranges meet without a gap. Consequently each case has $c < (1243/10)n$, actual AP-avoider differences at most 17, and avoiding-member sizes at least $\lfloor n/18 \rfloor + 3$. If there is no avoider, the family is a star. With no AP avoider, use the separate empty-class branch of P08. An actual one-sided AP avoider permits O10: its hypothesis that the entire family has no actual triples was established above. In the remaining case the AP-avoider class is nonempty and consists entirely of straddling

members, so the nonempty branch of P08 applies. Every branch closes the original nonstar contradiction at the same selected root.

There is therefore no covering set of at most eight ambient points. Any actual nonempty member of size at most eight would itself be such a covering set, a contradiction. □

Remark D.2 (Local domains and closed covers)**.** This argument uses the independent tail only to truncate the ambient range. The small-covering-set results do not pass through finite high-range ROOT/MASS, minimum size fifteen, or the final extremal theorems at 10000 or 4000. New complete rational subdivision trees were generated for O08, both parameter sets of O09, and O10. Shared subdivision faces, zero faces and truncation faces are retained. Endpoint sampling does not replace a continuous-domain proof. These numerical tests support only the combinatorial statements for which they were proved; they do not enlarge any drawup, label or one-sided ambient domain.

E The minimum-size-fifteen reduction: joint feedback from an actual shortest member and the AP reservoir

This section supplies the finite improvement obtained during the present revision. It uses sufficient integer inequalities, all of which have been recomputed on the full stated ranges. The separate finite foundations of the minimum-size-nine reduction are recorded in the computational supplement and are not included in the scope of this particular calculation.

Write

$$n = N - 1, \qquad B(N) = \binom{N}{2} + 1 + \left\lfloor \frac{n}{4} \right\rfloor .$$

The family $\mathcal{F}$ consists of distinct subsets of $[N]$, and the intersection of any two distinct members is a nonempty AP. We use the following input results, whose full proofs are included in the technical supplement [32].

1. The common-AP-core theorem: an admissible family of distinct non-AP integer sets containing a fixed AP C of at least two points has size at most $|(\bigcup A) \setminus C|$. Thus the number of non-AP members containing a fixed ambient pair is at most $N - 2$. Ambient points avoided by every member of a class may be removed from its union before this theorem is applied.

2. The self-closure bound: write $\mathrm{Hull}(T)$ for the AP between the extreme points of T whose difference is the gcd of their differences. Every AP containing T contains this hull. A member containing u, v is self-closed at that pair if $\mathrm{Hull}(u, v, y) \subseteq A$ for every $y \in A$. In an admissible family, the number of non-AP members self-closed at a fixed pair of positive distance g is at most

$$C_0(g) = 2\big(d_3(g) - \tau(g)\big).$$

   Here d_3 counts ordered factorizations into three positive factors, and τ counts positive divisors. This is a theorem for all g, rather than a finite-table assertion.

3. The independent minimum-size-nine reduction: the local results C03 and C04, combined with the independent tail T01 on their literal domains, imply that $N \geq 10000$, nonstarredness and $|\mathcal{F}| \geq B(N)$ force every member to have size at least nine. This reduction is given in Section D; it does not use the extremal theorems at 10000 or 4000, or the present section. Its remaining finite foundations are separately identified in the dependency list.

We handle $10000 \leq N \leq 11993$ by joint feedback, recompute the junction $11994 \leq N \leq 12499$, and give an analytic continuation from 12500.

E.1 A graph-moment bound requiring a size improvement only within a rooted subfamily

Define

$$P_0(v) = \sum_{g=1}^{v} C_0(g), \qquad P_2(v) = \sum_{g=1}^{v} C_0(g)^2, \qquad P_0(0) = P_2(0) = 0.$$

The divisor-convolution identity $C_0(g) = 2\sum_{d|g}(\tau(d) - 1)$ has nonnegative divisor coefficients, so

$$P_0(u) + P_0(v) \le P_0(u + v) \qquad (u, v \ge 0).$$

For an integer $a \ge 3$, put $h = a - 2$ and

$$K^{\mathrm{num}}(n) = \binom{n}{2} + P_0(n), \qquad K_a(n) = \frac{K^{\mathrm{num}}(n)}{h}.$$

Set

$$Q_a^{\mathrm{num}}(n) = hn(n-1)^2 + \frac{(h-1)^2 n(n-1)(2n-1)}{6} + \max_{0 \le t \le n} \left\{ 2h(h+1)(n-1)\big(P_0(t) + P_0(n-t)\big) - h^2\big(P_2(t) + P_2(n-t)\big) \right\}, \quad (44)$$

$$Q_a(n) = Q_a^{\mathrm{num}}(n)/h^2. \quad (45)$$

The maximum is over all ambient root positions. Only the symmetry $t \leftrightarrow n - t$ is used computationally; no endpoint maximizer is assumed.

Lemma E.1 (Generalized rooted graph-moment bound)**.** *Let $\mathcal{E}$ be a subfamily of non-AP members containing a fixed point p, and suppose every member of $\mathcal{E}$ has at least a points. No such size assumption is imposed on members outside $\mathcal{E}$. If*

$$h \max_{1 \le g \le n} C_0(g) \le (h+1)(n-1), \quad (46)$$

then $|\mathcal{E}| \le K_a(n)$. If d_v is the degree of a nonroot point v in $\mathcal{E}$, then $\sum_{v \ne p} d_v^2 \le Q_a(n)$.

Proof. For $A \in \mathcal{E}$, form a graph on $A \setminus \{p\}$ by joining x, y when $\mathrm{Hull}(p, x, y) \not\subseteq A$. An edge cannot belong to the graphs of two members: their AP intersection would contain the missing hull.

Each graph has exactly one nontrivial connected component. Here is the geometric reason. Translate p to zero, and choose a nonzero $z \in A$ of minimum absolute value. If x is a nonneighbor of z, the hull of $0, z, x$ lies in A. Its difference must be $|z|$, since a smaller difference would put a nonzero point of smaller absolute value into A. Thus x is a multiple of z, and the whole $|z|$-grid segment from zero to x lies in A. Any two nonneighbors of z therefore have their hull with zero inside A, so these nonneighbors form an independent set. If z were isolated, the same description would make A an AP. Hence z lies in a nontrivial component, and every vertex outside that component is isolated.

Let that component have $m \ge 2$ vertices and let there be t isolated vertices. Then $m + t \ge h + 1$. For any selection of ambient nonroot vertices, give an edge weight $(h+1)/h$ when both endpoints are selected, weight one when exactly one is selected, and weight zero otherwise. If a proper subset of the nontrivial component is selected, a spanning tree pays at least one per selected vertex: each component of the selected induced forest has an outgoing edge. If the whole component is selected, the payment is at least $(h+1)(m-1)/h$, whose deficit from m is at most t/h.

Let u_v, s_v count the nonisolated and isolated occurrences of v, and put $T = \sum_v s_v$. Order the u_v nonincreasingly. Edge privacy and the preceding weighted argument give, for every k,

$$\sum_{i\le k} u_i \le \sum_{i\le k} v_i + T/h, \qquad v_i = n - 1 - \frac{h-1}{h}(i-1).$$

Also $u_i \le n-1$, by choosing a distinct incident edge for each nonisolated occurrence. Abel summation with the nonincreasing weights $u_i + v_i$ yields

$$\sum_i u_i^2 \le \sum_i v_i^2 + \frac{2(n-1)}{h} T.$$

The common-pair bound gives $d_v = u_v + s_v \le n-1$. Hence

$$\sum_v d_v^2 \le \sum_i v_i^2 + \frac{2(h+1)}{h}(n-1)\sum_v s_v - \sum_v s_v^2.$$

Isolation at v is exactly self-closure at the pair p, v, so $s_v \le C_0(|p-v|)$. Guard (46) makes every quadratic term in s_v nondecreasing throughout the required interval. Substitute the upper bounds coordinatewise and use

$$h^2 \sum_i v_i^2 = hn(n-1)^2 + \frac{(h-1)^2 n(n-1)(2n-1)}{6}$$

to obtain (44). Finally, every member contributes at least $m - 1 + t \ge h$ edges plus isolated vertices. Edge privacy and self-closure give $h|\mathcal{E}| \le \binom{n}{2} + P_0(p-1) + P_0(N-p) \le K^{\mathrm{num}}(n)$. □

Let $S = \{s_1, \dots, s_k\}$ be an ambient set meeting every non-AP member, and define $\mathcal{E}_i = \{A \in \mathcal{F} : A \text{ is non-AP},\ A \cap S = \{s_i\}\}$ and $u_i = |\mathcal{E}_i|$. If these classes have member sizes at least a, the lemma gives $u_i \le K = K_a(n)$. Members from two different classes intersect outside S. Counting their intersections, and applying Cauchy–Schwarz to their degree vectors on the same ambient coordinates, gives

$$u_i u_j \le Q = Q_a(n) \quad (i \ne j). \tag{47}$$

Consequently any $j \ge 2$ classes have total size at most

$$M_j = \left\lfloor \max\{j\sqrt{Q},\ K + (j-1)Q/K\} \right\rfloor, \qquad M_1 = \lfloor K \rfloor, \quad M_0 = 0. \tag{48}$$

If their largest size is at most $\sqrt{Q}$, use the first expression. Otherwise let its size be x, with $\sqrt{Q} < x \le K$. The remaining sizes are at most Q/x, and the convex function $x + (j-1)Q/x$ is bounded by its endpoint maximum. If $K < \sqrt{Q}$, the second expression is unnecessary but remains harmless.

E.2 Two finite steps from minimum size nine to minimum size fourteen

For an integer $b \ge 2$, define the AP reservoir

$$\mathcal{R}_b(n) = \sum_{d=1}^{\lfloor n/(b-1) \rfloor} f_b(\lfloor n/d \rfloor + 1), \tag{49}$$

where

$$f_b(v) = \begin{cases} 0, & v < b, \\ (v-b+1)(v-b+2)/2, & b \le v \le 2b-2, \\ \lfloor (v+1)^2/4 \rfloor - b(b-1)/2, & v \ge 2b-1. \end{cases}$$

AP members with a fixed difference lie on one residue chain. Their intervals intersect pairwise, so have a common chain position. A central position maximizes the number of intervals of each length simultaneously. Thus the number of actual AP members of size at least b is at most $\mathcal{R}_b(n)$.

Lemma E.2 (The finite minimum-size-nine to thirteen step)**.** *For* $10000 \le N \le 11993$*, if all members have size at least nine and* $|\mathcal{F}| \ge B(N)$*, no twelve ambient points meet every non-AP member. In particular every member has size at least thirteen.*

Proof. Use $a = 9$, $j = 12$ in (48), and put $Z_{12} = B(N) - \mathcal{R}_9(n) - \binom{12}{2}(n-1)$. Enlarge any smaller covering set to twelve points. The common-pair bound and a union bound pay for members meeting it at least twice; the singleton classes have total size at most M_{12}. The density assumption would require $Z_{12} \le M_{12}$. Exact checking at every integer and every ambient root split verifies (46), whereas the minimum of $Z_{12} - M_{12}$ is 347467, at $N = 10001$. Every actual member itself meets every non-AP member, so none can have size at most twelve. The entire step is checked by `independent_joint_check.py`, without importing a historical low-range table. □

Lemma E.3 (Excluding a thirteen-point covering set)**.** *For* $10000 \le N \le 11993$*, minimum member size thirteen and* $|\mathcal{F}| \ge B(N)$ *exclude a thirteen-point ambient covering set for the non-AP members. In particular every member has size at least fourteen.*

Proof. Take $a = r = 13$, $h = 11$, and enlarge a smaller covering set to size r. At most $\binom{r}{2}(n-1)$ non-AP members meet it at least twice, and the singleton classes have total size at most M_r. Put

$$Z_r = B(N) - \mathcal{R}_{13}(n) - \binom{r}{2}(n-1).$$

It suffices to verify

$$\begin{aligned} Z_r &> 0, \\ h^2 Z_r^2 - r^2 Q_{13}^{\mathrm{num}}(n) &> 0, \\ hK^{\mathrm{num}}(n) Z_r - (K^{\mathrm{num}}(n))^2 - (r-1) Q_{13}^{\mathrm{num}}(n) &> 0. \end{aligned} \tag{50}$$

These inequalities imply $Z_r > M_r$. Over the full integer range their respective minimum left sides are

$$8130839, \qquad 154216087113429, \qquad 1405809013620514.$$

The minimum gap in guard (46) is 97584. Every maximum is computed over all root splits at each integer; no endpoint interpolation is used. An actual nonempty member meets every non-AP member, including itself when appropriate, and hence cannot have size at most thirteen. □

E.3 Contact counting from an actual fourteen-point member and short APs

Assume now that all members have size at least fourteen, and let $S = \{s_1, \dots, s_{14}\}$ be an actual fourteen-point member. Every non-AP member meets S. Order the points of S and partition the members meeting S at least twice by their first two points of intersection. Their total number is at most

$$Z = \sum_{j=2}^{14} (j-1)(N-j). \tag{51}$$

For a fixed first point and second point s_j, the class contains their pair and avoids the other $j-2$ earlier points. The common-pair bound on its actual union is therefore $N - j$. Given a proved bound A_P for the total number of actual AP members, put

$$\delta = B(N) - A_P - Z, \qquad \sum_{i=1}^{14} u_i \ge \delta. \tag{52}$$

Let an actual member A avoid s_i, and write $r = |A \setminus S| \le N - 14$. Order $A \setminus S$ and partition $\mathcal{E}_i$ by the first contact with it. The ℓth class contains $\{s_i, t_\ell\}$ and avoids the other thirteen points of S and the preceding $\ell - 1$ contact points. Its size is at most $N - 14 - \ell$. Hence

$$u_i \le \Phi(r), \qquad \Phi(r) = r(n-14) - r(r-1)/2, \quad 0 \le r \le N - 14. \tag{53}$$

The function is nondecreasing on this integer domain, with final increment zero. For $0 < q \le \Phi(N-14)$ define

$$I(q) = \min\{r \in \{0, \dots, N-14\} : \Phi(r) \ge q\}, \qquad I(0) = 0.$$

If $u_i \ge q > 0$, invert only on the actual domain to obtain $r \ge I(q)$. Since S is an actual member and A avoids $s_i \in S$, $A \ne S$ and admissibility gives $A \cap S \ne \varnothing$. Thus $|A| \ge r + 1 \ge 1 + I(q)$. We never evaluate $\Phi(|A| - 1)$ outside its domain. If $q > \Phi(N-14)$, the positive class and an avoiding member already contradict one another.

Lemma E.4 (Short APs containing fixed points)**.** *For any $m \ge 2$ distinct integer points H and integer $b \ge 2$, the number of finite APs containing H and having cardinality strictly less than b is at most*

$$E(m,b) = \sum_{e=1}^{\lfloor (b-2)/(m-1) \rfloor} \binom{b - (m-1)e}{2}. \tag{54}$$

Proof. Let $G = \gcd\{x - \min H : x \in H\}$ and $R = (\max H - \min H)/G \ge m - 1$. The difference d of a containing AP divides G. Write $e = G/d$; its two extreme prescribed points are Re steps apart. There are at most $(\ell - Re)_+$ placements of an AP of cardinality ℓ containing them. Requiring the other prescribed points, or containment in $[N]$, only decreases this count. Summing over $\ell < b$ gives $\binom{b-Re}{2}$, interpreted as zero when its upper parameter is less than two. Now sum over $e \mid G$, discard that divisibility restriction, and replace R by its lower bound $m - 1$ to obtain (54). □

If at least $m \ge 2$ classes have size at least $q > 0$, set $b = \max(14, 1 + I(q))$. Every actual AP member of size less than b contains the corresponding fixed m points of S, since otherwise (53) forces its size to be at least b. Thus

$$A_P \le \mathcal{R}_b(n) + E(m,b). \tag{55}$$

Only actual AP members have been divided into long members and short exceptions; the global minimum member size has not been raised to b. The condition $m \ge 2$ is essential, as is the strict cutoff below b.

E.4 The largest class, ordered large classes and causal feedback

An incoming parameter a bounds member sizes only within the singleton classes $\mathcal{E}_i$, not within multiple-hit classes. Compute Q, K from Lemma E.1 and M_j from (48). For $u_{(1)} \ge \cdots \ge u_{(14)}$, inequality (52) implies

$$u_{(m)} \ge q_m := \max\left(0, \left\lceil \frac{\delta - M_{m-1}}{15 - m} \right\rceil\right), \quad 2 \le m \le 14. \tag{56}$$

Indeed the first $m - 1$ classes total at most M_{m-1}, and each remaining class has size at most $u_{(m)}$. Every positive q_m supplies a candidate in (55); retain the minimum of all valid candidates.

The actual largest class has integer size $x \le \lfloor K \rfloor$. If $x > \sqrt{Q}$, the product bounds and (52) give

$$\delta x \le x^2 + 13Q. \tag{57}$$

An improved integer upper bound $X \le \lfloor K \rfloor$ is accepted only after checking

$$\begin{gathered} X \ge \lfloor \sqrt{Q} \rfloor, \\ (X+1)^2 - \delta(X+1) + 13Q < 0, \qquad \lfloor K \rfloor^2 - \delta \lfloor K \rfloor + 13Q < 0. \end{gathered} \tag{58}$$

Convexity makes the quadratic negative at every integer $X < x \le \lfloor K \rfloor$, contradicting (57). The branch $x \le \sqrt{Q}$ is covered by the first guard. If no improvement is certified, retain $X = \lfloor K \rfloor$. In particular the large-root branch of the quadratic inequality is not discarded.

For any nonempty $\mathcal{E}_i$, the other thirteen classes total at least $\delta - u_i \ge \delta - X$. Some other root therefore has a class of size at least

$$q = \max\left(0, \left\lceil \frac{\delta - X}{13} \right\rceil\right).$$

Every member of $\mathcal{E}_i$ avoids that other root, so (53) gives size at least $1 + I(q)$. The other root may depend on i; the resulting size bound applies uniformly to all singleton classes. Empty classes cause no difficulty. This improvement may be used only in the next state, and not for multiple-hit classes or for the whole family.

Start from $a = 14, A_P = \mathcal{R}_{14}(n)$. Each state uses only its already proved a, A_P to compute Q, K, δ, M_j. The strict terminal condition is

$$M_{14} < \delta. \tag{59}$$

Otherwise the largest-class argument supplies a new singleton size $a' \ge a$, and (56)–(55) supply $A'_P \le A_P$. These values are used only in the next state. Neither a fixed point nor an iteration limit constitutes a contradiction.

Proposition E.5 (Finite exclusion of an actual fourteen-point member)**.** *For every* $10000 \le N \le 11993$*, no admissible family of size at least* $B(N)$ *can have minimum member size fourteen and contain an actual fourteen-point member.*

Proof. The causal feedback reaches (59) for every one of the 1994 integers, in at most four states. Each state maximizes (44) over every ambient root split and checks (46) and each used instance of (58). The minimum terminal gap $\delta - M_{14}$ is 150, at $N = 11294$. For example, at $N = 10000$ the states are

state	a	A_P	δ	M_{14}
0	14	41077084	8011326	8635167
1	58	40799119	8289291	8345748
2	61	40744836	8343574	8341900

The first AP update uses nine large classes, $b = 53$ and $E(9, 53) = 2360$; the second uses eight large classes, $b = 61$ and $E(8, 61) = 4392$. The last state is strictly contradictory. All states for all integers are retained; this example does not replace the full-range certificate.

All implementation decisions are integer decisions. For $Q = Q^{\mathrm{num}}/h^2$, the square-root branch of (48) is computed as $\mathrm{isqrt}(\lfloor j^2 Q^{\mathrm{num}}/h^2 \rfloor)$, and the other branch as

$$\left\lfloor \frac{(K^{\mathrm{num}})^2 + (j-1)Q^{\mathrm{num}}}{hK^{\mathrm{num}}} \right\rfloor.$$

Divisor convolution constructs C_0, and $\mathcal{R}_b$ includes all possible ambient differences. There are no floating-point root comparisons, unlisted integer interpolations, inversions outside the physical domain, or unpaid truncations of the large classes.

The primary programs are `joint_ap_feedback_probe.py` and `joint_ap_feedback_full band.py`; their complete states are in `joint_ap_feedback_fullband.json`. The separately

written `independent_joint_feedback_audit.py` constructs C_0 by prime factorization, considers every root split and uses a different interval bisection for the largest class, obtaining the same states. A further selected-state checker, `independent_joint_check.py`, uses prime factorization and quotient-block AP summation. It checks 3392 states and both finite entrance steps. Their minimum terminal integer gaps are respectively 347467 and 78757, both at $N = 10001$. The implementations, exact states, execution records and file hashes accompany the supplement. They verify sufficient finite scalar conditions proved above; they do not enumerate all admissible set families. □

E.5 The junction at 11994 and the analytic improvement from 12500

We first supply the finite junction, rather than appeal to a historically cited table. For every integer $11994 \le N \le 12499$, take $a = 9$ and a covering set of size $r = 14$. Recompute all ambient root positions, $Q_9, K_9, \mathcal{R}_9$, and the guard in (46). The resulting strict inequality is

$$B(N) - \mathcal{R}_9(n) - 91(n-1) - M_{14} > 0.$$

The minimum over these 506 integers is 1697, attained at $N = 12001$. Thus the minimum-size-nine hypothesis implies minimum size fifteen throughout this junction. The complete output is `check_booster_join.result.json`.

We next give the analytic continuation, including its unbounded quantifier. Put $L(x) = (\log x)^2 + 4\log x + 2$ and $D_j(x) = \sum_{m \le x} d_j(m)$. The identity $D_{j+1}(x) = \sum_{a \le x} D_j(x/a)$, followed by the first-term-plus-integral bound, gives

$$\Pi_0(z) = 1, \qquad \Pi_{j+1}(z) = \Pi_j(z) + \int_0^z \Pi_j(t)\,dt, \qquad D_j(x) \le x\Pi_{j-1}(\log x).$$

Indeed, $P(\log(x/t))/t$ is nonnegative and decreasing whenever P, P' are nonnegative on the nonnegative real axis. Since $\Pi_2(z) = 1 + 2z + z^2/2$,

$$P_0(n) = \sum_{g \le n} C_0(g) \le 2D_3(n) \le nL(n).$$

Here P_0 continues to denote the self-closure prefix; the auxiliary polynomials are Π_j.

In the root-graph second-moment argument, discard the negative sum of squared isolated degrees before substituting the self-closure bounds. This requires no finite guard (46) and yields

$$Q^*(n) = \frac{n(n-1)(19n-13)}{49} + \frac{16}{7}(n-1)P_0(n), \qquad K^*(n) = \frac{\binom{n}{2} + P_0(n)}{7}.$$

All AP members have size at least nine. Summing over their actual differences and using $\sum_{d \ge 1} d^{-2} < 329/200$ and $H_n \le 1 + \log n$, we obtain

$$\#\{\text{AP members}\} < \frac{329}{800}n^2 + n(1 + \log n).$$

For each difference the reservoir is at most $n^2/(4d^2) + n/d$: the unrestricted central-interval count has an additional constant at most one, which is removed by excluding even one short interval. If fourteen ambient points meet every non-AP member, the singleton and multiple-hit classes therefore give

$$|\mathcal{F}| < \frac{329}{800}n^2 + n(1 + \log n) + 91(n-1) + \max\{14\sqrt{Q^*(n)}, K^*(n) + 13Q^*(n)/K^*(n)\}.$$

For $n_0 = 12499$ we have

$$\frac{Q^*(n)}{n^4} \le A(n) = \frac{19}{49n} + \frac{16L(n)}{7n^2},$$
$$\frac{K^*(n) + 13Q^*(n)/K^*(n)}{n^2} \le H(n) = \frac{1}{14} + \frac{L(n)+494}{7n} + \frac{416L(n)}{n^2}.$$

The second estimate uses $K^*(n) \ge n(n-1)/14 > 0$; in particular, no choice of a root of a squared inequality is involved. The functions A, H and $(\log n + 92)/n$ decrease for real $n \ge n_0$, as follows from

$$(L/n)' = \frac{-(\log n)^2 - 2\log n + 2}{n^2} < 0, \qquad (L/n^2)' = \frac{-2(\log n)^2 - 6\log n}{n^3} < 0.$$

The positive exponential series gives

$$e^{10} > \sum_{k=0}^{10} \frac{10^k}{k!} = \frac{7281587}{567} > 12499,$$

so $\log n_0 < 10$ and $L(n_0) < 142$. Set

$$b = \frac{1}{2} - \frac{329}{800} - \frac{102}{n_0} > 0,$$
$$A_0 = \frac{19}{49n_0} + \frac{16 \cdot 142}{7n_0^2},$$
$$H_0 = \frac{1}{14} + \frac{636}{7n_0} + \frac{416 \cdot 142}{n_0^2}.$$

Exact rational arithmetic gives

$$b^2 - 196A_0 = \frac{694777241}{99984000640000} > 0, \qquad b - H_0 = \frac{1324101897}{874860005600} > 0.$$

Both branches consequently give $|\mathcal{F}| < n^2/2 < B(N)$, a contradiction. Monotonicity proves this for every real $n \ge 12499$, rather than for sampled large integers. Thus, for every $N \ge 12500$, no set of at most fourteen ambient points meets every non-AP member: a smaller covering set could be enlarged to fourteen points. Every actual member itself meets every non-AP member, with its own intersection also nonempty when it is non-AP. Hence every member has size at least fifteen.

Theorem E.6 (Restoring minimum size fifteen)**.** *Using the input results stated at the beginning of this section, for every $N \ge 10000$, a nonstar admissible family with $|\mathcal{F}| \ge B(N)$ has every member of size at least fifteen.*

Proof. Apply the independent minimum-size-nine reduction. For $N \ge 11994$, use the finite junction and analytic continuation in Section E.5. For $10000 \le N \le 11993$, Lemma E.2 first gives minimum size thirteen, and Lemma E.3 gives minimum size fourteen. Any member still smaller than fifteen is an actual fourteen-point member, contrary to Proposition E.5. No extremal theorem at 10000 or 4000 is used in this chain. □

Remark E.7 (A strict-excess version already suffices for the extremal formula)**.** Even the weaker implication

$$\mathcal{F} \text{ nonstar}, \quad |\mathcal{F}| > B(N) \implies \min_{A \in \mathcal{F}} |A| \ge 15$$

would suffice: a strict-excess counterexample would meet the downstream hypotheses $|\mathcal{F}| \ge B(N)$ and minimum size fifteen. The upper bound would then combine with the construction to give equality. This weaker statement alone would not classify equality families as stars or establish uniqueness. The joint feedback above actually retains the stronger density hypothesis $|\mathcal{F}| \ge B(N)$.

F A uniform finite proof for $10000 \le N \le 74999$

This section uses a uniform specification of original members, ambient intervals and prefix certificates. For $10000 \le N \le 34999$ we replay the sufficient certificates identified as T019. For $35000 \le N \le 74999$ we use the multiround global ordinary-population feedback constructed on 2 November 2026. The latter proof does not depend on the older numerical transitions through the thresholds 49000, 49100, 45000, 40000 and 35000. The minimum-fifteen premise is supplied by the preceding supplementary argument, constructed on 1 November 2026; a later revision date does not alter the date of that argument.

F.1 The original family, same-integer inventories and full resources

Put $B(N) = \binom{N}{2} + 1 + \lfloor (N-1)/4 \rfloor$. Throughout an integer box $L \le N \le H$, retain the same original family $\mathcal{F}$. For each non-AP member, its globally maximum AP core, core step and original defect are fixed once. Embedding the family in $[H]$ changes none of these objects. It does not make H a new near-extremal-density parameter for the original family.

For integers $b \ge 2$, define

$$f_b(v) = \sum_{\ell=b}^{v} \min(\ell, v+1-\ell), \qquad \mathcal{R}_b(N) = \sum_{d=1}^{\lfloor (N-1)/(b-1) \rfloor} f_b\Big(\Big\lfloor \frac{N-1}{d} \Big\rfloor + 1\Big),$$

with an empty sum equal to zero. Let O denote the number of original ordinary members, namely AP members of step one, and let C denote the number of all original non-AP members. If every original member has size at least b, the AP inventory is at most $\mathcal{R}_b(N)$, and the inventory of AP members of step greater than one is at most $\mathcal{R}_b(N) - f_b(N)$. Therefore, with

$$\begin{aligned} D_b &= \min_{L \le N \le H} \{B(N) - \mathcal{R}_b(N)\}, \\ E_b &= \min_{L \le N \le H} \{B(N) - \mathcal{R}_b(N) + f_b(N)\}, \end{aligned} \tag{60}$$

we have

$$C \ge D_b, \qquad O \ge E_b - C, \qquad O \le f_b(H). \tag{61}$$

The two minima are computed independently at every actual integer. In particular, $B(L) - \mathcal{R}_b(H)$ is not substituted for a same-N difference. This argument needs only a lower bound b on all member sizes, not an actual member of size b.

Put $\Gamma = H - 2$. Let $D^*(H)$ bound the full-ambient rooted sum of C^*; let $F_t(H)$ be the full-ambient rooted bound for the complete shifted residual cost W_{t-1}; and let $\mathrm{BetaHost}_q(H)$ be the independent full-ambient rooted bound for truncated original-defect cost. The complete event partition and left–right reflection control every ambient point and endpoint position; the central point alone is not sufficient. The rank account uses $F_2(H)$, not $\mathrm{BetaHost}_2(H)$. Only in the truncated crossing account may we take

$$A_q(H) = \min\{F_q(H), \mathrm{BetaHost}_q(H)\}.$$

For the first-hit account use the universal gap multisets defining $\mathrm{TopC}(j)$ and $\mathrm{TopBeta}_q(j)$: each gap g occurs once, and occurs a second time only when $2g \le H - 1$; the function is the sum of the j largest entries. These distinct resources are not interchanged.

F.2 Prefix capacity certificates and their sufficiency

Suppose a completed step has proved $O \geq P$, and all original ordinary members have length at least b. When $0 < P \leq f_b(H)$, the first inverse

$$\ell = \min \left\{ v \in \{b, \dots, H\} : \sum_{j=b}^{v} \min(j, H+1-j) \geq P \right\}$$

proves the existence of an actual original ordinary member of length at least ℓ, not necessarily of length exactly ℓ. No owner is inferred when $P = 0$, and $P > f_b(H)$ is already a contradiction. Physical unions of subfamilies of long ordinary members receive length lower bounds by the corresponding full-inventory inversion. An actual ordinary member and an interval union are different objects; a union does not receive a reserve specific to an actual owner.

For a fixed context (H, r, J, b, P, ℓ), each capacity option is used only when all its original hypotheses hold. It bounds the same original subfamily within $[H]$, with original size at least $r+1$, fixed-core step d, and original defect between l and u. Only complete candidate bounds are minimized. Adding small- and large-class bounds requires a disjoint partition of the original members. An improvement for the support of the small class does not change the hypotheses for the whole class or the large class. Signed quadratic constants are not clipped to zero, and negative numerators use mathematical floor division.

Let $X_{d,j}$ count the corresponding original members of defect at most j, with $X_{d,0} = 0$. A selected record $0 \leq \pi < j \leq u \leq J$ gives the sufficient inequality

$$X_{d,j} \leq X_{d,\pi} + \mathrm{Cap}(H, r, d, \pi+1, u; b, P, \ell). \tag{M1}$$

Once the required parent has been proved, this edge suffices. It need not be optimal, and unused alternatives need not be searched again. The strict physical emptiness criterion is $d(r-j) > H-1$; equality does not imply emptiness. All possible steps are represented up to $\lfloor (H-1)/(r-J) \rfloor$, and larger steps cannot accommodate a tracked core. Summing over steps gives the complete prefix bound Π_j.

The reduction in computation removes only prefixes not used by the final scalar inequalities and edges unrelated to their ancestry. For every required j, its π-ancestors are recursively closed at each physical step, and every required capacity is recomputed. The binding context still includes H, r, J, b, P, ℓ, the phase, and the conditional hypothesis. A population-dependent capacity is not cached by H alone. This reduces work without narrowing the parameter domain covered by the proof.

F.3 A crossing account for all actual minimum sizes

Let $\mathcal{T}_r$ consist of all original non-AP members of size strictly greater than r. Using the preceding original-object prefixes, every available q gives the full upper bound

$$|\mathcal{T}_r| \leq U_r(q) := \frac{rq + (r+1)\{(H-r-1)\Gamma + A_q(H)\} + r\sum_{j<q} \Pi_j}{rq}. \tag{62}$$

The ordinary population P may be zero or may be a previously proved global population of actual original ordinary members. In the latter case the new context retains the same original family and the same ordinary-member length lower bound b.

Formula (62) does not require an actual $(r+1)$-point member. If the tail is nonempty, choose a member of its actual minimum size $s \geq r+1$. The truncated crossing account, restoring this member exactly once, gives

$$|\mathcal{T}_r| \leq 1 + \frac{s\{(H-s)\Gamma + A_q(H)\}}{(s-1)q} + \frac{\sum_{j<q} \Pi_j}{q}.$$

The identity

$$\frac{s\{(H-s)\Gamma + A_q\}}{s-1} = (H-1-s)\Gamma + A_q + \frac{(H-1)\Gamma + A_q}{s-1}$$

shows that this term is nonincreasing for $s \geq r+1$, since $\Gamma > 0$ and $A_q \geq 0$. Substitution of $r+1$ is therefore valid in this crossing account and handles every remaining actual minimum size without enumerating it separately. An empty tail also satisfies the nonnegative bound (62). This monotonicity does not automatically extend to a different mixed account with different reserved terms.

F.4 The small-member contradiction and its discharge

Fix an attained minimum size $15 \leq a \leq 132$ of the full original family, choose an actual member of size a, and temporarily assume that $K_{132} = \{A \in \mathcal{F} : |A| \leq 132\}$ has no common point. The first-hit construction covers all original members by at most $131a+1$ distinct ambient pairs, with at most a incident vertices, each of degree at most 132. For completeness, order an actual a-point member S so that $S \cap T$ comes first, where $T \in K_{132}$ and $1 \leq t = |S \cap T| \leq a-1$. Such a T exists, since otherwise all members of K_{132} would contain S. For each of the first t vertices use a member of K_{132} avoiding that vertex, which exists by the temporary no-common-point assumption; for the remaining vertices use T. Delete previously ordered vertices of S from each partner list. The number of distinct covering pairs is at most $132t + (a-t)(132-t) = 132a - t(a-t) \leq 131a+1$. The universal incident-resource functions are nondecreasing and concave; with at most a degrees, each at most 132 and total at most $131a+1$, their sum is at most the value for $a-1$ degrees 131 and one degree 132. Thus the truncated original-defect account gives

$$qC \leq (131a+1)\Gamma + \min\begin{Bmatrix}(q-1)((a-1)\mathrm{TopC}(131) + \mathrm{TopC}(132)),\\ (a-1)\mathrm{TopBeta}_q(131) + \mathrm{TopBeta}_q(132)\end{Bmatrix} + \sum_{j<q}\Pi_j. \qquad (63)$$

Here the prefixes count all original non-AP members, with $r = a-1$. The minimum is taken after aggregating each complete cover-error bound; coordinatewise savings from different accounts are not added together.

Start with the conditional population $P = 0$. The valid incoming prefixes in (63) yield an integer upper bound u on C. If $u < D_a$, there is a contradiction. Otherwise only the next state may use

$$P_{\mathrm{next}} = \max(P, 0, E_a - u).$$

If $P_{\mathrm{next}} > f_a(H)$, the same original ordinary inventory gives a contradiction. Otherwise, first obtain the actual-owner and union information by inversion, then evaluate the next state. Iteration continues only when $P_{\mathrm{next}} > P$; stagnation or an iteration limit is not a proof.

After every branch has reached a contradiction, retain only $\mathrm{STAR}(K_{132})$, the assertion that this subfamily has a common point. All populations, owners, prefixes, tail bounds and reserves obtained only under its contrary assumption are discarded. The attained member of size $a \leq 132$ guarantees that K_{132} is nonempty, so a common ambient point p can now be chosen and fixed. Its existence does not come from a conditional population or a conditional tail mass.

F.5 Minimum size at least 133: global feedback without small members

Suppose now that every original member has size at least 133. No actual member of size 133 is assumed. All non-AP members lie in $\mathcal{T}_{132}$. At any previously justified global ordinary population state P, formula (62) therefore yields $C \le U_{132}(P)$, while (61) gives

$$C \ge D_{133}, \qquad O \ge E_{133} - \lfloor U_{132}(P) \rfloor. \tag{64}$$

Start at $P = 0$. Even when $U_{132}(0) \ge D_{133}$, the second inequality can give a genuine new ordinary population; it does not require a prior contradiction on C. Set

$$P_{\text{next}} = \max\{P, 0, E_{133} - \lfloor U_{132}(P) \rfloor\}.$$

Either $U_{132}(P) < D_{133}$ or $P_{\text{next}} > f_{133}(H)$ is a contradiction. Otherwise the next capacity context uses only this genuine population and the valid lower bound 133 on original ordinary lengths. The argument does not treat a tail subfamily as a new near-extremal family, create a 133-point member, require K_{132} to be nonempty, or seek a common point of it. Each crossing step already handles all remaining actual minimum sizes.

F.6 Attained minimum size 15 through 132: restarting globally

After the conditional hypothesis in Section F.4 has been discharged, restart the global chain with $P = 0$. For the fixed common point p, the complete original-member rank account gives

$$C_p \le T_a := \frac{(H-1)(H-2) + F_2(H)}{2(a-2)}. \tag{65}$$

It holds for any ambient point and original subfamily, without a near-extremal-density requirement on that subfamily. Every non-AP member avoiding p has size greater than 132, hence lies in $\mathcal{T}_{132}$. Consequently

$$C \le C_p + C_{\neg p} \le \lfloor T_a \rfloor + \lfloor U_{132}(P) \rfloor.$$

If

$$D_a - U_{132}(P) > T_a,$$

there is a strict contradiction. Otherwise the complete inventory of nonordinary AP members in the same original family gives the genuine global update

$$P_{\text{next}} = \max\{P, 0, E_a - \lfloor T_a \rfloor - \lfloor U_{132}(P) \rfloor\}. \tag{66}$$

If $P_{\text{next}} > f_a(H)$, this contradicts the ordinary inventory with the same length lower bound a. Otherwise the new population licenses an owner, a physical union and new crossing prefixes only in the next round. The same fixed p is used throughout. The crossing bound is still an upper bound for the entire original tail, not an unproved density statement about the members avoiding p. The new global population follows from (65) and the independent global tail bound, not from the discarded conditional ROOT branch.

F.7 Complete numerical coverage and adjacent endpoints

For the 174 selected boxes in $10000 \le N \le 34999$, the present verification reevaluates every selected prefix used by a final scalar inequality and all of its ancestors. It reconstructs the complete ambient resources, recomputes both independent same-integer inventory minima at every actual integer, and checks every conditional ROOT discharge and global conclusion with exact integer and rational arithmetic. There are 369447453 required capacity evaluations, 410016001 required strict-span zero checks, 60501 required contexts and 1339014 used prefixes. All 350 verification jobs exit successfully. Optimality of unused search choices and unsuccessful parent boxes are not necessary computations for these sufficient certificates.

For $35000 \le N \le 74999$, use the forty adjacent boxes $[35000 + 1000i, 35999 + 1000i]$, $0 \le i \le 39$. The pre-existing certificates for these boxes had already proved all 118 conditional ROOT exclusions per box, although their old single-round GLOBAL criterion failed. We reuse only those ROOT certificates and verify them completely. The multiround feedback in Sections F.5 and F.6 then generates new global contexts, without importing numerical conclusions from the unsuccessful GLOBAL criterion.

All forty boxes now pass the selected-ancestor capacity checks, full-ambient resource reconstruction, integer-by-integer inventory computation and independent exact scalar checks. This requires 136787449 capacity evaluations, 84298004 strict-span zero checks, 85890234 full-ambient resource event intervals, and 8000000 same-integer inventory pairs. All 82 verification jobs exit successfully. The new global certificates use 4883 contexts. The independent scalar verification covers 5516 conditional ROOT steps, 80 global steps for minimum size at least 133, and 9523 global steps for the small attained minima. The minimum conditional ROOT integer margin is 33125; the minimum rational margin for the minimum-at-least-133 branch is 420069025/6; and the minimum rational rank-contradiction margin for the small-minimum branches is 1681299343/14190. Exact execution counts, strict margins and phase timings are recorded in the frozen computation report.

The two ranges meet at the adjacent integers 34999 and 35000. The second ends at 74999, immediately before the independent high-range proof beginning at 75000. The branches exhaust all attained minimum sizes: $15, \dots, 132$ are treated separately, and Section F.5 covers every remaining minimum. No unhandled minimum-size tail remains beyond a finite cutoff. Together with the minimum-fifteen premise, the common-point upper bound and the standard construction, this establishes the exact extremal formula throughout the stated finite range.

G A new finite connection from 75000 to the independent tail

We give a newly generated and independently checked finite certificate for $75000 \le N \le 1025000$. A common initial cutoff 160 and three mass stages replace the finite acceptance assertions in the earlier G modules. The resulting interval meets the independent tail, which assumes no minimum member size. We verify sufficient inequalities; neither the entire historical search matrix nor its optimality is a premise.

Fix a nonstar admissible family $\mathcal{F} \subseteq 2^{[N]}$ with

$$|\mathcal{F}| \ge B(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor .$$

The preceding minimum-size theorem gives size at least fifteen for every member. For each non-AP member A, fix once and for all an AP subset of maximum cardinality, over all differences. Its difference and the defect $\delta(A)$ remain fixed. Embedding the family in a larger ambient interval changes none of these objects. The capacity statements used below have complete proofs in the technical supplement [32]; their numerical interfaces are stated explicitly here.

G.1 Common-pair resources and the complete AP reservoir

Write

$$C_0(g) = 2(d_3(g) - \tau(g)), \qquad P_0(n) = \sum_{g=1}^{n} C_0(g).$$

The self-closure bound and the common-AP-core theorem on the actual union of the sets give, for every ambient pair $e = \{x, y\} \subseteq [H]$ and integer $q \geq 1$,

$$\sum_{A \supseteq e} \min(\delta(A), q) \leq H - 2 + qC_0(|x - y|). \tag{67}$$

The sum is over non-AP original members only. The common-AP-core theorem also bounds their unweighted number through a fixed pair by $H - 2$.

For clarity, delete from each original member through e the points outside e that are private to it within this pair class. Denote the retained support by Q_A and the number deleted by b_A. Deleted points are distinct and lie in no retained support. Each Q_A is self-closed: a retained point is shared with another original member, and their AP intersection contains its hull with e. Furthermore $Q_A \cap Q_B = A \cap B$, so the retained non-AP supports still have AP pair intersections and are distinct. If there are c such non-AP supports, the common-core theorem applied on their actual union gives $\sum_A b_A + c \leq H - 2$, and the self-closure bound gives $c \leq C_0(|x - y|)$. Let $\alpha_e(Q_A)$ be the deficit from a largest AP subset of Q_A containing e. Then $\delta(A) \leq b_A + \alpha_e(Q_A)$, with $\alpha_e(Q_A) = 0$ when Q_A is an AP. Consequently the left side of (67) is at most $\sum_A b_A + qc \leq H - 2 + (q - 1)C_0(|x - y|)$, a slightly stronger inequality. Only labels and resource payments are transferred to the retained supports; neither density nor the original minimum member size is transferred.

For an integer $a \geq 3$, define

$$f_a(v) = \begin{cases} 0, & v < a, \\ (v - a + 1)(v - a + 2)/2, & a \leq v \leq 2a - 2, \\ \lfloor (v + 1)^2/4 \rfloor - a(a - 1)/2, & v \geq 2a - 1, \end{cases}$$

$$R_a(H) = \sum_{d=1}^{\lfloor (H-1)/(a-1) \rfloor} f_a\left(\left\lfloor \frac{H - 1}{d} \right\rfloor + 1 \right).$$

Actual AP members with one fixed difference lie on the same residue chain. Their intervals intersect pairwise, and hence share a chain point. The central point simultaneously maximizes the interval count at each length, giving f_a. The total number of actual AP members of size at least a is therefore at most $R_a(H)$. Simultaneous attainment for different differences is not required.

G.2 Fixed maximum-core capacities and selected prefix certificates

Fix H, r, d, l, u, with $1 \leq l \leq u$, and count original non-AP members of size greater than r, fixed maximum-core difference d, and defect in $[l, u]$. Put $k = r + 1 - u \geq 4$ and $M = \lceil H/d \rceil$. If $d(k - 1) > H - 1$, this class is empty. Otherwise define

$$\mathrm{Pack}(d, t) = (d - v) \left\lfloor \frac{h}{t} \right\rfloor + v \left\lfloor \frac{h + 1}{t} \right\rfloor, \qquad H = hd + v, \quad 0 \leq v < d,$$

$$C = \min \left\{ \mathrm{Pack}(d, \lfloor k/2 \rfloor), \ \left\lfloor \frac{2(H - 1)}{k} \right\rfloor, \ \mathrm{Pack}(d, k - 1) + d - 1 \right\}.$$

When $k > 2u+2$, also include $\mathrm{Pack}(d, k-u-1)$ in this minimum. Set

$$A = 4H + luC + \min(4uH, 2lCM), \qquad \Delta = A^2 - 16H^2, \quad Z = \left\lfloor \frac{A+\sqrt{\Delta}}{2l} \right\rfloor.$$

The fixed maximum-core capacity theorem bounds the class by Z. If both

$$10000r + 9999 > 10000(3u+2), \qquad d(20000r - 30000u - 2) \geq 10000H,$$

hold, also use the upper bound $2H + (u+2)d + \lfloor 2H/d \rfloor + 2(u+1)$. Only for $l = u = 1$, put $Q_d = \max(0, \lfloor (H-d+1)/(r-1) \rfloor)$ and also use $2H + 3Q_d + 82$; if $M \leq 2r-3$, also use $H + 3Q_d$. Let $T_d(H; r, l, u)$ be the minimum of these individually qualified bounds. Failure of an optional qualification removes that bound; it does not declare the original class empty.

Fix $J = q-1$ with $r - J \geq 3$. For each $1 \leq d \leq \lfloor (H-1)/(r-J) \rfloor$, let S_j count members of size greater than r, with this fixed core difference and defect at most j. For every $j = 1, \ldots, J$, the certificate supplies (π_j, ν_j) satisfying

$$0 \leq \pi_j < j \leq \nu_j \leq J. \tag{68}$$

Define

$$D_0 = 0, \qquad D_j = D_{\pi_j} + T_d(H; r, \pi_j + 1, \nu_j).$$

The actual counts satisfy

$$S_j \leq S_{\pi_j} + \#\{A : \pi_j < \delta(A) \leq \nu_j\},$$

within this fixed size and core-difference class. Induction on j gives $S_j \leq D_j$. In particular $\nu_j > j$ is legitimate: the recurrence uses only the earlier bound D_{π_j}, not an unproved D_{ν_j}. The certificate is a sufficient witness, not a claim of optimality for unused matrix entries. Put

$$\mathrm{CDF}(H; r, q) = \sum_{d=1}^{\lfloor (H-1)/(r-J) \rfloor} \sum_{j=1}^{J} D_{d,j}.$$

All possible differences and all prefixes are present; there is no sampling of the difference range.

G.3 A mass bound not assuming that a short member exists

Let $\mathcal{T}_r$ be all original non-AP members of size greater than r. The following bound is automatic if this class is empty. Otherwise take an actual shortest member S of this class, of size $s \geq r+1$. Assign every member of $\mathcal{T}_r$ to its first contact with S and count its at least $s-1$ other ambient points. The selected member S itself is included. Summing (67) over these pairs, using $P_0(L) + P_0(R) \leq P_0(L+R)$ and $s/(s-1) \leq (r+1)/r$, yields

$$\sum_{A \in \mathcal{T}_r} \min(\delta(A), q) \leq \frac{r+1}{r}\big((H-1)(H-2) + qP_0(H-1)\big).$$

The identity

$$q|\mathcal{T}_r| = \sum_{A \in \mathcal{T}_r} \min(\delta(A), q) + \sum_{j=1}^{q-1} \#\{A \in \mathcal{T}_r : \delta(A) \leq j\}$$

therefore gives

$$|\mathcal{T}_r| \leq U(H; r, q) := \frac{r+1}{r}\left(\frac{(H-1)(H-2)}{q} + P_0(H-1)\right) + \frac{\mathrm{CDF}(H; r, q)}{q}. \tag{69}$$

There is no assumption of a short member, a common point, or an actual global minimum equal to $r+1$.

For an integer ambient box $[L,H]$, $B(N) \geq B(L)$ and the AP upper bound applies to the same original family embedded in $[H]$. Thus

$$|\{A \in \mathcal{F} : A \text{ is non-AP},\ |A| \leq r\}| \geq M(L,H;r,q) := B(L) - R_{15}(H) - U(H;r,q). \tag{70}$$

The density and reservoir are compatible bounds for this one family; no family at a different point of the box is substituted.

G.4 ROOT160 covers every possible actual minimum

First verify $M(L,H;160,41) > 180(H-2) > 0$. There is consequently an actual non-AP member of size at most 160. This excludes the entire unbounded branch with global minimum at least 161; it does not invent an actual member of size 160.

Suppose all actual members of size at most 160 have empty total intersection. Choose their shortest member S, of size $s \in [15,160]$; it is also a shortest member of the whole family. Choose another such member T omitting a point of S, and put $t = |S \cap T| \in [1, s-1]$. Order $S \cap T$ first, followed by the remaining points of S. For each of the first t anchors choose an actual member of size at most 160 omitting it; for all later anchors use T. Remove only earlier anchors from the corresponding partner sets. Every original member has a first contact with S and meets the selected witness at that contact, giving an ambient covering pair. The number of pairs is at most

$$160t + (s-t)(160-t) = 160s - t(s-t) \leq 159s + 1.$$

Form a multiset containing one copy of $C_0(g)$ for every $1 \leq g \leq H-1$ and a second copy when $2g \leq H-1$. The complete two-sided gap multiset of any actual anchor embeds in it. Let $F_H(j)$ be the sum of its j largest entries. This function is nondecreasing and discretely concave. Each anchor has at most 160 partners, and there are at most $159s+1$ partners in total. Concave balancing gives

$$\sum_{\text{cover}} C_0(g) \leq (s-1)F_H(159) + F_H(160).$$

For $a \leq s \leq b$ the right side may safely be enlarged by replacing s by b.

Define

$$U_1(H,a) = \sum_d T_d(H; a-1, 1, 1),$$

$$U_t(H,a) = \sum_d \min\{T_d(H;a-1,1,t), T_d(H;a-1,1,1) + T_d(H;a-1,2,t)\} \quad (t \geq 2),$$

with differences up to $\lfloor (H-1)/(a-t-1) \rfloor$ and zero contribution from physically empty classes. Defect one and defects $2,\dots,t$ are disjoint; the minimum is taken before summing over differences. Put $C_q(H,a) = \sum_{t=1}^{q-1} U_t(H,a)$. Summing (67) over the covering pairs and restoring the low-defect contribution once gives the sufficient contradiction

$$\begin{aligned} q\,[B(L) - R_a(H)] > {} & (159b+1)(H-2) \\ & + q[(b-1)F_H(159) + F_H(160)] + C_q(H,a), \\ & 1 \leq q \leq \min(12, a-3). \end{aligned} \tag{71}$$

In each successful ambient box the adjacent minimum-size rectangles cover all integers $15,\dots,160$. All actual members of size at most 160 must therefore have one actual common point p.

G.5 Three mass stages retain the same common point

Every ambient box verifies

$$M(L,H;160,41) > 180(H-2), \tag{72}$$
$$M(L,H;180,46) > 250(H-2), \tag{73}$$
$$M(L,H;250,65) > \rho H^2. \tag{74}$$

First (72) gives nonemptiness, and then ROOT selects p; the order is important. Every member A avoiding p meets each rooted short non-AP member, which can be charged to a pair $\{p,x\}$ with $x \in A$. Each pair supports at most $N-2 \le H-2$ non-AP members. Thus (72) forces $|A| > 180$, rooting every member of size at most 180 at the same p. Next (73) forces $|A| > 250$, again without changing p. Finally (74) supplies more than $\rho H^2 \ge \rho N^2$ actual non-AP members at this root. The three mass estimates concern the same original family; their masses are not added together and their roots are not reselected.

G.6 Five exact parameter choices for the final counting inequality

Write $n = N-1$, $\gamma = 17/15$ and $a = \rho/\gamma$. The unweighted common-pair theorem already gives at most $N-2 < \gamma n$ original non-AP members through any pair; this weaker coefficient is therefore valid throughout the present range. The strict rooted mass forces every avoiding member to have size greater than an. If there are $c > 0$ non-AP avoiders, counting all their ambient nonroot pairs gives

$$\frac{c}{n} < \frac{\gamma(n-1)}{a(an-1)}.$$

For $c = 0$ any positive upper bound suffices; no division by an empty count is used. Since $0 < a < 1$, the right side decreases with n: the derivative numerator of $(n-1)/(an-1)$ is $a - 1 < 0$.

Define

$$U_d(n) = \left\lfloor \frac{(\lfloor n/d \rfloor + 2)^2}{4} \right\rfloor, \qquad R(n) = \sum_{d=1}^{n} U_d(n).$$

Here $U_1(n)$ means the $d = 1$ term. This complete-difference reservoir $R(n)$ differs from the minimum-size reservoir $R_a(H)$. The full visible-four-edge counting theorem, valid when all members have size at least thirteen, gives the sufficient condition

$$3B(N) + \lfloor n/4 \rfloor > 4R(n) - U_1(n) + 4c.$$

Its separate empty-AP-avoider branch is included. No fictitious AP witness is used to invoke a nonempty-class statement. The left side minus the right side without $4c$ is strictly greater than

$$\frac{21}{200}n^2 - 4nH_n - \frac{n}{2}.$$

For each row of the table, put $n_s = L_s - 1$ and $c_s = \gamma(n_s-1)/[a_s(a_s n_s - 1)]$ and verify

$$E_s = \left(\frac{21}{200} - \frac{4}{h_s}\right) n_s - \frac{1}{2} - 4c_s > 0.$$

L_s	upper endpoint	ρ_s	h_s	E_s
75000	99999	31/1000	3500	2350945245560169/13615353171000
100000	149999	1/40	5000	14826385435271/13493745000
150000	299999	1/50	7000	4458306935381/4048443000
300000	449999	3/200	10000	1930543640537/351405000
450000	1025000	3/250	10000	3621331792512439/546634035000

Each ambient box lies within a single table row.

For the full harmonic continuation, choose j with $2^j \le n_s < 2^{j+1}$. The program checks $n_s \ge h_s(j+2)$ and $2^{j+1} \ge h_s(j+3)$ exactly. The bound $H_n \le j+2$ handles the initial partial dyadic block. The second guard handles the next full block, and doubling preserves $H_n \le n/h_s$ on all later blocks. Thus for every $n \ge n_s$ the final gap is strictly greater than

$$n\left[\left(\frac{21}{200} - \frac{4}{h_s}\right)n - \frac{1}{2} - 4c_s\right] > 0.$$

The three mass stages therefore close the nonstar contradiction in every certified box. Analytic continuation applies only to this final inequality; it does not supply mass on an uncomputed interval.

G.7 Complete certificate scope and justified optimizations

There are 1492 disjoint adjacent successful integer boxes covering all 950001 integers $75000, \dots, 1025000$, and 29168 minimum-size rectangles for ROOT. The minimum strict integer ROOT margin is 6299. A separately implemented checker reconstructs $C_0, P_0, F_H(159), F_H(160)$ and the AP reservoir, recomputes every selected prefix capacity and ROOT correction, and does not call the producer's capacity implementation. It checks all 349860653 prefix witnesses and 426516217 capacity calls, together with the five endpoint inequalities, all box junctions and the dyadic guards.

The producer may reuse the choices (π_j, ν_j) from an earlier ambient box as a candidate template. At every new H, d, j it still recomputes the qualified capacity, the earlier value D_{π_j} and every strict margin. A template contributes no old numerical bound, density, capacity or acceptance state, and need not be optimal. Parameter caches and quotient-block summation likewise reduce search without reducing the verified mathematical scope. Square roots have exact integer brackets and adjacent-root checks. Checked integer arithmetic covers capacity intermediates; large scalar cross-products use arbitrary-precision integers.

The independent T01 theorem covers all $N \ge 1025001$ without assuming minimum size nine, thirteen or fifteen. The minimum-size reduction is therefore acyclic. The present finite certificates meet that tail at adjacent integers, completing the upper bound for $N \ge 75000$. The exact star construction supplies the lower bound, giving $t(N) = B(N)$. This is a computer-assisted proof with explicit analytic inputs and complete finite checks, not a claim of full formal verification.

H The two complementary branches for 7000 to 9999

This appendix treats only $7000 \le N \le 9999$ and does not use the minimum-fifteen conclusion. Original members, maximum AP cores and original defects remain fixed throughout. The complete selected certificates and small-member numerical gates are described in the computation report. Write $B(N) = \binom{N}{2} + 1 + \lfloor (N-1)/4 \rfloor$.

H.1 Stocks, prefixes and complete tail identities

Fix an integer box $L \le N \le H$. For each non-AP member A, choose a fixed maximum AP core I_A (an AP subset of maximum cardinality), retaining its common difference d_A and original defect $j_A = |A| - |I_A| \ge 1$. Embedding in $[H]$ changes none of these objects. For an actual

whole-family minimum $a \ge 2$, put

$$f_a(v) = \sum_{h=a}^{v} \min(h, v+1-h), \qquad f_a(v) = 0 \ (v < a),$$
$$R_a(N) = \sum_{d=1}^{\lfloor (N-1)/(a-1) \rfloor} f_a\Big(\Big\lfloor \frac{N-1}{d} \Big\rfloor + 1\Big),$$
$$D_a = \min_{L \le n \le H} \{B(n) - R_a(n)\},$$
$$E_a = \min_{L \le n \le H} \{B(n) - R_a(n) + f_a(n)\}.$$

Every integer difference is evaluated at the same n; no endpoint monotonicity is assumed. If C counts non-AP members and O counts ordinary intervals, then

$$C \ge D_a, \qquad O \ge E_a - C, \qquad O \le f_a(H). \tag{75}$$

Indeed, actual APs of one common difference lie in one residue chain and form pairwise intersecting intervals there. They have a common chain point, and the central point simultaneously maximizes every length stock. Separating the common-difference-one term gives the second inequality. D_a, E_a need not attain their minima at the same integer.

For non-AP members of size at least $r+1$, core difference d, and defect in $[\ell, u]$, let $\mathrm{Cap}_H(r, d, \ell, u; P, b)$ denote a complete upper bound furnished by the local capacity lemmas. Its basic guards include

$$1 \le \ell \le u \le J, \qquad r + 1 - u \ge 4.$$

Here P must be a previously proved lower bound for the number of actual ordinary members of lengths at least b. An actual interval owner and an inverse bound for an interval union are different objects; the latter does not inherit the owner's reservations. The class is physically empty if $d(r-u) > H-1$. Complete legal alternatives combine by minimum; a sum requires a proved disjoint, exhaustive partition of original rows.

Let $X_{d,j}$ count applicable original members with core difference d and defect at most j. For $0 \le p < j \le u \le J$,

$$X_{d,j} \le X_{d,p} + \mathrm{Cap}_H(r, d, p+1, u; P, b). \tag{76}$$

This is just a defect partition; allowing $u > j$ gives a safe upper bound. Any legal selected path is sufficient, without a claim of search optimality. The three forward choices are $\mathrm{Cap}(1, j)$, $B_{d,j-1} + \mathrm{Cap}(j, j)$ and $B_{d,1} + \mathrm{Cap}(2, j)$. Some earlier records instead use paths in the full interval-constraint graph. Its reverse zero edges express $X_{d,j-1} \le X_{d,j}$, so summing any path from zero is valid. Every $d \le \lfloor (H-1)/(r-J) \rfloor$ must be covered; greater differences are excluded by span. Let Π_j be the resulting prefix bound summed over all differences.

Put $\Gamma = H - 2$, and let $\mathcal{B}_q$ be a valid complete-vertex bound for the truncated pair error on the full physical host; it may be the minimum of separate complete truncated-resource bounds. If the non-AP tail $\mathcal{T}_r = \{A : |A| > r\}$ is nonempty, choose its own actual minimum member S, of size $s \ge r+1$. Every other tail member gives at least $s-1$ cut pairs relative to S. Sum the truncated pair account, restore S once, and use

$$q|\mathcal{T}_r| = \sum_{A \in \mathcal{T}_r} \min(j_A, q) + \sum_{j=1}^{q-1} \#\{A \in \mathcal{T}_r : j_A \le j\}$$

to obtain

$$|\mathcal{T}_r| \le U_r^{\mathrm{cross}}(q) := \left\lfloor 1 + \frac{(r+1)((H-r-1)\Gamma + \mathcal{B}_q)}{rq} + \frac{\sum_{j<q} \Pi_j}{q} \right\rfloor. \tag{77}$$

Before substituting $s \geq r+1$, use

$$\frac{s((H-s)\Gamma+\mathcal{B}_q)}{s-1} = (H-1-s)\Gamma+\mathcal{B}_q+\frac{(H-1)\Gamma+\mathcal{B}_q}{s-1}.$$

This is nonincreasing for $s \geq 2$. Thus the bound covers every actual tail minimum, without postulating a member of size exactly $r+1$. The empty tail satisfies the same nonnegative bound. Defects greater than J remain paid by the truncated tail fee.

H.2 The small-minimum branch

Fix $7000 \leq N \leq 9999$, put $n = N-1$, and suppose $|\mathcal{F}| > B(N)$. An empty member forces at most one row, and a singleton forces a star, contrary to the prescribed-root bound. Thus the actual minimum k lies between two and ten.

For a fixed root p and an actual avoider A of size a, the contact-pair injection and physical selfclosed cost give

$$\#\{\text{rooted non-AP members}\} + \#\{\text{rooted AP triples}\} \leq K_N(a) := an - \frac{a(a+1)}{2} + T_a. \quad (78)$$

Here T_a bounds the sum of the a largest selfclosed costs at any physical root. Rows with an intrinsic pair touching A use their private pairs, in a pool of size $\binom{n}{2} - \binom{n-a}{2} = an - a(a+1)/2$. Other rows are selfclosed at an actual contact point and use the complete cost there. This distinguishes intrinsic privacy from an unjustified privacy claim for arbitrary non-AP triples.

The physical resources are

Host H	Maximum full vertex sum $D^*(H)$	Maximum pair cost
7500	365931	798
8500	429118	886
10000	527789	1004

Use the respective host for $N \leq 7500$, $7501 \leq N \leq 8500$ and $8501 \leq N \leq 9999$. Above table length 2048 use

$$T_a \leq \min\left\{D^*, T_{2048} + (a-2048)\left\lfloor \frac{T_{2048}}{2048} \right\rfloor\right\}.$$

This follows from decreasing order statistics at each fixed root; the worst root need not be the same for different a. If all avoiders have size at least $r+1$, the arbitrary-root rank bound and their contacts with the $k-1$ points of an actual minimum member other than p give

$$c \leq (k-1)\left\lfloor \frac{n(n-1)+D^*}{2(r-1)} \right\rfloor, \quad (79)$$

where c counts non-AP avoiders and the root stays fixed.

First handle a nonempty avoider class whose true intersection has at least two points. The original integer-by-integer entrance uses the complete $H = 10000$ host throughout; the smaller hosts used later must not be silently substituted into this entrance record. A non-AP avoider with a private point can be injected into a point outside the true intersection; one without a private point is selfclosed at two fixed core points. If b_p counts actual pairs through p, then $c + b_p \leq n + C^*_{\max}$. With $a_0 = \lfloor n/15 \rfloor + 1$, (78), the once-only global stock of APs of size at least four, and this budget yield the entrance bound

$$|\mathcal{F}| \leq R_4(N) + a_0 n - \frac{a_0(a_0-1)}{2} + N - 2 + D^* + C^*_{\max}$$

if an avoider has size at most a_0. All 3000 integer records compare this with $\binom{N}{2}+1$. Hence the avoiders are long and the fixed-root budget is licensed. The owner, hole and closed-simplex arguments then give $|\mathcal{F}| \le \binom{N}{2}+1$. The uniform CORE substitution is proved in Section H.4. This true-core conclusion precedes the unique-pair reduction: if two actual pairs meet at p, every avoider contains their other two endpoints and the true-core conclusion excludes the counterexample. Thus a counterexample has at most one actual pair.

Choose an actual minimum member S of size k. Assign non-AP members and AP triples disjointly to their first actual contact with S; pay other AP members once globally. Put $\epsilon = 1$ for $k = 2$, and zero otherwise. Choose the largest integer r such that

$$R_{\max(4,k)}(N) + kK_N(r) + \epsilon \le B(N).$$

If every point of S had an actual avoider of size at most r, (78) would contradict $|\mathcal{F}| > B(N)$. Thus one fixed $p \in S$ has all avoiders of size at least $r+1$. At that root iterate (79) and

$$m = B(N) + 1 - \epsilon - R_{\max(3,k)}(N) - c, \qquad r_{\text{new}} = \max\left\{r, \min\{a : K_N(a) \ge m\} - 1\right\},$$

using a completed state's length bound only in the next state. The 27000 channels cover every (N,k) and yield

Actual minimum	Exception count $c + b_p$	All avoider sizes	AP differences
2	$\le 6n$	$> n/12 + 1$	≤ 11
3	$\le 59n/5$	$> n/12 + 1$	≤ 11
4–10	$\le 63n$	$> n/13 + 1$	≤ 12

For $k \ge 4$ there are no actual triples. The exhaustive fixed-root cases are an empty AP-avoider class, all AP avoiders straddling the root, a one-sided witness with zero hull span, and a one-sided witness with positive span. The first two use explicit rational quadratics with positive leading coefficient, value and derivative at $n = 6999$. The zero-span lemma requires that every actual pair contain the same root. For $k = 2$ this follows from the unique pair and $p \in S$; for $k \ge 3$ there are no pairs. This scope correction is essential.

At positive span the complete closed simplex is covered by owner, common adjacent-pair, bad-trace packing or inventory terminals. The small-exception branch deletes at most 1681 bad avoiders, applies the common-pair theorem to the remaining family, and restores them once:

$$|\mathcal{F}| \le \binom{N}{2} + 1 + 1681 \le B(N).$$

The larger-minimum branch uses the inventory contradiction with no rooted triples. The reconstructed trees have 4613 and 263 nodes; they must not be mixed with the differently sized original certificate sets. Passage from the base 6999 to the whole actual range uses complete normalized real-ray monotonicity, not tests at box corners. This proves the small-minimum branch.

H.3 The large-minimum branch

We use the complete V97 reconstruction; its 816 certificates are not mixed with the 544 certificates of V62. Suppose $|\mathcal{F}| \ge B(N)$ and the actual minimum is at least 11. For a window $a \in [A, A']$, take

$$R = 49 \quad (11 \le a \le 31), \qquad R = 79 \quad (32 \le a \le 79), \qquad R = 132 \quad (80 \le a \le 132).$$

Let K_R contain all original members of size at most R, and temporarily suppose it has no common point. Choose an actual minimum member S and $T \in K_R$ with $1 \le t = |S \cap T| \le a-1$,

ordering these intersection points first. For each root point choose an actual small partner omitting it and remove earlier roots from the partner. Each original member supplies a partner point at its first contact with S, and unordered pairs are not repeated. Each outgoing bucket has at most R pairs and the total is at most

$$Rt + (a - t)(R - t) \le (R - 1)a + 1.$$

There is no assertion about total undirected degrees. For the single fixed dominating error multiset, let $\mathrm{Top}_q(s)$ be its largest-s sum. Discrete concavity bounds the complete error by $(A' - 1)\,\mathrm{Top}_q(R - 1) + \mathrm{Top}_q(R)$, giving

$$C \le \left\lfloor \frac{((R-1)A' + 1)\Gamma + (A' - 1)\,\mathrm{Top}_q(R-1) + \mathrm{Top}_q(R) + \sum_{j<q} \Pi_j}{q} \right\rfloor. \tag{80}$$

Start at $P = 0$ and update ordinary populations by (75) only after completing a state. A terminal upper bound below D_A contradicts the temporary hypothesis. The actual minimum belongs to K_R, so K_R is nonempty; fix one common point p. Discard all populations, prefixes and owners justified only under the no-common-point hypothesis.

All non-AP members of size at most R contain p and number at most

$$\mathrm{Rank}_A(H) = \left\lfloor \frac{(H-1)(H-2) + F_2(H)}{2(A-2)} \right\rfloor.$$

Here F_2 is the complete rank resource, not a truncated β resource. For the remaining tail start new unconditional prefixes at $P = 0$. The actual minimum member is outside the tail, so cutting against it needs no restoration term. Define

$$\begin{aligned} g_t(k, j) &= \max\{(k-1)\min(j, t) + j\min(\max(j, \lfloor k/2 \rfloor), t),\ 2(k + j - 2)\}, \\ K_t &= \max_{A \le a \le A'} a\{(H - a)\Gamma + (t - 1)D^*\}, \\ v_j &= g_t(\max(R + 1 - j, 2), j), \qquad w_j = \min_{z \ge j} v_z. \end{aligned}$$

The two fees act on the same account and combine by maximum, not addition. For $z \ge \max(R - 1, t)$, $v_z = t(z + 1)$ and increases, so its finite head and linear tail determine the suffix over all defects. Nonnegative telescoping gives

$$|\mathcal{T}_R| \le U_{\mathrm{mix}}(t, Q) = \left\lfloor \frac{K_t + \sum_{j<Q}(w_{j+1} - w_j)\Pi_j}{w_Q} \right\rfloor.$$

Take its minimum with (77), obtaining U and $C \le \mathrm{Rank}_A + U$ from the fresh global stock. For $A \le 31$, if the next population counts only ordinary lengths at least 31, subtract the full shorter-length bound $\sum_{h=A}^{30} h$. For $A \ge 32$ no trim is needed. Every state uses only the population supplied by the preceding completed state. Each global window terminates with $\mathrm{Rank}_A + U < D_A$.

For actual minimum at least 133, use six independent width-500 boxes with $r = 132, J = 45, P = 0$ and (77). The V97 strict comparisons are

N-range	q	Integer whole-tail upper bound	$\min D_{133}$
7000–7499	43	4386942	4966482
7500–7999	43	5026205	5653852
8000–8499	42	5708671	6386733
8500–8999	42	6444600	7162695
9000–9499	42	7226589	7981219
9500–9999	42	8059110	8849362

These integers are certificate targets. Replay derives the upper bounds from resources and prefixes; comparing the two unverified table columns alone would not suffice. ROOT and GLOBAL use the same radius at every actual (N, a). Each width-100 box has 18 ROOT and nine GLOBAL windows, giving 810 certificates; the six unbounded-minimum certificates bring the total to 816. The adjacent small- and large-minimum branches give the upper bound throughout $7000 \le N \le 9999$, and the construction gives equality.

H.4 A common true-core gate for the two low ranges

We replace only the numerical gate of the true-core branch, without changing the original family, root or counting identity. Put $n = N - 1$, $b(N) = \binom{N}{2} + 1$ and $B(N) = b(N) + \lfloor n/4 \rfloor$. After the true-core entrance has been established, fix the same root p. Let H_3 count rooted members of size at least three, let I count all AP avoiders, and let e_{eff} be the remaining contribution in the identity

$$\widehat{F} = |\mathcal{F}| + \lfloor n/4 \rfloor = H_3 + I + e_{\mathrm{eff}}.$$

In both ranges the entrance must first prove that the avoider class is nonempty, every actual pair contains this p, every actual avoider has size strictly greater than $n/15 + 1$, and the asserted bound for e_{eff}. These are geometric and counting hypotheses, not conclusions drawn from the numerical tree.

For $4000 \le N \le 6999$ the entrance gives $e_{\mathrm{eff}} \le (3/2)n$; for $7000 \le N \le 9999$ it gives $e_{\mathrm{eff}} \le (71/50)n$. Since $71/50 < 3/2$, both satisfy the same, weaker budget $e_{\mathrm{eff}} \le Cn$ with $C = 3/2$. No member is charged a second time, and $\widehat{F}$ is unchanged. An AP of common difference d has at most $\lfloor n/d \rfloor + 1$ points, so the identical length hypothesis gives $d \le 14$ in both ranges. We may therefore use the common CORE gate $C = 3/2, D = 14$.

Its base is $n_0 = 3999$ and its domain is the whole closed simplex $p, q, x \ge 0$, $p + q + x \le n_0$. The V130 rational replay traverses all 455 nodes: 227 closed splits, 198 OS leaves and 30 COMMON/HOLE leaves. We prove that every leaf persists on its complete normalized real ray. This allows the earlier CORE tree used in V86 to be replaced explicitly.

Persistence of OS leaves. Write $a = 129/200$, $\gamma = 16641/131600$, $c = 46/63$, and

$$g(z) = \begin{cases} z/2, & 0 \le z \le 2, \\ \min\{z/2, \log z + 3/5\}, & z > 2, \end{cases} \qquad s(z) = \min\{9/2, 1 + \log \max(1, z)\}.$$

Fix arbitrary real $P, Q, X \ge 0$, $P + Q + X \le 1$, and put $(p, q, x) = n(P, Q, X)$, $W = P + Q + X$. The quadratic form

$$Q_0 = \tfrac{1}{2}\{(p+q)^2 + x^2 - a^2 p^2\} - a(p+q)x$$

is constant after division by n^2. The complete error is

$$\begin{aligned} E &= pg(q) + (2 - \gamma)pg(p) + ps(q) + 2\sqrt{n+1} + \max(0, Z) - \tfrac{1}{2} + (nW + 1)(H_b - 1), \\ Z &= -n/2 + (329/200 + 3/2)(p + q) + (c + 1)q + (9/2)x, \\ H_b &= \log(n/2) + 3/5 + 1/n. \end{aligned}$$

The OS test is $Q_0 - E - Cn - (n+1)/4 > 0$. The function $g(z)/z$ is nonincreasing on $z > 0$: on its logarithmic branch the derivative has sign $2/5 - \log z < 0$, and the branches join continuously. Likewise $s(z)/z$ is nonincreasing, by its three branches; the logarithmic branch has derivative numerator $-\log z \le 0$. At zero coordinates use $g(0) = 0, s(0) = 1$ directly. The first three terms of E/n^2 are therefore nonincreasing. So are $\sqrt{n+1}/n^2$ and $\max(0, Z)/n^2 = \max(0, Z_0)/n$, where $Z = nZ_0$.

The negative constant $-1/2$ must be retained. With $h(n) = \log(n/2) - 2/5$, the remaining terms satisfy the exact identity

$$\frac{(nW+1)(H_b-1) - 1/2 + Cn + (n+1)/4}{n^2} = \frac{Wh(n)}{n} + \frac{h(n) + W - 1/4}{n^2} + \frac{1}{n^3} + \frac{C+1/4}{n}.$$

For $n \geq 16$, $h(n) > 1$. The derivatives of the first two terms are $W(1-h(n))/n^2 \leq 0$ and $(3/2 - 2h(n) - 2W)/n^3 < 0$; the other two terms decrease. Thus the complete normalized error is nonincreasing, including zero coordinates, truncation faces and the face $Z = 0$. Every strict base OS comparison persists for every real $n \geq 3999$.

Persistence of COMMON/HOLE leaves. Let $B_D = \sum_{d=2}^{D} d^{-2}$, $H_D = \sum_{d=1}^{D} d^{-1}$ and $T = P + Q$. The complete inventory and exception budget give the hole-bound quantity

$$W_D = (B_D(p+q) + q)(x+1) + (C + 47/20)n + (p+q+x)(H_D - 1) + H_D + D - 9/4.$$

Hence

$$\frac{W_D}{n^2} = (B_D T + Q)X + \frac{B_D T + Q + C + 47/20 + W(H_D - 1)}{n} + \frac{H_D + D - 9/4}{n^2}.$$

For $D = 14$ these coefficients are nonnegative, so this expression is nonincreasing. The COMMON threshold $T_K = (n - p - q - x)/2$ satisfies $T_K/n = (1-W)/2$. Once $T_K > 1$ at the base, $\beta T_K(T_K - 1)/n^2$ is nondecreasing, with $\beta = 7/80$. Thus the strict comparison and its guard persist on the whole ray. This leaf excludes every actual avoider whose trace in the actual interval kernel K has size at most $\lceil |K|/2 \rceil$. An actual ordinary-interval witness then gives a common adjacent pair, and the true-core count yields $|\mathcal{F}| \leq b(N)$.

Other branches and physical scope. The empty AP-avoider class, the case in which all AP avoiders straddle the root, and the one-sided zero-span case use the complete V130 CORE argument. The rational CORE cross-step and empty-class polynomials have positive leading coefficient, value at 3999 and derivative there, so they are positive for all real $n \geq 3999$. The zero-span comparison uses $H_N \leq 1 + \log 10000$ on its finite physical domain, after checking that all actual pairs contain p. No scalar ray enlarges the domains of the combinatorial lemmas: GFL, the owner ledger and the strengthened structure lemma are used only at actual integers $4000 \leq N \leq 9999 \leq 10000$. The single gate with the larger budget therefore legitimately covers the true-core branch in both low ranges.

I Exact verification and reproducibility

This appendix records the finite computations used in this version. Counts refer to the completed verification campaign accompanying the proof, not to new independent executions implied merely by the English translation. The electronic companion retains inputs, programs, selected certificates, and detailed outputs. The computations check necessary inequalities for arbitrary putative counterexamples; they do not search all set families.

I.1 Arithmetic and acceptance conditions

Integer and rational quantities are evaluated exactly. Square-root ceilings are checked by integer squaring, and continuous gates use rational interval bounds on closed subdivisions. Displayed decimal margins are illustrative only. Fixed-width integer operations are used only with explicit bounds; larger scalar expressions use arbitrary-precision integers. The optimization selects sufficient paths and their complete ancestor closure. No proof of optimality of an unused search is needed. An accepted state must satisfy its literal hypotheses, use only previously established information, and retain the correct root, original family, maximum cores, and physical host. In particular, a conditional population bound is discarded when its premise is discharged. A numerical host upper bound does not replace the actual N in a progression inventory unless the particular monotonic bound has been proved.

The runs used Python 3.12.14, NumPy 2.3.5, and C#/.NET on Windows. The portable Python reproduction programs and native source files are included in the companion. A second implementation is used for several important checks, but we do not claim that every formula was independently reimplemented. Repeating a program, checking a file hash, or verifying a dependency graph is not a mathematical proof of its underlying lemma.

I.2 Structural foundations and the fifteen-element reduction

The starred calculation checks 8001 finite rectangles; a proved analytic continuation covers the remaining rectangles. The low-range finite gap-loss calculation checks all 50005000 nonnegative pairs (L, R) with $L + R \le 9999$, including the axes. It gives

$$\max(40M - 33P) = 12406, \qquad J = \frac{285139977338150579}{2079149797771200},$$

and the resulting constant $298481188540515779/83165991910848000 < 18/5$. The associated owner calculation checks 49985001 states and 308451 labels. Its minimum scaled slack is zero, as permitted by the non-strict acceptance inequality. These are full-domain reruns of reviewed algorithms, not claims of independent reconstruction of each algorithm.

The high-range primitive-pair calculation has its own domain and analytic continuation: 536887296 cases with $L + R < 32768$, followed by a tail proof. It does not extrapolate a low-range physical lemma beyond its stated domain. Other foundational checks include the coefficient functions through 1400000, the prescribed arithmetic gates and prime grid, and all 4500000000 pairs $1 \le k \le 30000$, $1 \le j \le 150000$ in the RAW gate. The continuous trees include 8703 nodes for O08, 909 and 717 for the two O09 trees, and 1357 for O10. Their closed coverage and rational leaves are checked; the formulas and their geometric applicability are separate mathematical inputs.

For the minimum-fifteen reduction, the range $10000 \le N \le 11993$ contains 1994 integers. All 3392 selected feedback states pass an independently implemented arithmetic check. The two entrance margins have minima 347467 and 78757, both at $N = 10001$; the joint-feedback terminal margin has minimum 150, at $N = 11294$. The 506 integers $11994 \le N \le 12499$ are checked separately, with minimum terminal margin 1697 at $N = 12001$. For $N \ge 12500$, the real-ray proof uses $n_0 = 12499$ and $L(n_0) < 142$, with positive rational anchors

$$b^2 - 196A_0 = \frac{694777241}{99984000640000}, \qquad b - H_0 = \frac{1324101897}{874860005600}.$$

The logarithmic and zeta bounds used at these anchors are justified by finite sums and analytic remainder bounds, not floating-point comparisons.

I.3 The low range

The physical resources were recomputed over all pairs at four hosts.

Host H	Pairs checked	Maximum full-root sum D^*	Maximum pair value
5500	15122250	245873	709
7500	28121250	365931	798
8500	36120750	429118	886
10000	49995000	527789	1004

This is 129359250 physical pairs in total. The selected certificates need 192 host-parameter environments, containing 50641308 host-position values and 928740 top-resource values; the additional environments computed during preparation are not counted as required certificates.

The small-member checks cover 18000 parameter cases and 78168 causal states in the lowest range, and 27000 cases and 121789 states in the seven-thousand range. The minimum-eight argument checks 3000 integers and 18002 states. Its largest $c/(N-1)$ is $196021/4000 < 50$, and its smallest $14r-(N-1)$ is 74; both occur at $N = 4001$. The rational trees for the two CORE/SMALL gates have 455 nodes each; the NO3 gate has 487 nodes, the seven-thousand trace and population gates have 4613 and 263, and the minimum-eight bridge has 249. All split boundaries are included. The rays are justified analytically with their constant terms retained, including negative ones.

There are 1138 selected V145 certificates and 816 selected V97 certificates. Their ancestor closures have 11731 and 2936 states, respectively. Verification evaluates $14512299 + 6833859$ defect-prefix entries and $12837651 + 5882899$ capacity records. The minimum strict terminal margins are 68 and 409. The full branches beginning at actual minimum sizes 64 and 133 are analytic monotonicity branches, not substitutions of a fictitious member of size exactly 64 or 133.

I.4 The middle range

The certificates cover every one of the 65000 integers from 10000 through 74999 in 214 adjacent boxes. The first 174 boxes cover $10000 \le N \le 34999$ and the remaining 40 cover $35000 \le N \le 74999$. The completed exact verification has 70406 contexts, 506234902 selected capacity evaluations, and 494314005 strict-zero checks. Resource construction visits 225373301 events and 13000000 progression-stock parameter pairs. Selection reduces the first part's capacity evaluations from 1260816555 to 369447453 by retaining just the necessary ancestors. The new upper-middle certificates have minimum margins 33125 for ROOT, 420069025/6 for the minimum-at-least-133 branch, and 1681299343/14190 for the rank step. These margins are attached to their specified inequalities and should not be compared as interchangeable units.

The native semantic verification covers all selected records. The portable checks also verify every scalar/context transition and selected independent capacity and stock examples. The latter tests are cross-checks; they are not described as a second complete portable replay of every middle-range capacity evaluation.

I.5 The high finite range and the infinite tail

There are 1492 adjacent high-range boxes covering all 950001 integers $75000 \le N \le 1025000$, with 29168 ROOT length rectangles. An independent checker evaluates 349860653 selected prefix records and 426516217 capacities. It reconstructs the arithmetic resources and progression stocks and uses an independent integer-square-root procedure; it does not call the producer's capacity routine. The minimum margins in the four stages are

$$6299, \qquad \frac{41572833}{3280}, \qquad \frac{187427165009}{4140}, \qquad \frac{3880269}{5000}.$$

Each newly encountered host recomputes its capacities. A path template specifies recurrence edges only; it never transfers numerical capacities between hosts. Seven deliberately corrupted certificate controls are rejected by the checker. The analytic tail starts at 1025001. Its fixed rational obligations comprise five mass inequalities, four bridges, fourteen crossing inequalities, and the required positive-definite and ellipse conditions. Finite evaluation of these fixed constants is combined with the analytic argument; it is not a test of finitely many large values of N.

I.6 What the checks establish

The combination is: proved local implications, exact accepted finite certificates with complete coverage, and proved infinite continuations. The review of this version also checks the order of hypotheses, actual member versus auxiliary subfamily, fixed root and cores, empty cases, closed boundaries, and the endpoint joins. It does not turn an internal status label or a program's exit code into an additional theorem. Partial unpublished formalization is not used to justify any step.

The refined position-kernel interface is proved explicitly in Appendix B. This replaces the inherited arithmetic-interface assertion in the older local note; its formulas and constants are unchanged, so the frozen numerical computations still apply. As a separate regression check, all admissible families of non-AP selfclosed supports at every pair on $[N]$, $2 \le N \le 11$, were enumerated: 220 pair instances, 899 candidate supports, and 1821 admissible families. All 18210 comparisons for C^*, W_Q $(1 \le Q \le 5)$, and K_t $(2 \le t \le 5)$ passed. This finite check does not replace the universal combinatorial proof.

J Accompanying mathematical texts and certificate files

The source distribution includes ancillary mathematical texts and an index mapping their original identifiers to files. They supply the complete local arguments cited in the paper, including the sparse common-AP-core theorem, the capacity menu, the small-member structural inputs, and the F/P/I/O/C/T modules of the original high-range notes. The notes retain their original versions and language; several are in Chinese. Internal words such as "verified" in a filename are historical identifiers and do not mean external peer review or proof-assistant certification.

The full computational companion is the archive

`erdos272_full_proof_and_certificates.zip`.

It contains the low-, middle-, and high-range selected-certificate archives, the foundational checks, source programs, reproduction entry points, the detailed computational report, and the fixed research texts. Its SHA-256 identity is

`ae1267e775f4ca033675ba030016503b648af6fa6f87359019b8462985812b5d`.

The corresponding byte size is 503886077. The ancillary `COMPUTATIONAL_COMPANION.json` records this identity and the identities of the three component archives; `THEORY_INDEX.json` identifies the mathematical texts supplied with the article source. The mathematical texts, the selected certificates, and the programs are complementary proof material. The smaller article-source archive does not contain the large computational companion.

The companion's `reproduce_all.py` is the reproduction entry point. The component documentation distinguishes full semantic checks, independently implemented checks, and additional regression examples. The English article revises the organization and clarifies hypotheses; the original companion remains fixed so that these computation records retain an

unambiguous identity. Any future modification of a certificate or its verifier requires a new identity and an appropriate rerun.

In particular, the global minimum-fifteen proposition is justified by Appendix E and its recorded finite verification; it is not inherited from an old file that merely asserts it. The unified low CORE gate in Appendix H.4 replaces the older borrowed gate. The current middle and high certificates replace the earlier search matrices and historical threshold junctions. The accompanying source index states these replacements; superseded material is not a prerequisite of the main theorem. The inherited refined-position interface in the older local capacity note is replaced by the self-contained proof in Appendix B, including its typed weighted bounds and the safe fringe-inventory majorant. The small-domain regression script and its result are also supplied as ancillary files. This added derivation changes no formula evaluated by the frozen certificates.

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The following items are the author's fixed research notes and computational material, not additional peer-reviewed publications. The mathematical texts are indexed in the ancillary THEORY_INDEX.json; computational records are identified in COMPUTATIONAL_COMPANION.json.

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[11] P. Wu, Global 7000 theorem, research note T020, version V126, 2026. `ERDOS272_GLOBAL7000_VERIFIED_V126.md`.

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